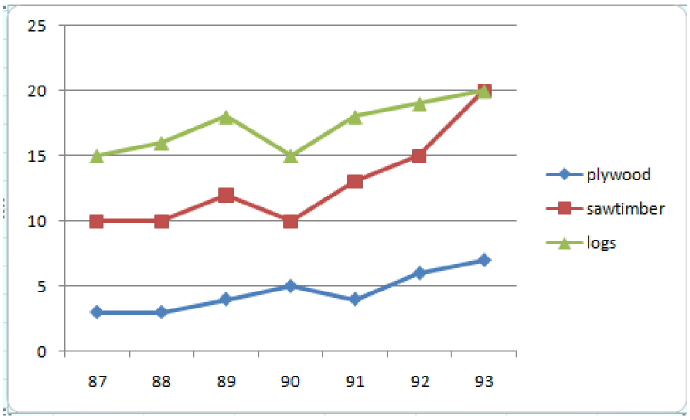
Top 163 CAT Charts Questions With Video Solutions

Questions

Instructions

Answer the questions based on the following information. In the following chart, the price of logs shown is per cubic metre that of plywood and saw timber is per tonne.



Question 1

If one cubic metre = 700 kg for plywood and 800 kg for saw timber, find in which year was the difference in the prices of plywood and saw timber (per cubic metre) the maximum?

A 1989

B 1990

C 1991

D 1992

VIEW SOLUTION

Question 2

In the previous question, if in 1994, prices increased by 5%, 1% and 10% while the volume sales break-up was 40%, 30% and 30% for plywood, saw timber and logs respectively, then what was the average realisation?

A 18.95

B 16.45

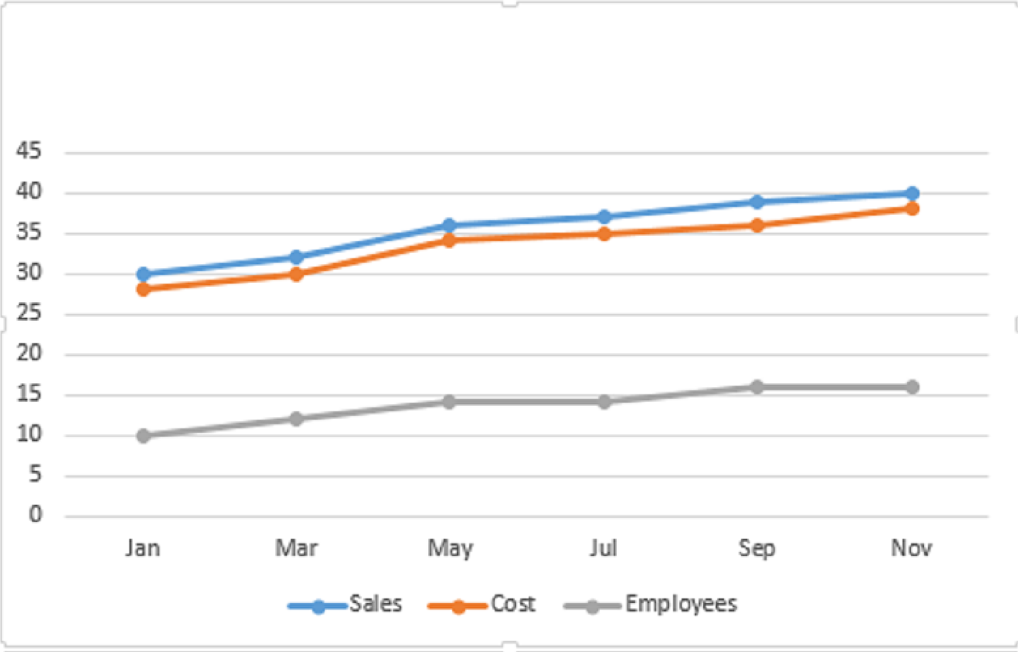
C 13.15

D 10.25

VIEW SOLUTION

Instructions

Answer the questions based on the following graph.



Employess in thousand Sales - cost = profit
(all values are integers)

Question 3

In which month is the total increase in the cost highest as compared to two months ago?

A March

B September

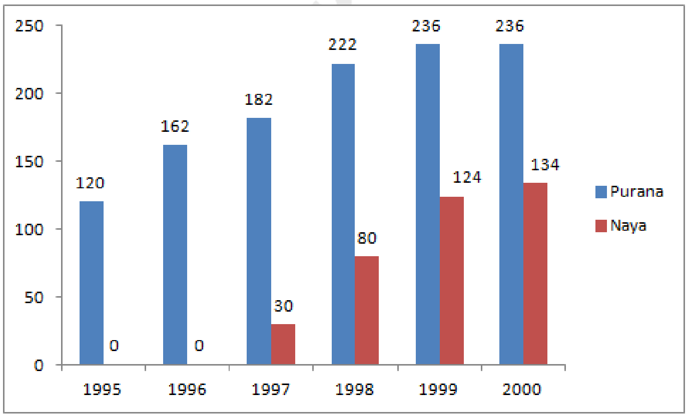
C July

D May

VIEW SOLUTION

Instructions

Directions for the following four questions: Answer the questions on the basis of the information given below.



Purana and Naya are two brands of kitchen mixer-grinders available in the local market. Purana is an old brand that was introduced in 1990, while Naya was introduced in 1997. For both these brands, 20% of the mixer-grinders bought in a particular year are disposed off as junk exactly two years later. It is knows that 10 Purana mixer-grinders were disposed off in 1997. The following figures show the number of Purana and Naya mixer-grinders in operation from 1995 to 2000, as at the end of the year.

Question 4

How many Naya mixer-grinders were purchased in 1999?

A 44

B 50

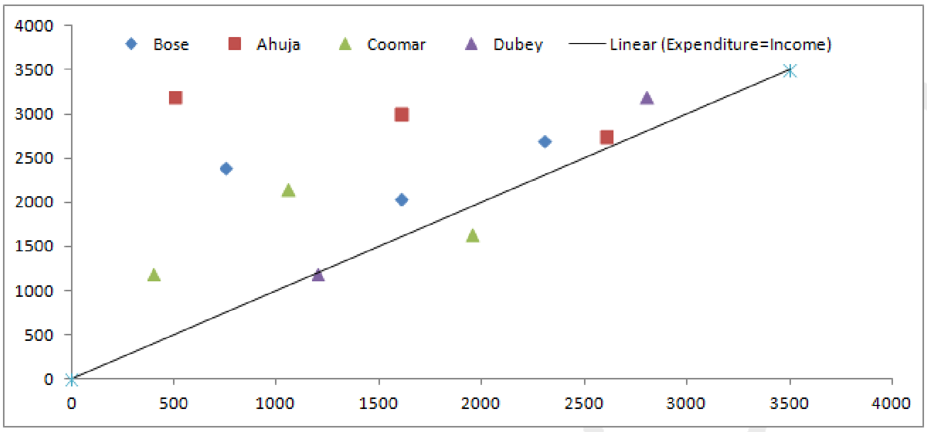
C 55

D 64

VIEW SOLUTION

Instructions

Directions for the following four questions: Answer the questions on the basis of the information given below.



The data points in the figure below represent monthly income and expenditure data of individual members of the Ahuja family, the Bose family, the Coomar family, and the Dubey family.

The X axis represents Expenditure and the Y axis represents the Income of the individual members.

For these questions, savings is defined as Savings = Income - Expenditure

Question 5

Which family has the highest average expenditure?

A Ahuja

B Bose

C Coomar

D Dubey

VIEW SOLUTION

Question 6

Which family has the lowest average income?

A Ahuja

B Bose

C Coomar

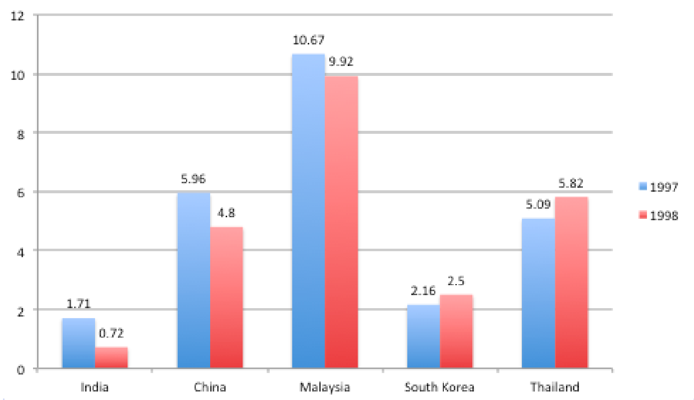
D Dubey

VIEW SOLUTION

CAT Previous Papers PDF

Instructions

Directions for the next 4 questions: Answer these questions based on the data presented in the figure below. FEI for a country in a year, is the ratio (expressed as a percentage) of its foreign equity inflows to its GDP. The following figure displays the FEIs for select Asian countries for the years 1997 and 1998.



Question 7

China's foreign equity inflows in 1998 were 10 times that into India. It can be concluded that:

A China's GDP in 1998 was 40% higher than that of India

B China's GDP in 1998 was 70% higher than that of India

C China's GDP in 1998 was 50% higher than that of India

D no inference can be drawn about relative magnitudes of China's and India's GDPs.

VIEW SOLUTION

Question 8

It is known that China's GDP in 1998 was 7% higher than its value in 1997, while India's GDP drew by 2% during the same period. The GDP of South Korea, on the other hand, fell by 5%.

Which of the following statements is/are true?

1. Foreign equity inflows to China were higher in 1998 than in 1997.
2. Foreign equity inflows to China were lower in 1998 than in 1997.
3. Foreign equity inflows to India were higher in 1998 than in 1997.
4. Foreign equity inflows to South Korea decreased in 1998 relative to 1997.
5. Foreign equity inflows to South Korea increased in 1998 relative to 1997,

A 1,3&4

B 2,3&4

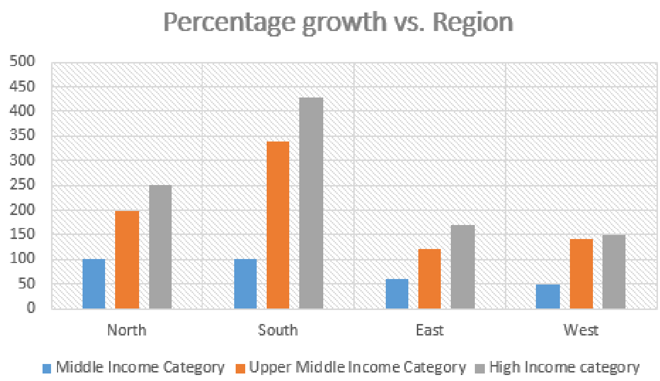
C 1,3&5

D 2&5

VIEW SOLUTION

Instructions

Answer the questions based on the following information. The following bar chart gives the growth percentage in the number of households in the middle, upper-middle and high-income categories in the four regions for the period between 1987-88 and 1994-95.



	Number of households in 1987-88	Average household income in 1987-88	Growth in average household income (1994-95 over 1987-88)
Middle Income	40	Rs. 30,000	50%
Upper-middle	10	Rs. 50,000	60%
High income	5	Rs. 75,000	90%

(Number of households in thousands)

Question 9

The average income for the northern region in 1987-88 was

A Rs. 37,727

B Rs. 37,277

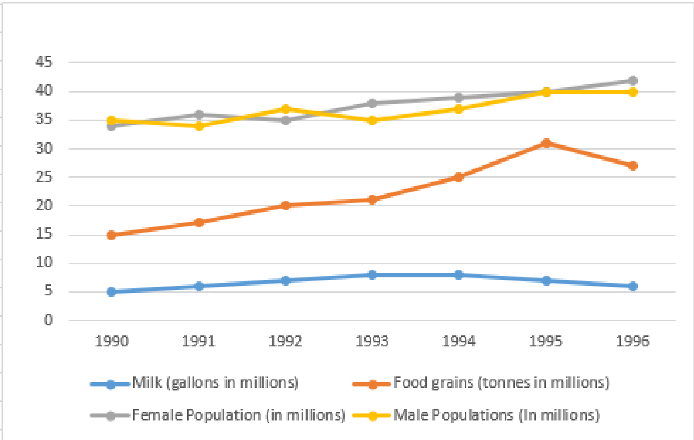
C Rs. 35,000

D Cannot be determined

VIEW SOLUTION

Instructions

The graph given below shows the quantity of milk and food grains consumed annually along with female and male population (in millions). Use the data to answer the questions that follow.



Question 10

When was the per capita production of foodgrains most?

A 1992

B 1993

C 1994

D 1995

VIEW SOLUTION

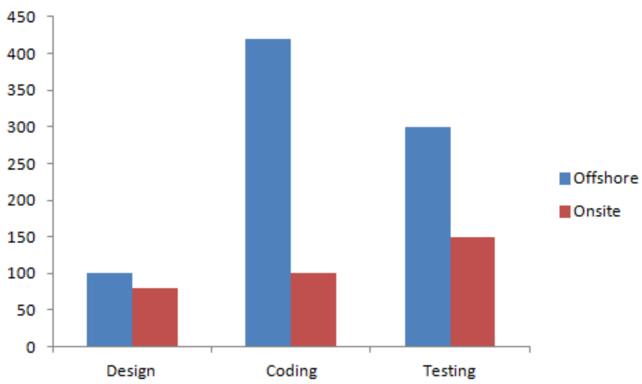
Instructions

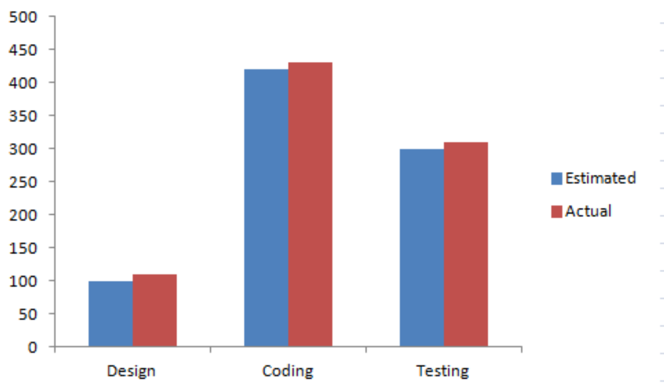
Directions for the following five questions: Answer the questions based on the two graphs shown below.

Figure I (the chart on the left) shows the amount of work distribution, in man-hours, for a software company between offshore and onsite activities.

Figure 2 (the chart on the right) shows the estimated and actual work effort involved in the different offshore activities in the same company during the same period.

[Note: Onsite refers to work performed at the customer's premise and offshore refers to work performed at the developer's premise.]





Question 11

Which work requires as many man-hours as that spent in coding?

A Offshore, design and coding

B Offshore coding

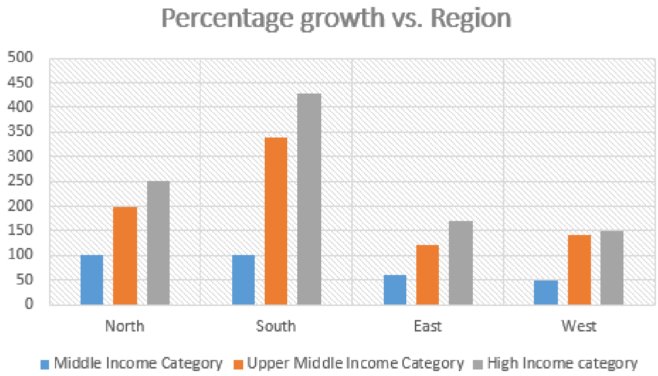
C Testing

D Offshore, testing and coding

VIEW SOLUTION

Instructions

Answer the questions based on the following information. The following bar chart gives the growth percentage in the number of households in the middle, upper-middle and high-income categories in the four regions for the period between 1987-88 and 1994-95.



	Number of households in 1987-88	Average household income in 1987-88	Growth in average household income (1994-95 over 1987-88)
Middle Income	40	Rs. 30,000	50%
Upper-middle	10	Rs. 50,000	60%
High income	5	Rs. 75,000	90%

(Number of households in thousands)

Question 12

What was the average income of the high-income group in 1987-88?

A Rs. 75,000

B Rs. 25,000

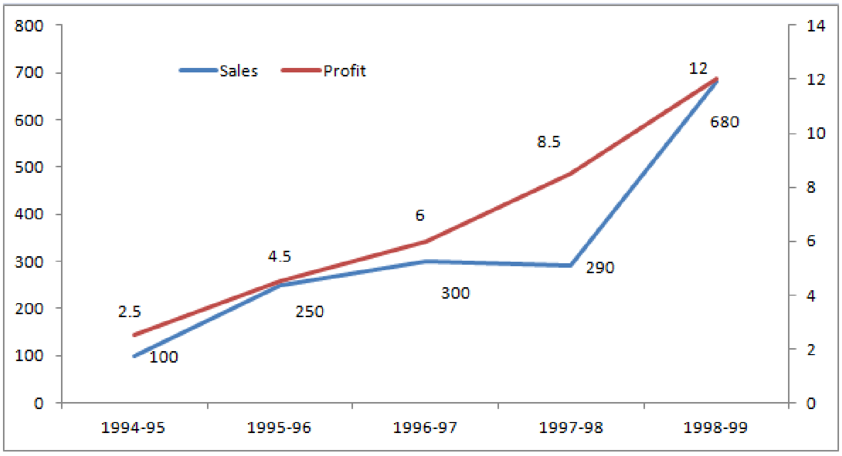
C Rs. 2,25,000

D Cannot be determined

VIEW SOLUTION

Instructions

These questions are based on the situation given below:
The figure below presents sales and net profit, in Rs. Crores, of IVP Ltd for the five years from 1994-95 to 1998-99. During this period, the sales increased from Rs.100 Crores to Rs. 680 Crores. Correspondingly, the net profit increased from Rs. 2.5 Crores to Rs. 12 Crores. Net profit is defined as the excess of sales over total costs.



Question 13

With regarded to profitability as defined in the earlier question, it can be concluded that

A Profitability is non-decreasing during the five years from 1994-95 to 1998-99.

B Profitability is non-increasing during the five years from 1994-95 to 1998-99.

C Profitability remained constant during the five years from 1994 - 95 to 1998 -99.

D None of the above.

VIEW SOLUTION

Question 14

The highest percentage growth in net profit, relative to the previous year, was achieved in

A 1998-99

B 1997-98

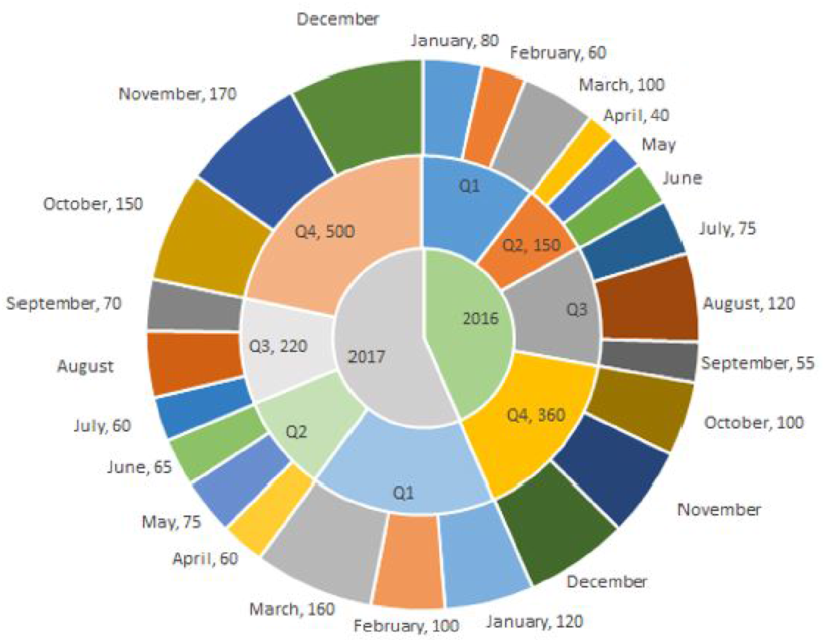
C 1996-97

D 1995-96

VIEW SOLUTION

Instructions

The multi-layered pie-chart below shows the sales of LED television sets for a big retail electronics outlet during 2016 and 2017. The outer layer shows the monthly sales during this period, with each label showing the month followed by sales figure of that month. For some months, the sales figures are not given in the chart. The middle-layer shows quarterwise aggregate sales figures (in some cases, aggregate quarter-wise sales numbers are not given next to the quarter). The innermost layer shows annual sales. It is known that the sales figures during the three months of the second quarter (April, May, June) of 2016 form an arithmetic progression, as do the three monthly sales figures in the fourth quarter (October, November, December) of that year.



Question 15

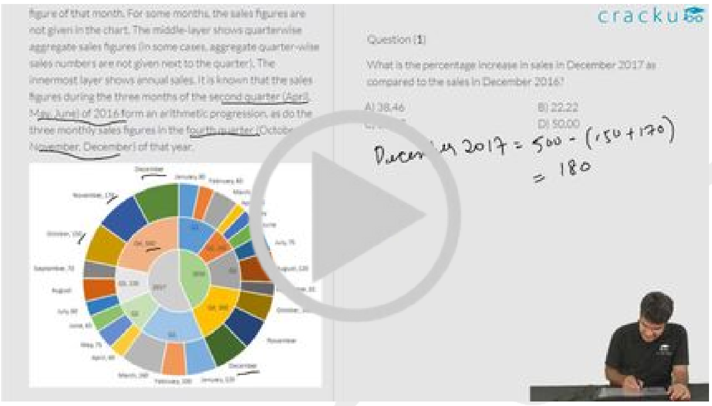
During which month was the percentage increase in sales from the previous month's sales the highest?

A March of 2017

B October of 2017

C March of 2016

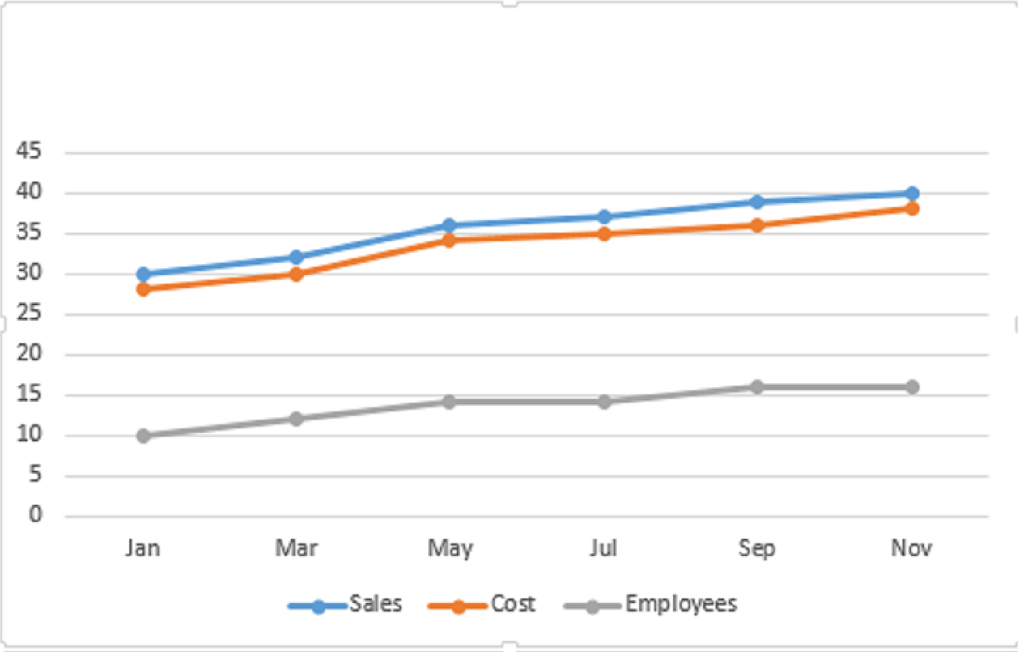
D October of 2016



VIDEO SOLUTION

Instructions

Answer the questions based on the following graph.



Employess in thousand Sales - cost = profit
(all values are integers)

Question 16

Which month records the highest profit percentage?

A September

B March

C July

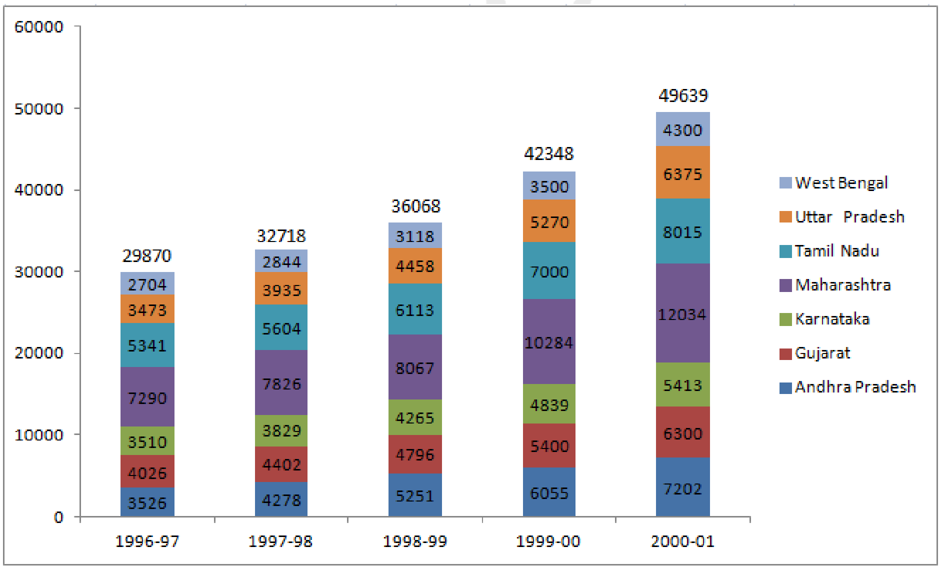
D May

VIEW SOLUTION

Instructions

The chart given below indicates the annual sales tax revenue collections (in rupees in crores) of seven states from 1997 to 2001.

The values given at the top of each bar represents the total collections in that year



Question 17

Identify the state whose tax revenue increased exactly by the same amount in two successive pair of years?

A Karnataka

B West Bengal

C Uttar Pradesh

D Tamil Nadu

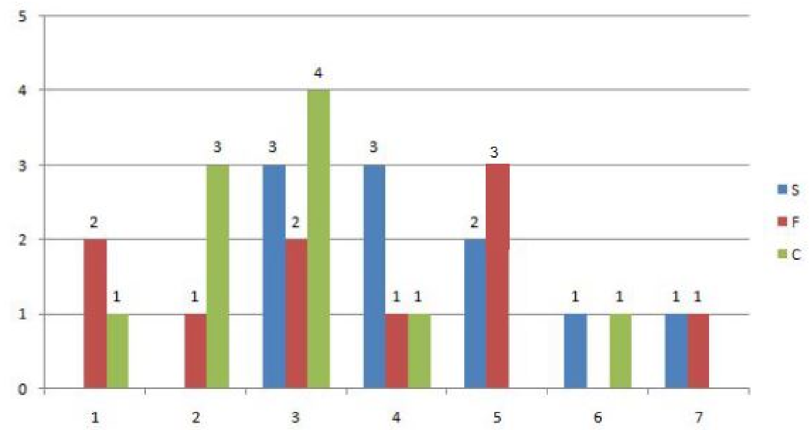
VIEW SOLUTION

Instructions

Simple Happiness index (SHI) of a country is computed on the basis of three, parameters: social support (S),freedom to life choices (F) and corruption perception (C). Each of these three parameters is measured on a scale of 0 to 8 (integers only). A country is then categorised based on the total score obtained by summing the scores of all the three parameters, as shown in the following table:

Toatal Score	0-4	5-8	9-13	14-19	20-24
Category	Very Unhappy	Unhappy	Neutral	Happy	Very Happy

Following diagram depicts the frequency distribution of the scores in S, F and C of 10 countries - Amda, Benga, Calla, Delma, Eppa, Varsa, Wanna, Xanda,Yanga and Zooma:



Further, the following are known.
1. Amda and Calla jointly have the lowest total score, 7, with identical scores in all the three parameters.
2. Zooma has a total score of 17.
3. All the 3 countries, which are categorised as happy, have the highest score In exactly one parameter.

Question 18

What is Zooma's score in S?

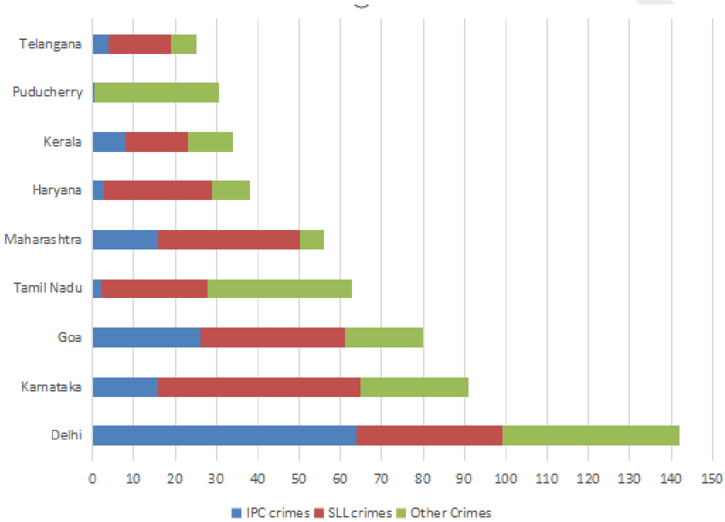
VIDEO SOLUTION

CAT Formulas PDF [Download Now]

Instructions

The Ministry of Home Affairs is analysing crimes committed by foreigners in different states and union territories (UT) of India. All cases refer to the ones registered against foreigners in 2016.

The number of cases - classified into three categories: IPC crimes, SLL crimes and other crimes - for nine states/UTs are shown in the figure below. These nine belong to the top ten states/UTs in terms of the total number of cases registered. The remaining state (among top ten) is West Bengal, where all the 520 cases registered were SLL crimes.



The table below shows the ranks of the ten states/UTs mentioned above among ALL states/UTs of India in terms of the number of cases registered in each of the three category of crimes. A state/UT is given rank r for a category of crimes if there are (r-1) states/UTs having a larger number of cases registered in that category of crimes. For example, if two states have the same number of cases in a category, and exactly three other states/UTs have larger numbers of cases registered in the same category, then both the states are given rank 4 in that category. Missing ranks in the table are denoted by *.

	IPC crimes	SLL crimes	OtherCrimes
Delhi	*	*	*
Goa	*	4	*
Haryana	8	6	*
Karnataka	3	2	*
Kerala	*	9	*
Maharashtra	3	4	8
Puducherry	13	29	*
Tamil Nadu	11	7	*
Telangana	6	9	8
WestBengal	17	*	16

Question 19

In the two states where the highest total number of cases are registered, the ratio of the total number of cases in IPC crimes to the total number in SLL crimes is closest to

A 3 : 2

B 19 : 20

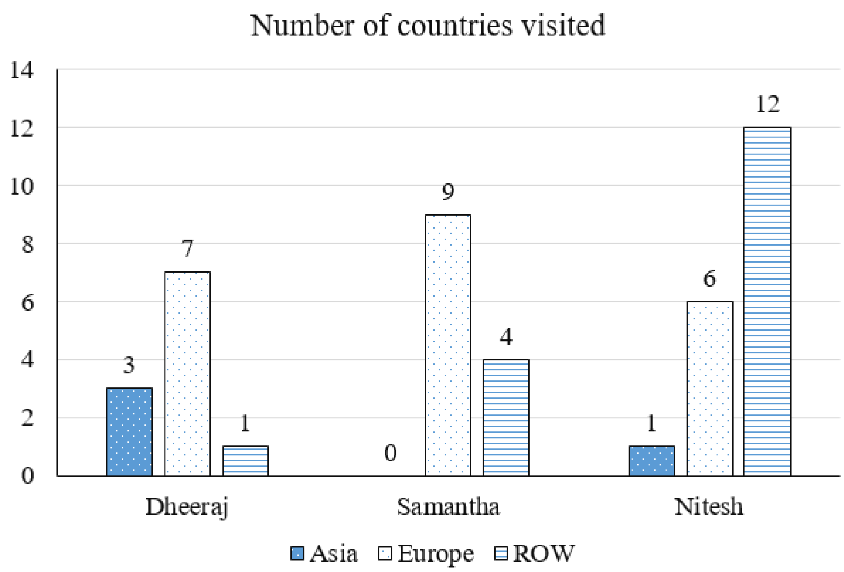
C 11 : 10

D 1 : 9

VIDEO SOLUTION

Instructions

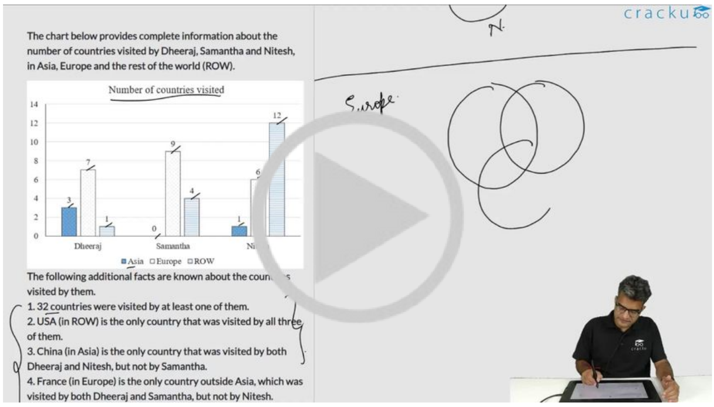
The chart below provides complete information about the number of countries visited by Dheeraj, Samantha and Nitesh, in Asia, Europe and the rest of the world (ROW).



The following additional facts are known about the countries visited by them.
1. 32 countries were visited by at least one of them.
2. USA (in ROW) is the only country that was visited by all three of them.
3. China (in Asia) is the only country that was visited by both Dheeraj and Nitesh, but not by Samantha.
4. France (in Europe) is the only country outside Asia, which was visited by both Dheeraj and Samantha, but not by Nitesh.
5. Half of the countries visited by both Samantha and Nitesh are in Europe.

Question 20

How many countries in the ROW were visited by both Nitesh and Samantha?



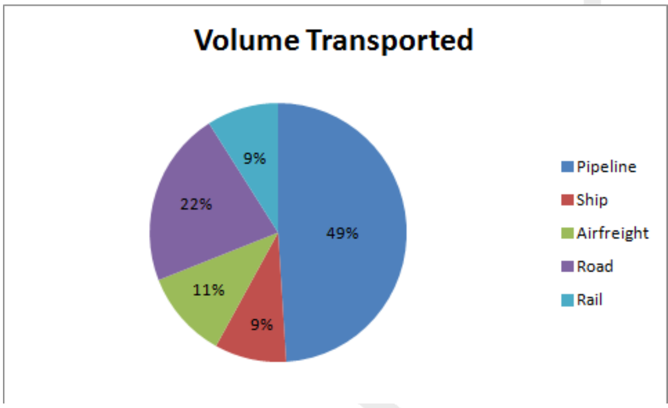
VIDEO SOLUTION

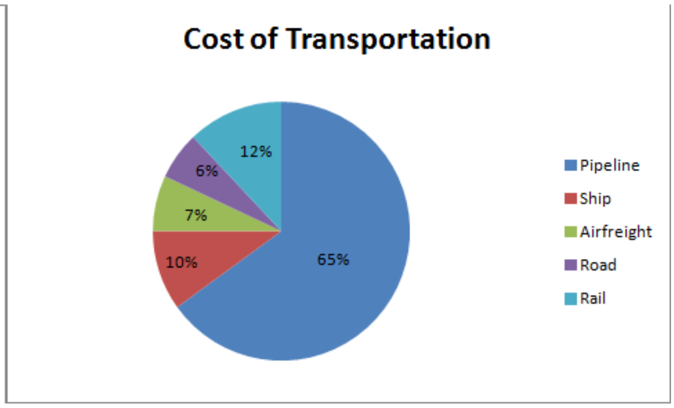
Instructions

Directions for the following three questions: Answer the questions based on the pie charts given below.

Chart 1 shows the distribution of 12 million tonnes of crude oil transported through different modes over a specific period of time.

Chart 2 shows the distribution of the cost of transporting this crude oil. The total cost was Rs. 30 million.





Question 21

The cost in rupees per tonne of oil moved by rail and road happens to be roughly

A Rs. 3

B Rs. 1.5

C Rs. 4.5

D Rs. 8

VIEW SOLUTION

Question 22

From the charts given, it appears that the cheapest mode of transport is

A road

B rail

C pipeline

D ship

VIEW SOLUTION

Instructions

Answer the questions based on the following information. The first table gives the percentage of students in MBA class, who sought employment in the areas of finance, marketing and software. The second table gives the average starting salaries of the students per month, (rupees in thousands) in these areas. The third table gives the number of students who passed out in each year.

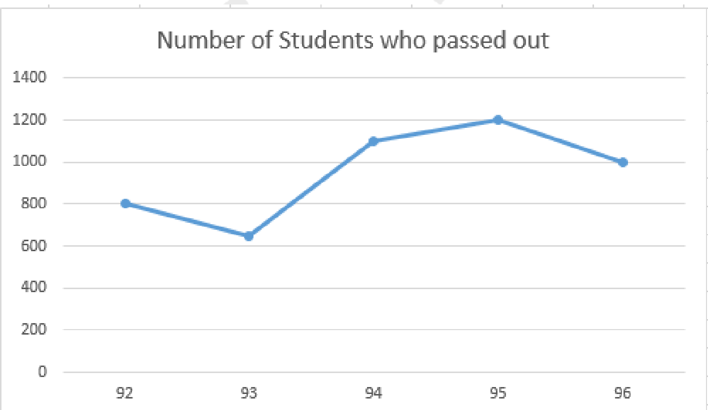
Table 1:

	Finance	Marketing	Software	Others
1992	22	36	19	23
1993	17	48	23	12
1994	23	43	21	13
1995	19	37	16	28
1996	32	32	20	16

Table 2:

	Finance	Marketing	Software
1992	5450	5170	5290
1993	6380	6390	6440
1994	7550	7630	7050
1995	8920	8960	7760
1996	9810	10220	8640

Table 3:



Question 23

The average annual rate at which the initial salary offered in software increases is

A 21%

B 33%

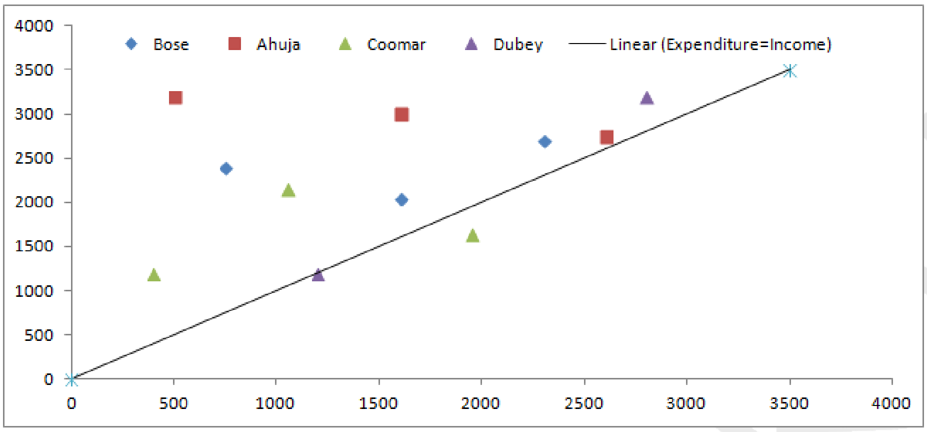
C 15.9%

D 65%

VIEW SOLUTION

Instructions

Directions for the following four questions: Answer the questions on the basis of the information given below.



The data points in the figure below represent monthly income and expenditure data of individual members of the Ahuja family, the Bose family, the Coomar family, and the Dubey family.

The X axis represents Expenditure and the Y axis represents the Income of the individual members.

For these questions, savings is defined as Savings = Income - Expenditure

Question 24

The highest amount of savings accrues to a member of which family?

A Ahuja

B Bose

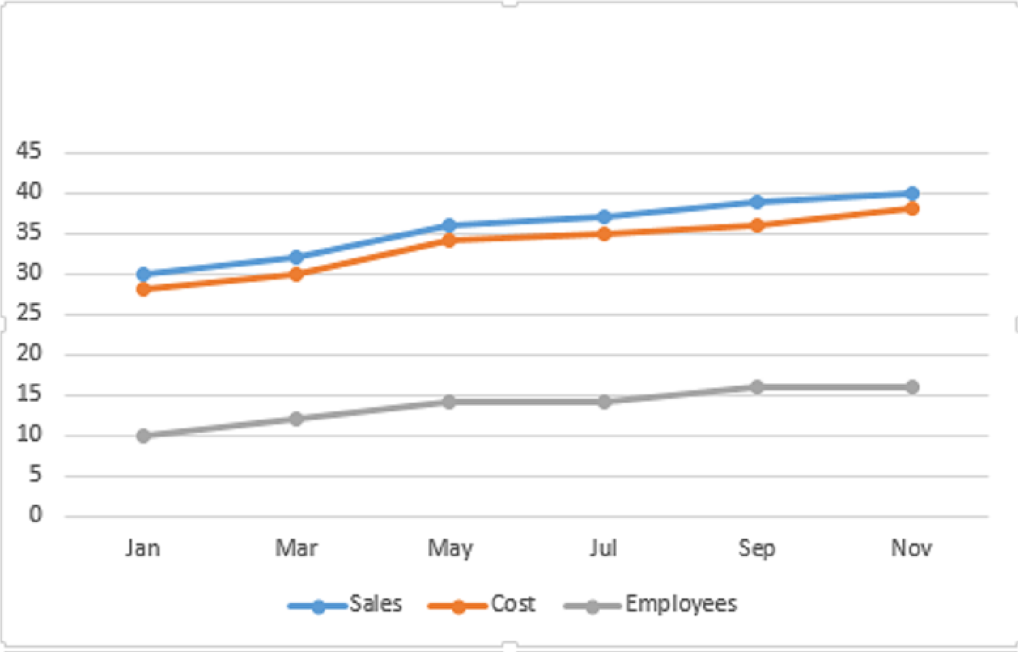
C Coomar

D Dubey

VIEW SOLUTION

Instructions

Answer the questions based on the following graph.



Employess in thousand Sales - cost = profit
(all values are integers)

Question 25

Which month has the highest profit per employee?

A September

B July

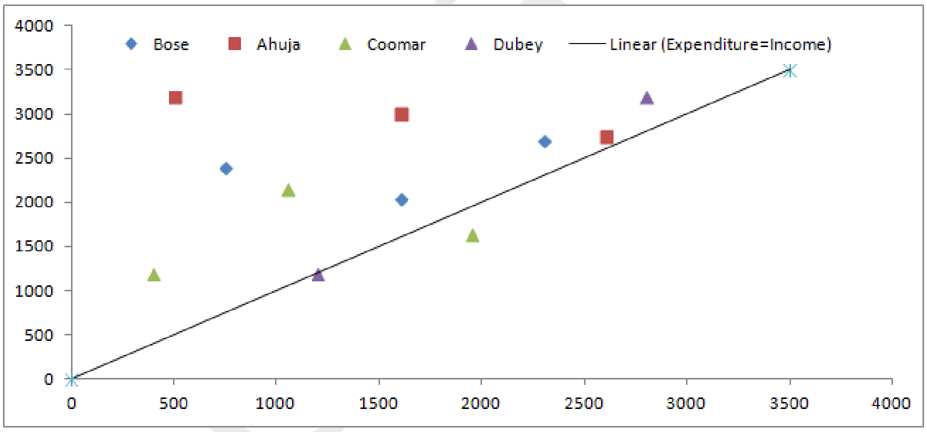
C January

D March

VIEW SOLUTION

Instructions

Directions for the following four questions: Answer the questions on the basis of the information given below.



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The X axis represents Expenditure and the Y axis represents the Income of the individual members.

For these questions, savings is defined as Savings = Income - Expenditure

Question 26

Which family has the lowest average savings?

A Ahuja

B Bose

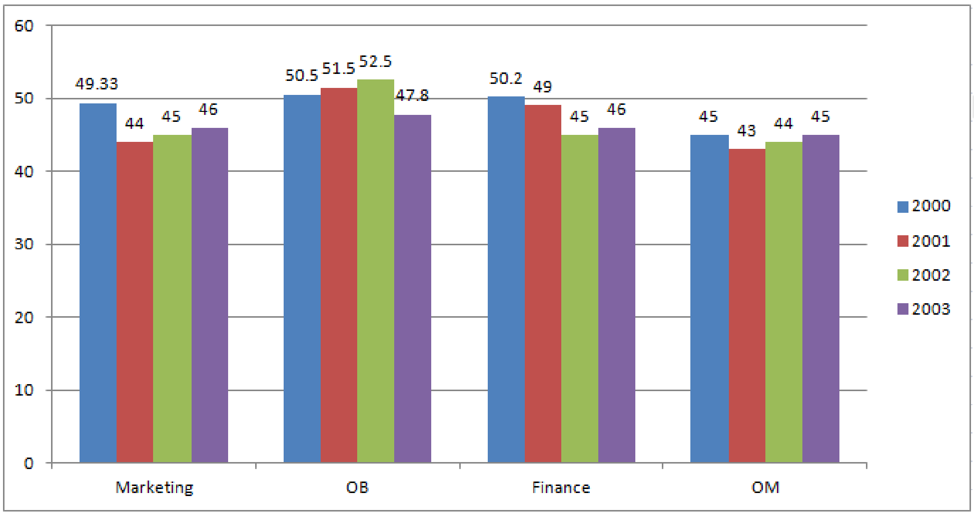
C Coomar

D Dubey

VIEW SOLUTION

Instructions

A management institute was established on January 1, 2000 with 3, 4, 5, and 6 faculty members in the Marketing, Organisational Behaviour (OB), Finance, and Operations Management (OM) areas respectively, to start with. No faculty member retired or joined the institute in the first three months of the year 2000. In the next four years, the institute recruited one faculty member in each of the four areas. All these new faculty members, who joined the institute subsequently over the years, were 25 years old at the time of their joining the institute. All of them joined the institute on April a) During these four years, one of the faculty members retired at the age of 60. The following diagram gives the area-wise average age (in terms of number of completed years) of faculty members as on April 1 of 2000, 2001, 2002, and 2003.



Question 27

From which area did the faculty member retire?

A Finance

B Marketing

C OB

D OM

VIEW SOLUTION

Free CAT Study Material

Instructions

Answer the questions based on the following information. The first table gives the percentage of students in MBA class, who sought employment in the areas of finance, marketing and software. The second table gives the average starting salaries of the students per month, (rupees in thousands) in these areas. The third table gives the number of students who passed out in each year.

Table 1:

	Finance	Marketing	Software	Others
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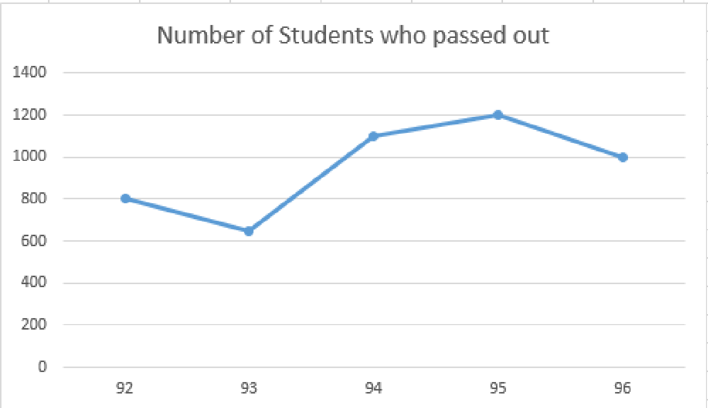


Question 28

The number of students who get jobs in finance is less than the students getting marketing jobs, in the 5 years, by

A 826

B 650

C 750

D 548

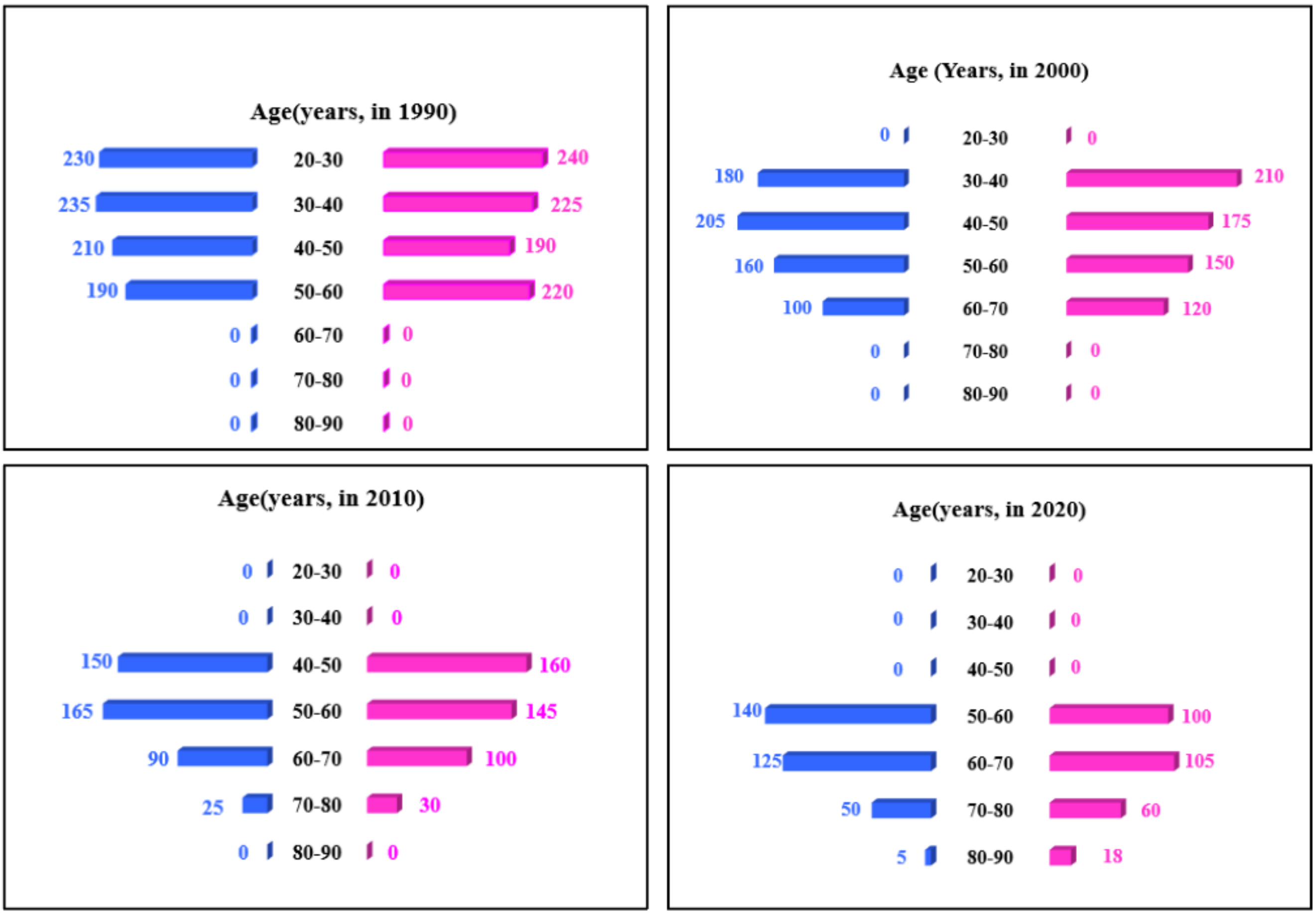
VIEW SOLUTION

Instructions

In the following, a year corresponds to 1st of January of that year.

A study to determine the mortality rate for a disease began in 1980. The study chose 1000 males and 1000 females and followed them for forty years or until they died, whichever came first. The 1000 males chosen in 1980 consisted of 250 each of ages 10 to less than 20, 20 to less than 30, 30 to less than 40, and 40 to less than 50. The 1000 females chosen in 1980 also consisted of 250 each of ages 10 to less than 20, 20 to less than 30, 30 to less than 40,and 40 to less than 50.

The four figures below depict the age profile of those among the 2000 individuals who were still alive in 1990, 2000, 2010, and 2020. The blue bars in each figure represent the number of males in each age group at that point in time, while the pink bars represent the number of females in each age group at that point in time. The numbers next to the bars give the exact numbers being represented by the bars. For example, we know that 230 males among those tracked and who were alive in 1990 were aged between 20 and 30.



Question 29

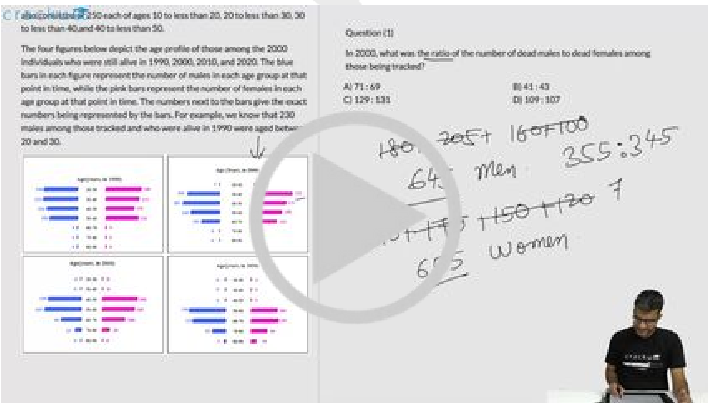
How many people who were being tracked and who were between 30 and 40 years of age in 1980 survived until 2010?

A 110

B 90

C 190

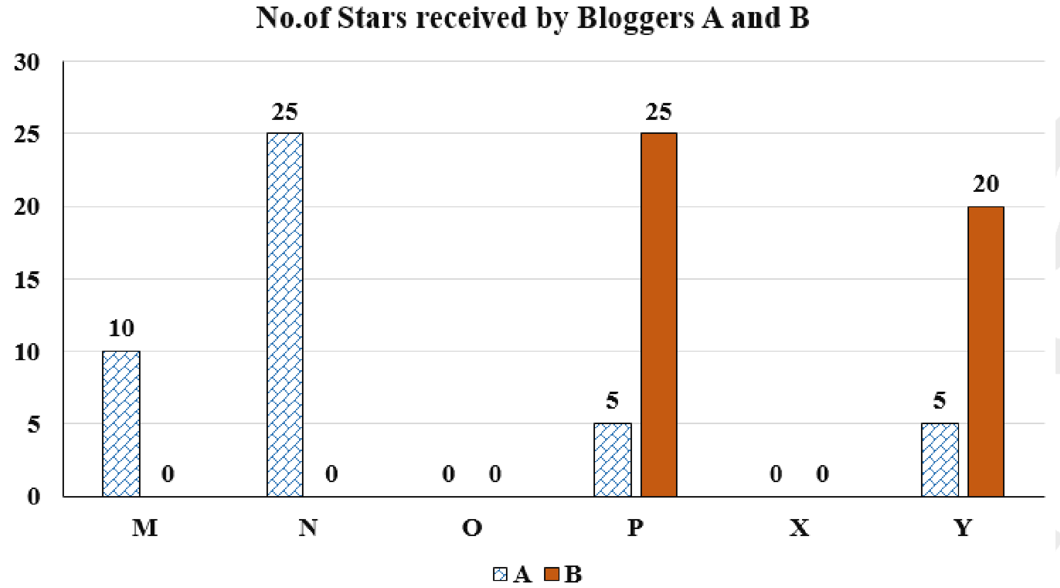
D 310



VIDEO SOLUTION

Instructions

Six web surfers M, N, O, P, X, and Y each had 30 stars which they distributed among four bloggers A, B, C, and D. The number of stars received by A and B from the six web surfers is shown in the figure below.

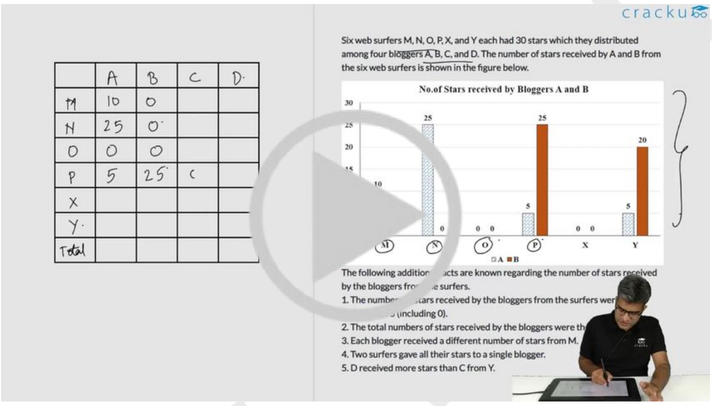


The following additional facts are known regarding the number of stars received by the bloggers from the surfers.

1. The numbers of stars received by the bloggers from the surfers were all multiples of 5 (including 0).
2. The total numbers of stars received by the bloggers were the same.
3. Each blogger received a different number of stars from M.
4. Two surfers gave all their stars to a single blogger.
5. D received more stars than C from Y.

Question 30

What was the total number of stars received by D?



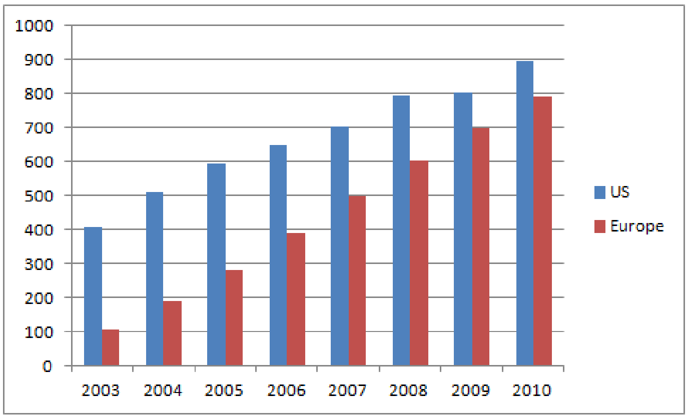
VIDEO SOLUTION

Free CAT Preparation Video Lectures

Instructions

Directions for the following four questions: Answer the following questions based on the information given below: The bar chart below shows the revenue received in million US Dollars (USD), from subscribers to a particular Internet service. The data covers the period 2003 to 2007 for the United States (US) and Europe. The bar chart also shows the estimated revenues from subscription to this service for the period 2008 to 2010.

The Y axis represents the subscription revenue in Million (USD) and the X axis represents the years.



Question 31

The difference between the estimated subscription in Europe in 2008 and what it would have been if it were computed using the percentage growth rate of 2007 (over 2006), is closest to:

A 50

B 80

C 20

D 10

E 0

VIEW SOLUTION

Question 32

While the subscription in Europe has been growing steadily towards that of the US, the growth rate in Europe seems to be declining. Which of the following is closest to the percent change in growth rate of 2007 (over 2006) relative to the growth rate of 2005 (over 2004)?

A 17

B 20

C 35

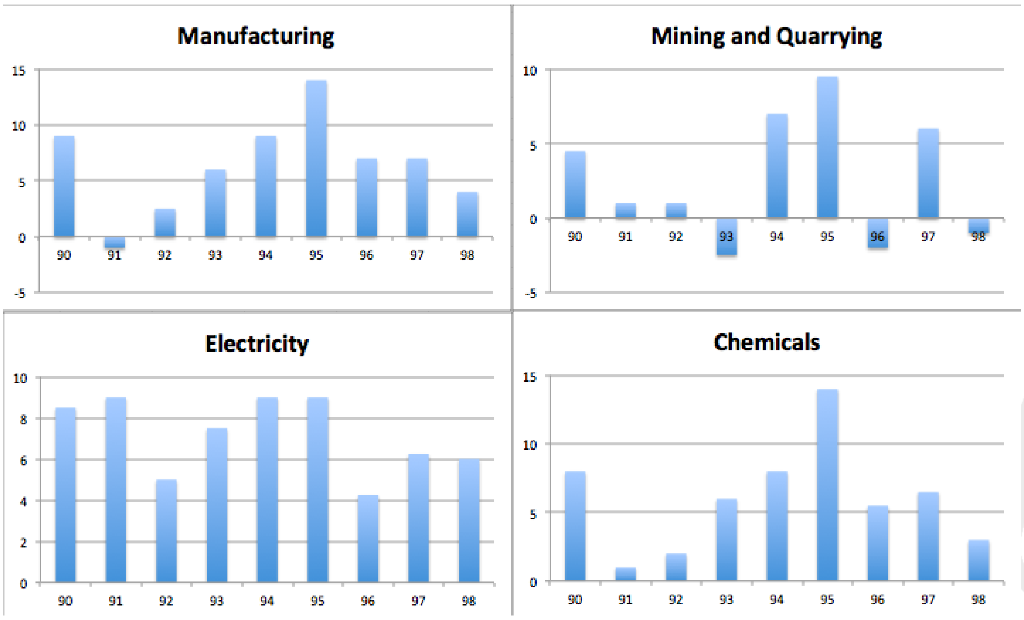
D 60

E 100

VIEW SOLUTION

Instructions

Answer these questions based on the data given below: The figures below present annual growth rate, expressed as the % change relative to the previous year, in four sectors of the economy of the Republic of Reposia during the 9 year period from 1990 to 1998. Assume that the index of production for each of the four sectors is set at 100 in 1989 Further, the four sectors: manufacturing, mining and quarrying, electricity, and chemicals, respectively, constituted 20%, 15%, 10%, 15 % of total industrial production 1989.



Question 33

It is known that the index of total industrial production in 1994 was 50 percent more that in 1989. Then, the percentage increase in production between 1989 and 1994 in sectors other than the four listed above is:

A 57.5

B 87.5

C 127.5

D 47.5

VIEW SOLUTION

Daily Free CAT Practice Tests

Question 34

The percentage increase of production in-the four sectors, namely, manufacturing, mining & quarrying, electricity and chemicals, taken together, in 1994, relative to 1989, is approximately:

A 25

B 20

C 50

D 40

VIEW SOLUTION

Question 35

Which is the sector with the highest growth during the period 1989 and 1998?

A Manufacturing

B Mining and quarrying

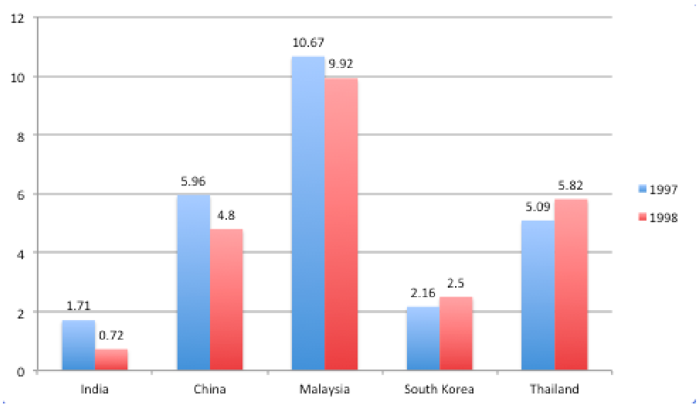
C Electricity

D Chemicals

VIEW SOLUTION

Instructions

Directions for the next 4 questions: Answer these questions based on the data presented in the figure below. FEI for a country in a year, is the ratio (expressed as a percentage) of its foreign equity inflows to its GDP. The following figure displays the FEIs for select Asian countries for the years 1997 and 1998.



Question 36

Based on the data provided, it can be concluded that

A absolute value of foreign equity inflows in 1998 was higher than that in 1997 for both Thailand and South Korea)

B absolute value of foreign equity inflows was higher in 1998 for Thailand and lower for China than the corresponding values in 1997.

C absolute value of foreign equity inflows was lower in 1998 for both India and China than the corresponding values in 1997.

D none of the above can be inferred,

VIEW SOLUTION

Free Videos for CAT Preparation

Question 37

The country with the highest percentage change in FEI in 1998 relative to its FEI in 1997, is:

A India

B China

C Malaysia

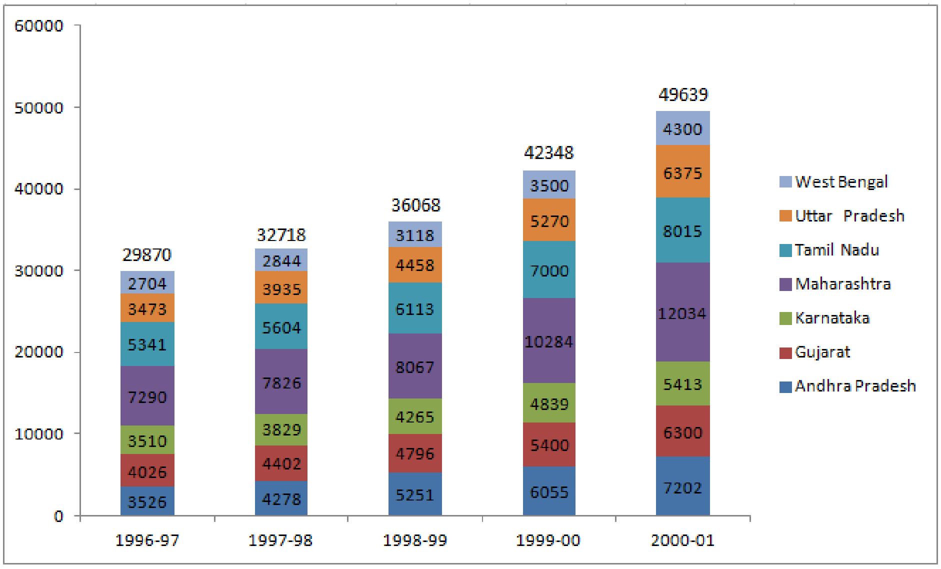
D Thailand

VIEW SOLUTION

Instructions

The chart given below indicates the annual sales tax revenue collections (in rupees in crores) of seven states from 1997 to 2001.

The values given at the top of each bar represents the total collections in that year



Question 38

The percentage share of sales tax revenue of which state has increased from 1997 to 2001?

A Tamil Nadu

B Karnataka

C Gujarat

D Andhra Pradesh

VIEW SOLUTION

Question 39

If for each year, the states are ranked in terms of the descending order of sales tax collections, how many states do not change the ranking more than once over the five years?

A 1

B 5

C 3

D 4

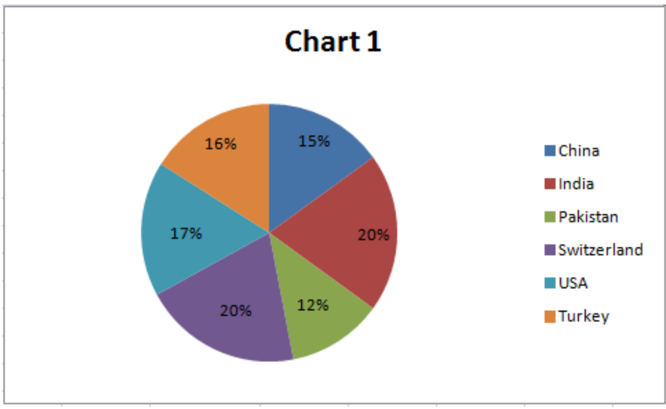
VIEW SOLUTION

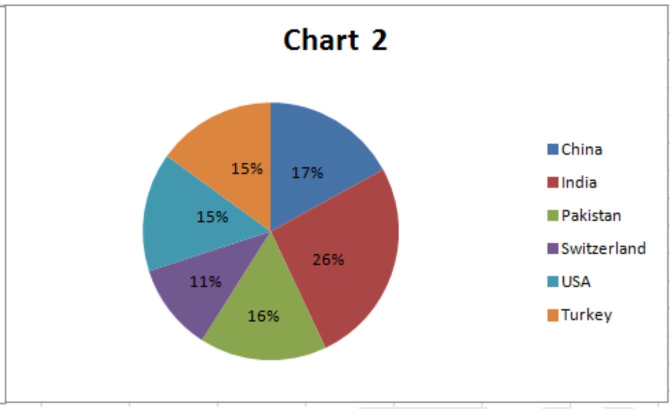
Instructions

Chart 1 shows the distribution by value of top 6 suppliers of MFA Textiles in 1995.

Chart 2 shows the distribution by quantity of top 6 suppliers of MFA Textiles in 1995.

The total value is 5760 million Euro (European currency). The total quantity is 1.055 million tonnes.





Question 40

The average price in Euro per kilogram for Turkey is roughly

A 6.20

B 5.60

C 4.20

D 4.80

VIEW SOLUTION

Question 41

The country which has the highest average price is

A USA

B Switzerland

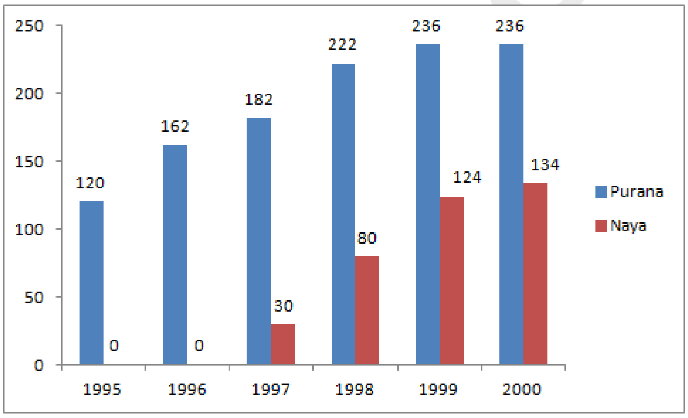
C Turkey

D India

VIEW SOLUTION

Instructions

Directions for the following four questions: Answer the questions on the basis of the information given below.



Purana and Naya are two brands of kitchen mixer-grinders available in the local market. Purana is an old brand that was introduced in 1990, while Naya was introduced in 1997. For both these brands, 20% of the mixer-grinders bought in a particular year are disposed off as junk exactly two years later. It is knows that 10 Purana mixer-grinders were disposed off in 1997. The following figures show the number of Purana and Naya mixer-grinders in operation from 1995 to 2000, as at the end of the year.

Question 42

How many Purana mixer-grinders were disposed off in 2000?

A 0

B 5

C 6

D Cannot be determined from the data

VIEW SOLUTION

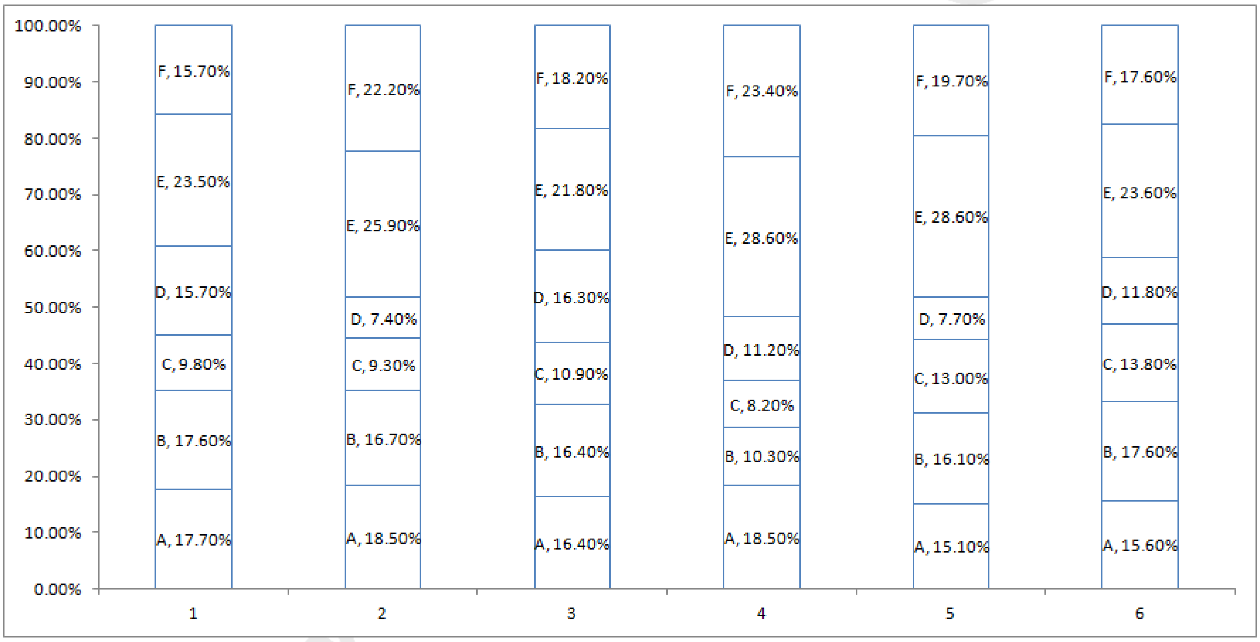
Important Verbal Ability Questions for CAT (Download PDF)

Instructions

Directions for the following three questions: Answer these questions based on the data given below:

There are six companies, 1 through 6. All of these companies use six operations, A through F. The following graph shows the distribution of efforts put in by each company in these six operations.

The Y axis represents the % distribution of effort and the X axis represents the company



Question 43

A new technology is introduced in company 4 such that the total effort for operations B through F get evenly distributed among these. What is the change in the percentage of effort in operation E?

A Reduction of 12.3

B Increase of 12.3

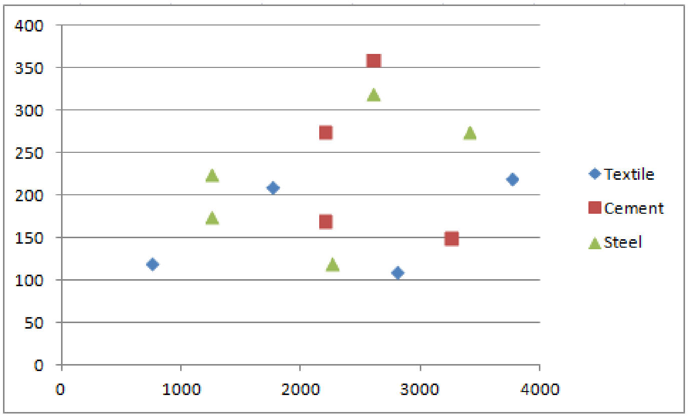
C Reduction of 5.6

D Increase of 5.6

VIEW SOLUTION

Instructions

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.



Each point in the graph below shows the profit and turnover data for a company. Each company belongs to one of the three industries: textile, cement and steel.

Question 44

An investor wants to buy stock of only steel or cement companies with a turnover more than 1000 and profit exceeding 10% of turnover. How many choices are available to the investor?

A 4

B 5

C 6

D 7

VIEW SOLUTION

Question 45

For how many companies does the profit exceed 10% of turnover?

A 8

B 7

C 6

D 5

VIEW SOLUTION

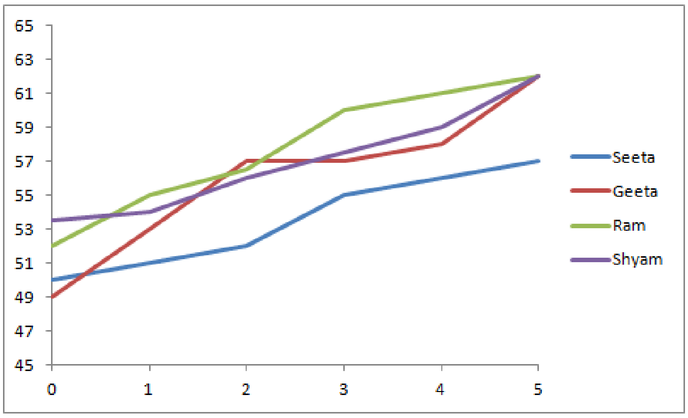
Data Interpretation for CAT Questions (download pdf)

Instructions

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

The length of an infant is one of the measures of his/her development in the early stages of his/her life.

The figure below shows the growth chart of four infants in the first five months of life.



Question 46

Among the four infants, who grew the least in the first five months of life? (in %)

A Geeta

B Seeta

C Ram

D Shyam

VIEW SOLUTION

Question 47

After which month did Seeta's rate of growth start to decline?

A Second month

B Third month

C Fourth month

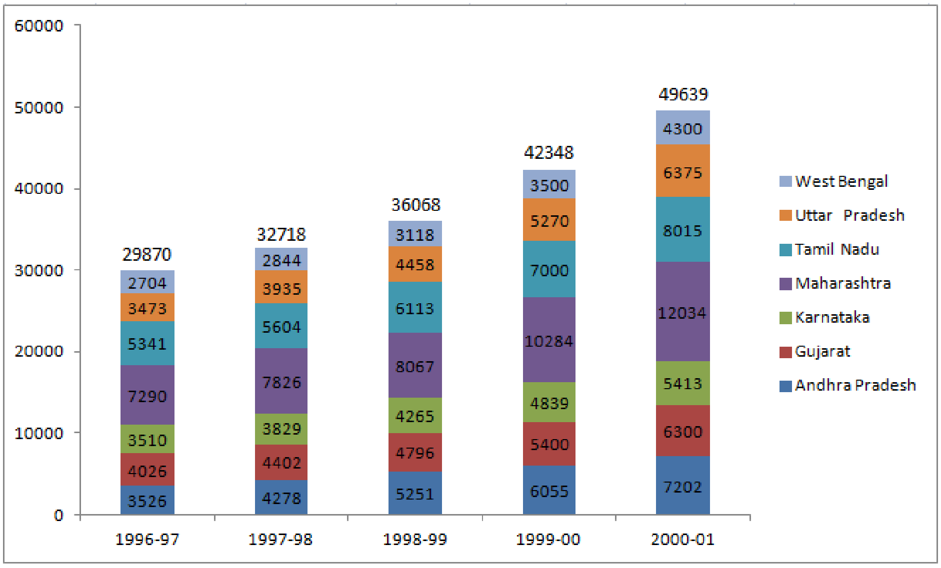
D Never

VIEW SOLUTION

Instructions

The chart given below indicates the annual sales tax revenue collections (in rupees in crores) of seven states from 1997 to 2001.

The values given at the top of each bar represents the total collections in that year



Question 48

Which pair of successive years shows the maximum growth rate of tax revenue in Maharashtra?

A 1997 to 1998

B 1998 to 1999

C 1999 to 2000

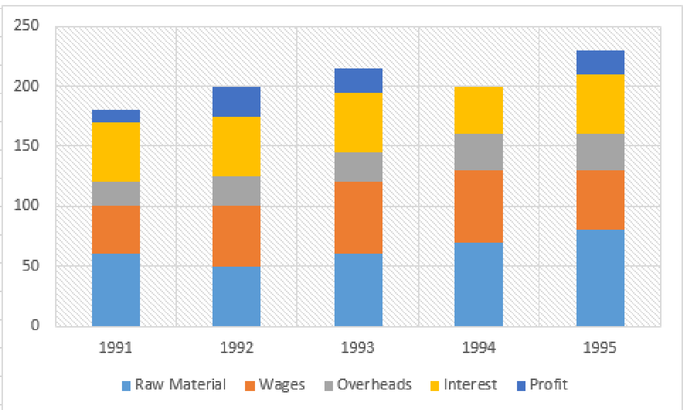
D 2000 to 2001

VIEW SOLUTION

Logical Reasoning for CAT Questions (download pdf)

Instructions

The graph given below gives the yearly details of money invested in producing a certain product over the years 1991 to 1995. It also gives the profit (in '000 rupees).



Question 49

In which year was the increase in raw material maximum?

A 1992

B 1993

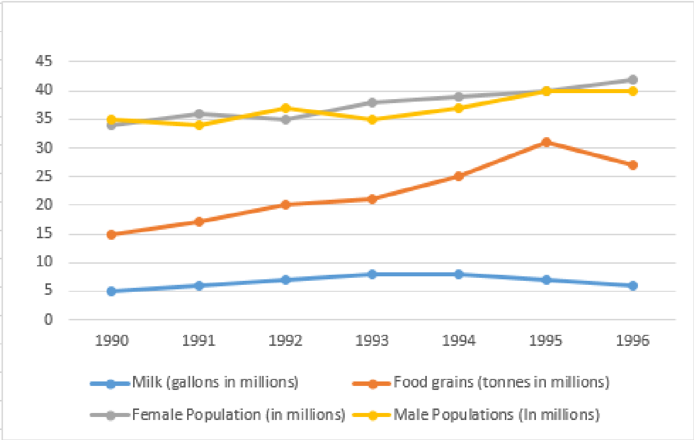
C 1994

D 1995

VIEW SOLUTION

Instructions

The graph given below shows the quantity of milk and food grains consumed annually along with female and male population (in millions). Use the data to answer the questions that follow.



Question 50

When was the percapita consumption of milk the least?

A 1990

B 1992

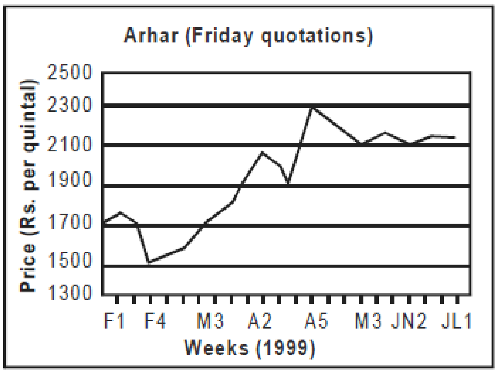
C 1994

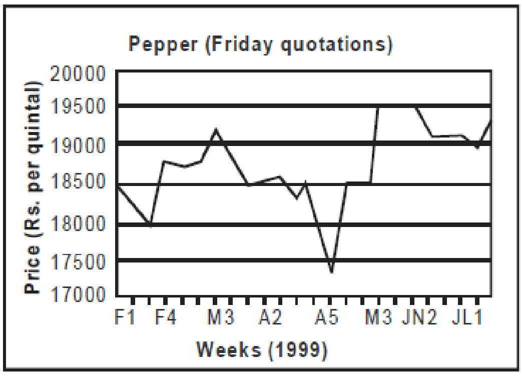
D 1996

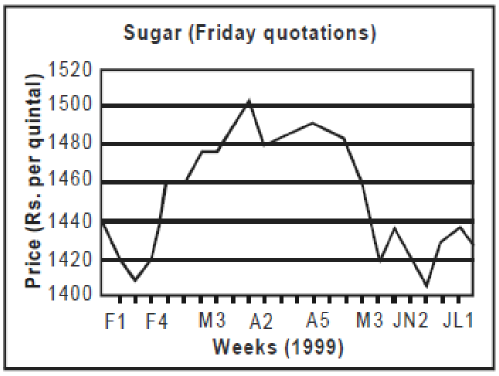
VIEW SOLUTION

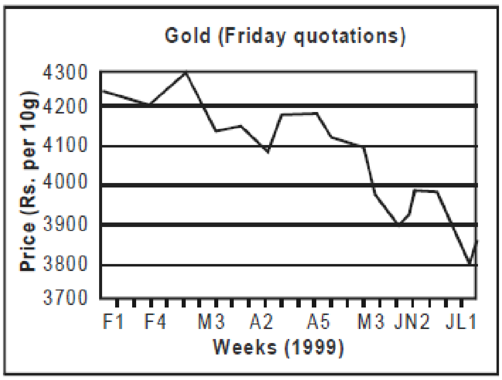
Instructions

These questions are based on the price fluctuations of four commodities - arhar, pepper, sugar and gold during February - July 1999 as described in the figures below:









Question 51

The price volatility of the commodity with the highest PV during the Febrauary-July period is approximately equal to:

A 3%

B 40%

C 20%

D 12%

VIEW SOLUTION

Question 52

Mr. X, a funds manager with an investment company invested 25% of his funds in each of the four commodities at the beginning of the period. He sold the commodities at the end of the period. His investments in the commodities resulted in:

A 17% profit

B 5.5% loss

C no profit, no loss

D 4.3% profit

VIEW SOLUTION

Question 53

Price volatility (PV) of a commodity is defined as follows: PV = (highest price during the period - lowest price during the period / average price during the period. What is the commodity with the lowest price volatility?

A Arhar

B Pepper

C Sugar

D Gold

VIEW SOLUTION

Question 54

Price change of a commodity is defined as the absolute difference in ending and beginning prices expressed as a percentage of the beginning. What is the commodity with the highest price change?

A Arhar

B Pepper

C Sugar

D Gold

VIEW SOLUTION

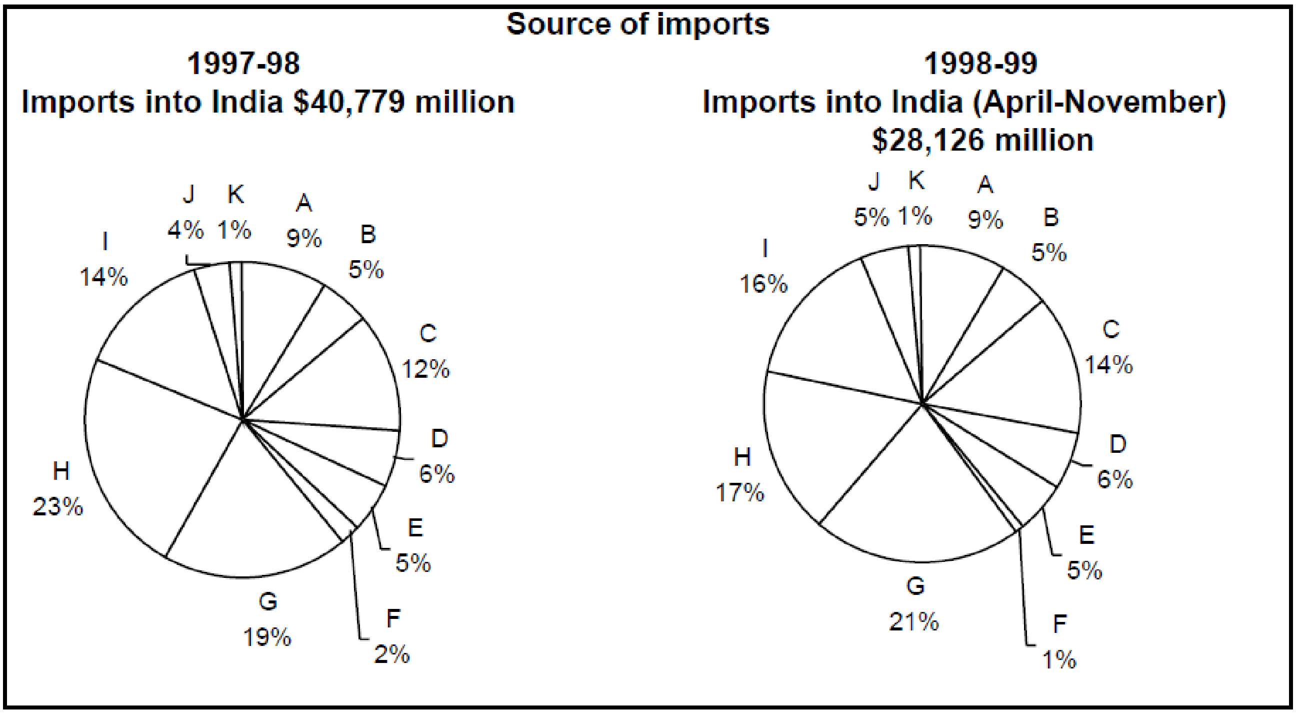
Instructions

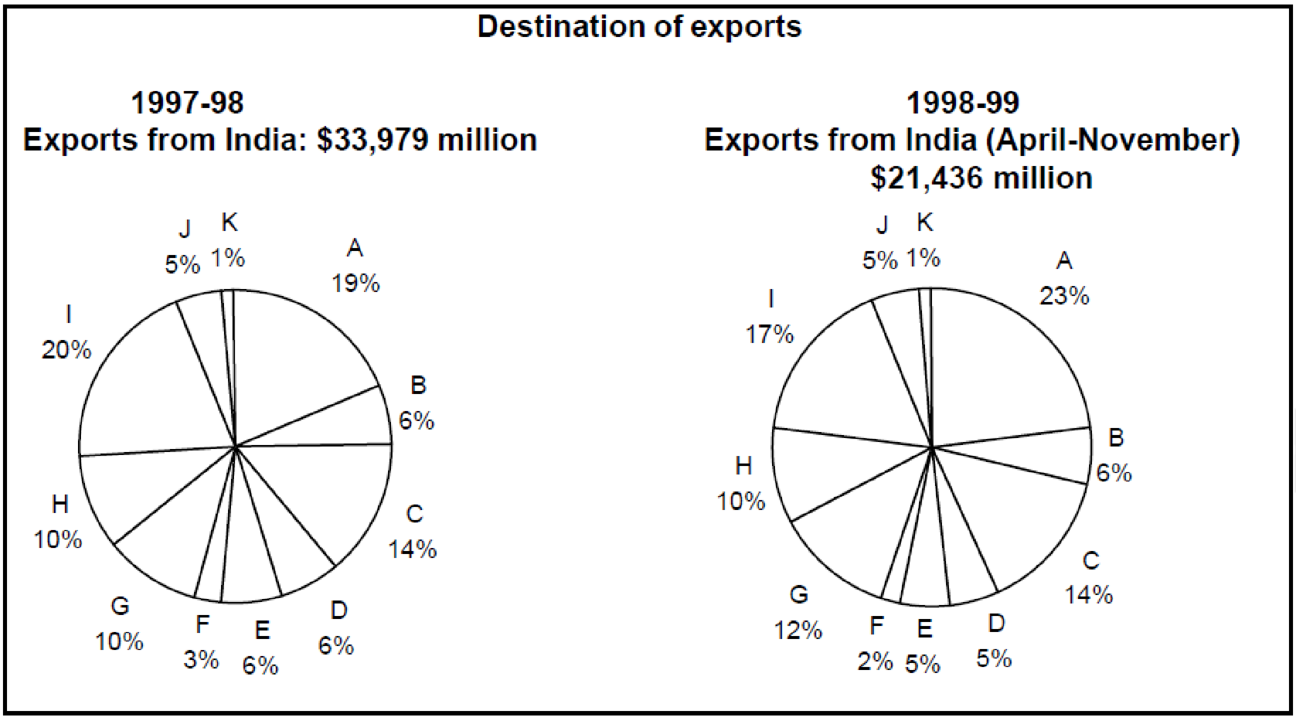
Consider the information provided in the figure below relating to India's foreign trade in 1997-98 and the first eight months of 1998-99.

Total trade with a region is defined as the sum of exports to and imports from that region.
Trade deficit is defined as the excess of imports over exports. Trade deficit may be negative.

A:USA. B:Germany C:Other EU. D:U.K. E:Japan F:Russia

G:Other East Europe H:OPEC I:Asia J:Other LDCs K:Others





Question 55

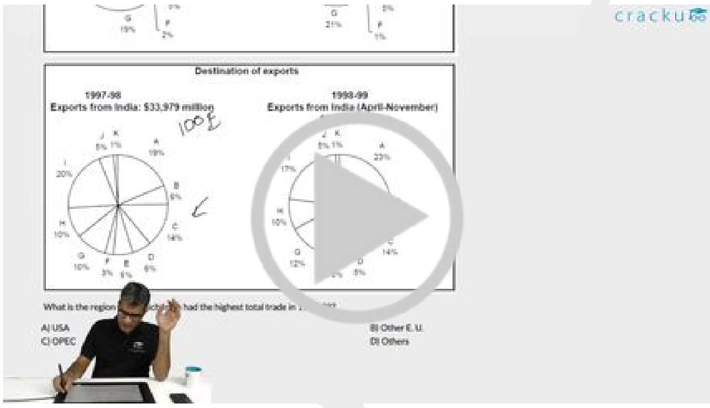
In 1997-98 the amount of Indian exports, in millions US $, to the region with which India had the lowest total trade, is approximately

A 750

B 340

C 220

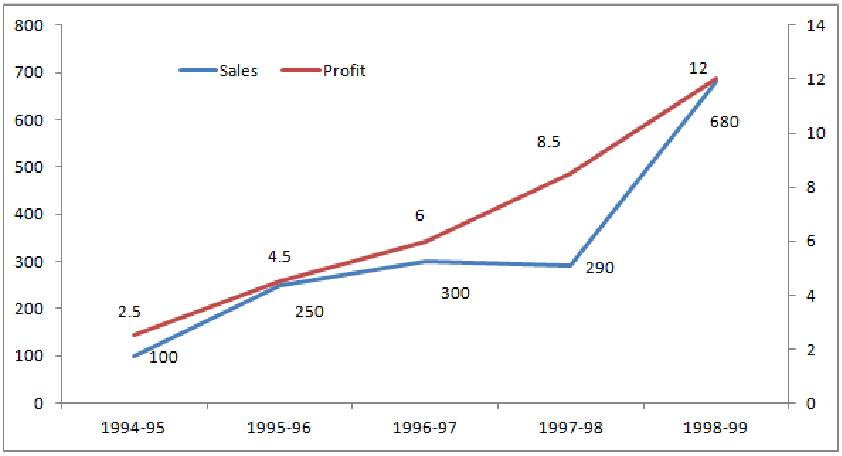
D 440



VIDEO SOLUTION

Instructions

These questions are based on the situation given below:
The figure below presents sales and net profit, in Rs. Crores, of IVP Ltd for the five years from 1994-95 to 1998-99. During this period, the sales increased from Rs.100 Crores to Rs. 680 Crores. Correspondingly, the net profit increased from Rs. 2.5 Crores to Rs. 12 Crores. Net profit is defined as the excess of sales over total costs.



Question 56

Defining profitability as the ratio of net profit to sales, IVP Ltd. recorded the highest profitability in

A 1998-99

B 1997-98

C 1994-95

D 1996-97

VIEW SOLUTION

Question 57

The highest percentage of growth in sales, relative to the previous year, occurred in

A 1995-96

B 1996-97

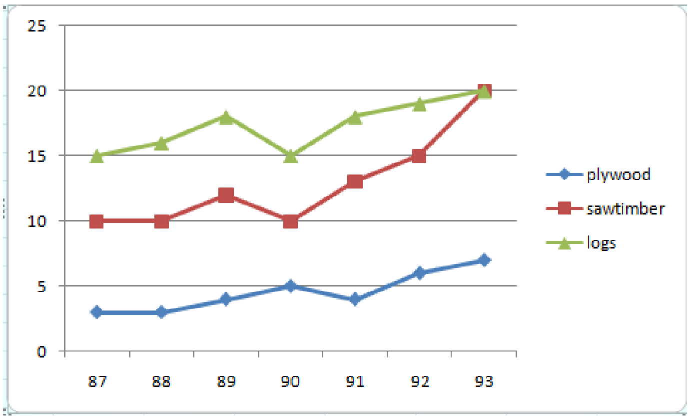
C 1997-98

D 1998-99

VIEW SOLUTION

Instructions

Answer the questions based on the following information. In the following chart, the price of logs shown is per cubic metre that of plywood and saw timber is per tonne.



Question 58

If the prices of sawtimber increased by 33.33%, in which year was the difference in prices of one tonne of saw timber and one cubic metre of logs the least?

A 1989

B 1990

C 1991

D 1992

VIEW SOLUTION

Question 59

Which product shows the maximum percentage increase in price over the period?

A Saw timber

B Plywood

C Logs

D Cannot be determined

VIEW SOLUTION

Question 60

What is the maximum percentage increase in price per cubic metre or per tonne over the previous year?

A 33.33%

B 85%

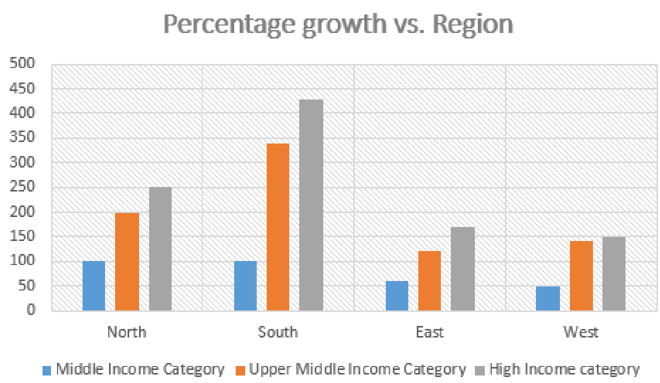
C 50%

D Cannot be determined

VIEW SOLUTION

Instructions

Answer the questions based on the following information. The following bar chart gives the growth percentage in the number of households in the middle, upper-middle and high-income categories in the four regions for the period between 1987-88 and 1994-95.



	Number of households in 1987-88	Average household income in 1987-88	Growth in average household income (1994-95 over 1987-88)
Middle Income	40	Rs. 30,000	50%
Upper-middle	10	Rs. 50,000	60%
High income	5	Rs. 75,000	90%

(Number of households in thousands)

Question 61

What is the percentage increase in total number of households for the northern region (upper-middle class) over the given period?

A 100%

B 200%

C 240%

D Cannot be determined

VIEW SOLUTION

Question 62

What was the total household income in northern region for upper-middle class?

A Rs. 50 lakh

B Rs. 500 million

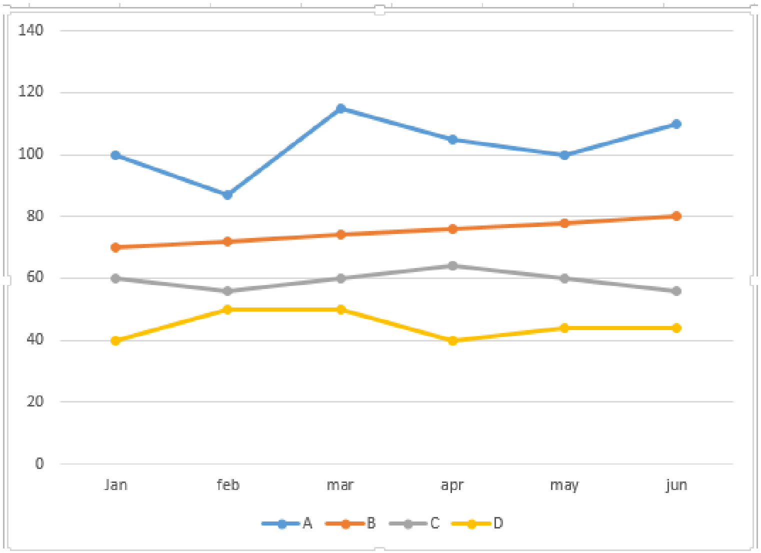
C Rs. 300 million

D Cannot be determined

VIEW SOLUTION

Instructions

The graph below shows the end of the month market values of 4 shares for the period from January to June. Answer the following questions based on this graph.



Question 63

Which share showed the greatest percentage increase in market value in any month during the entire period?

A A

B B

C C

D D

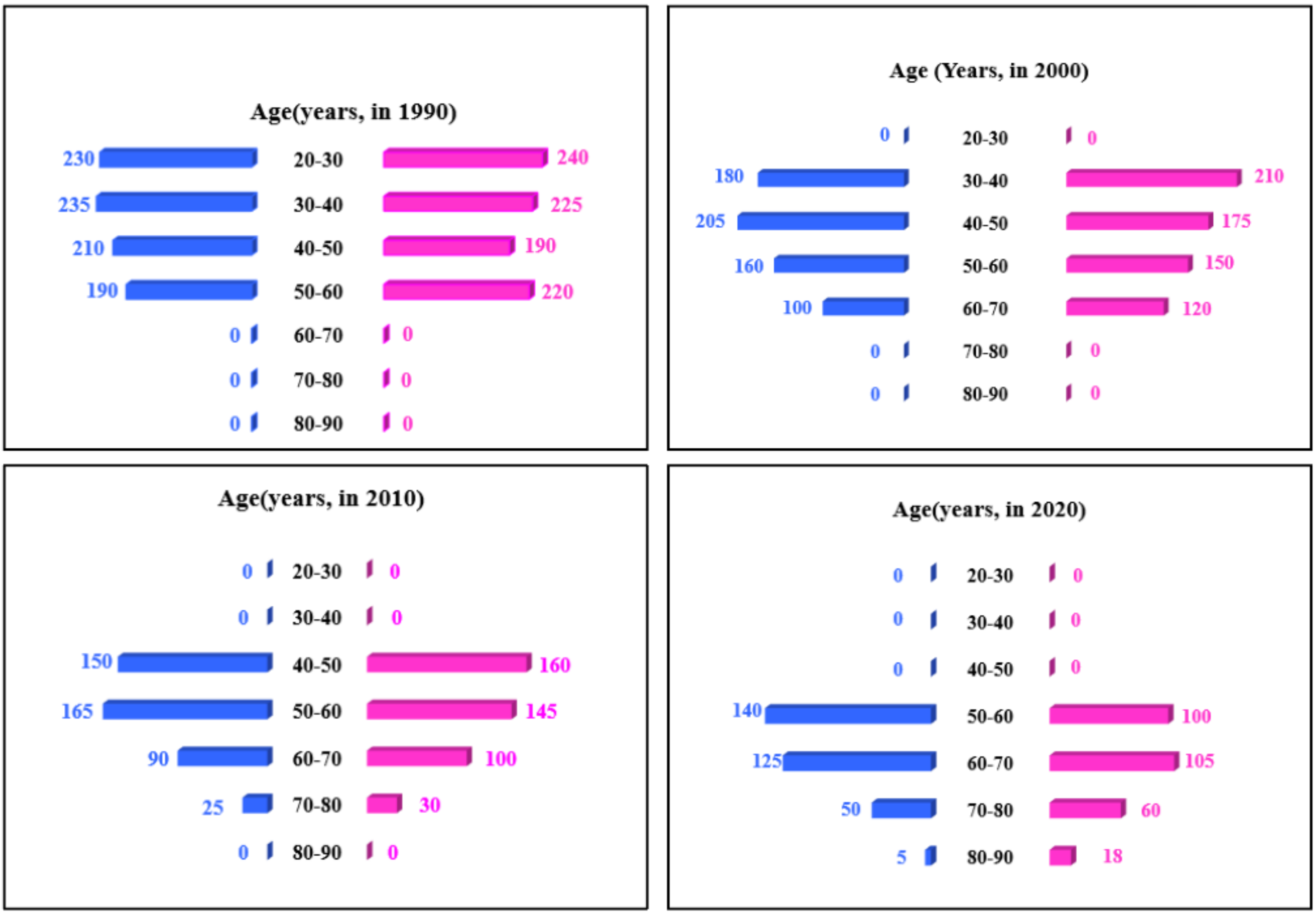
VIEW SOLUTION

Instructions

In the following, a year corresponds to 1st of January of that year.

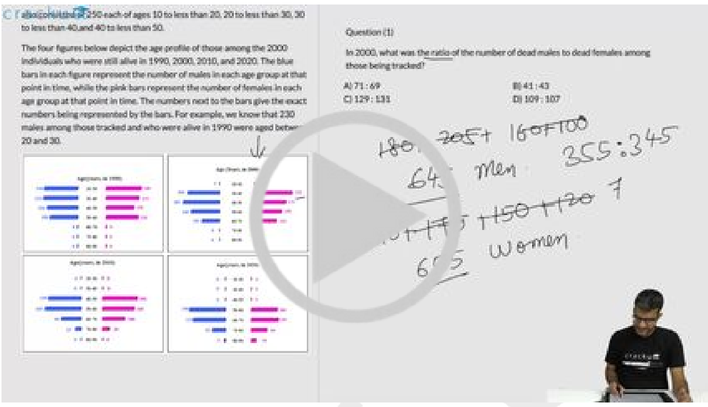
A study to determine the mortality rate for a disease began in 1980. The study chose 1000 males and 1000 females and followed them for forty years or until they died, whichever came first. The 1000 males chosen in 1980 consisted of 250 each of ages 10 to less than 20, 20 to less than 30, 30 to less than 40, and 40 to less than 50. The 1000 females chosen in 1980 also consisted of 250 each of ages 10 to less than 20, 20 to less than 30, 30 to less than 40,and 40 to less than 50.

The four figures below depict the age profile of those among the 2000 individuals who were still alive in 1990, 2000, 2010, and 2020. The blue bars in each figure represent the number of males in each age group at that point in time, while the pink bars represent the number of females in each age group at that point in time. The numbers next to the bars give the exact numbers being represented by the bars. For example, we know that 230 males among those tracked and who were alive in 1990 were aged between 20 and 30.



Question 64

How many of the females who were being tracked and who were between 20 and 30 years of age in 1980 died between the ages of 50 and 60?



VIDEO SOLUTION

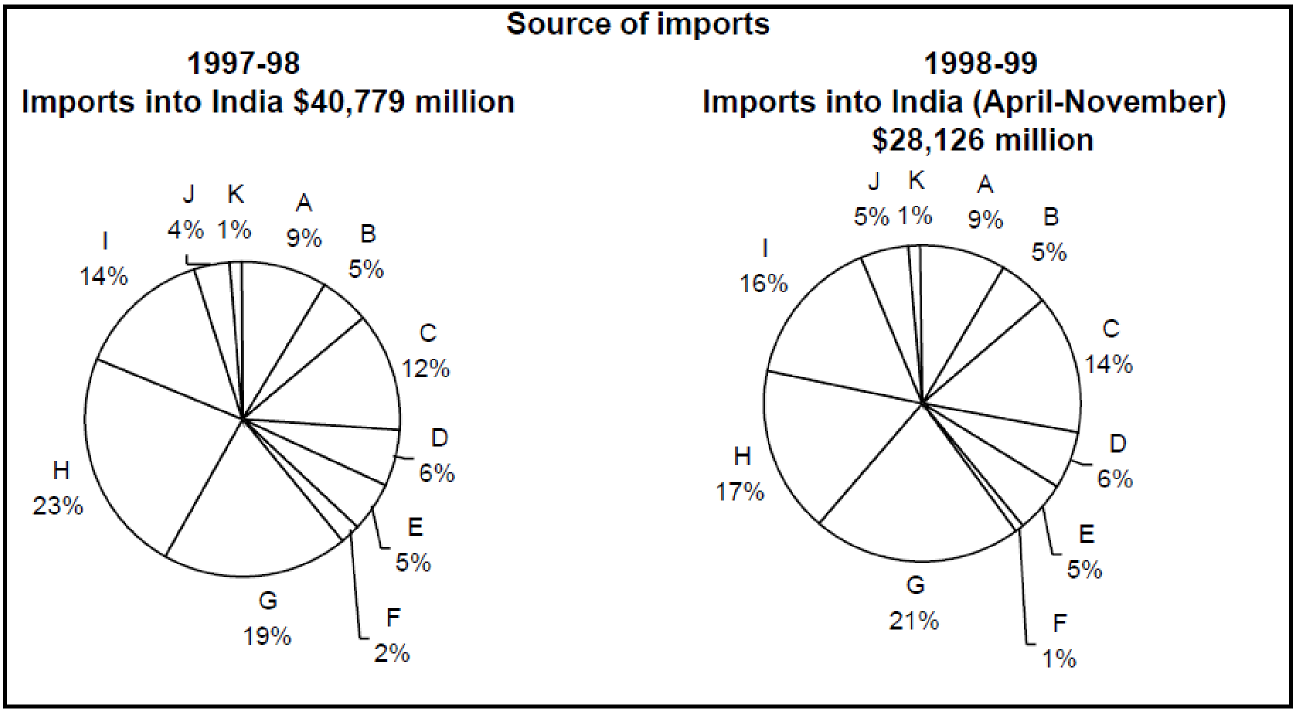
Instructions

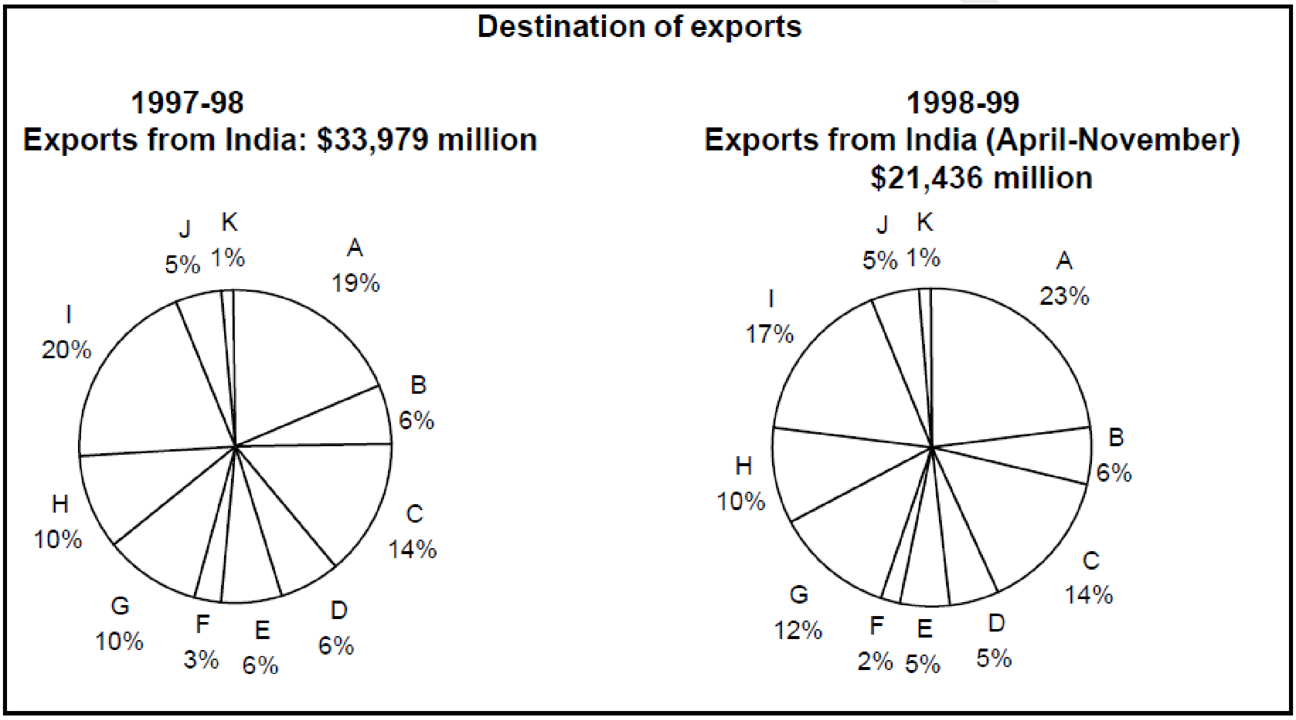
Consider the information provided in the figure below relating to India's foreign trade in 1997-98 and the first eight months of 1998-99.

Total trade with a region is defined as the sum of exports to and imports from that region.
Trade deficit is defined as the excess of imports over exports. Trade deficit may be negative.

A:USA. B:Germany C:Other EU. D:U.K. E:Japan F:Russia

G:Other East Europe H:OPEC   I:Asia J:Other LDCs K:Others





Question 65

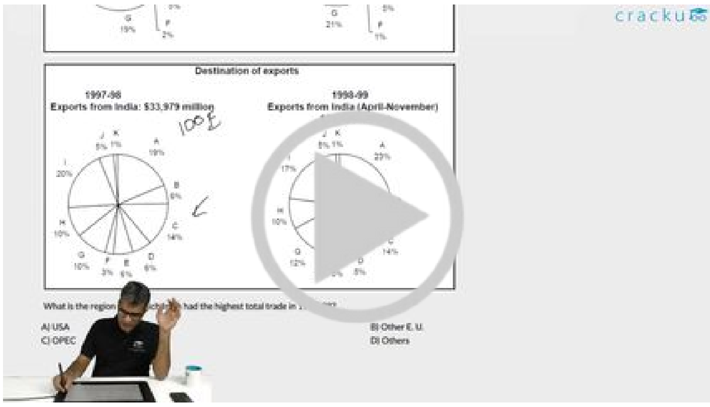
In 1997-98, the trade deficit with respect to India, in billions of US $, for the region with the highest trade deficit with respect to India, is approximately equal to

A 6.0

B 3.0

C 4.5

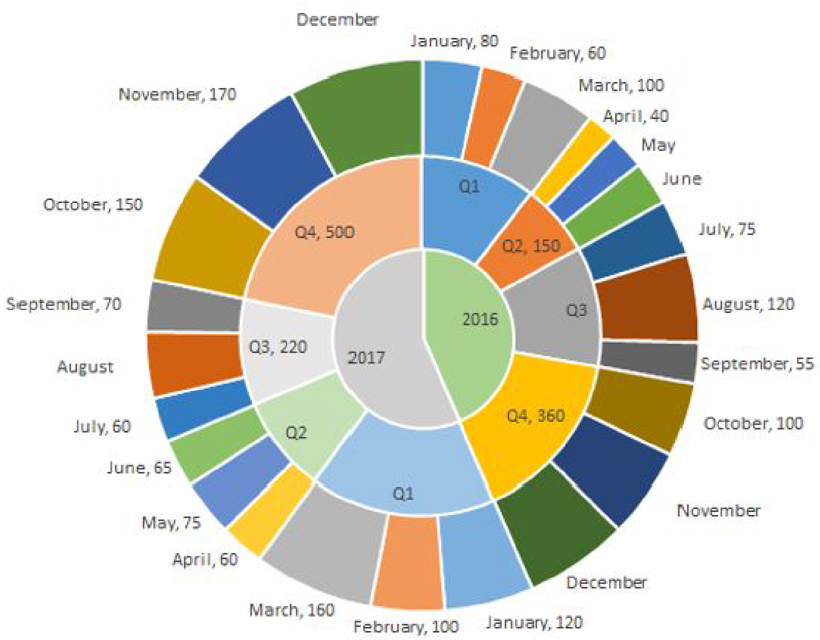
D 7.5



VIDEO SOLUTION

Instructions

The multi-layered pie-chart below shows the sales of LED television sets for a big retail electronics outlet during 2016 and 2017. The outer layer shows the monthly sales during this period, with each label showing the month followed by sales figure of that month. For some months, the sales figures are not given in the chart. The middle-layer shows quarterwise aggregate sales figures (in some cases, aggregate quarter-wise sales numbers are not given next to the quarter). The innermost layer shows annual sales. It is known that the sales figures during the three months of the second quarter (April, May, June) of 2016 form an arithmetic progression, as do the three monthly sales figures in the fourth quarter (October, November, December) of that year.



Question 66

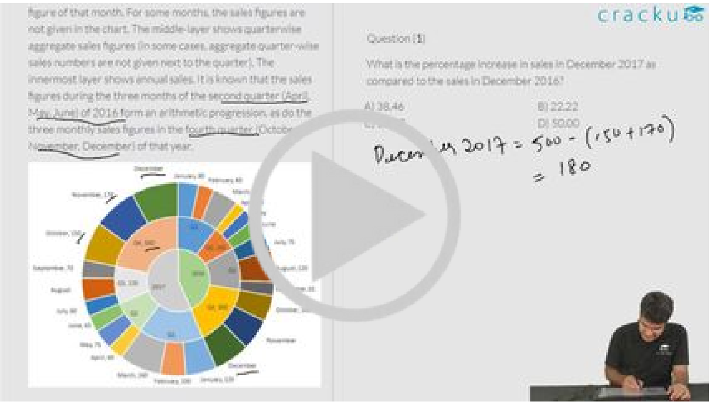
During which quarter was the percentage decrease in sales from the previous quarter's sales the highest?

A Q2 of 2017

B Q4 of 2017

C Q2 of 2016

D Q1 of 2017



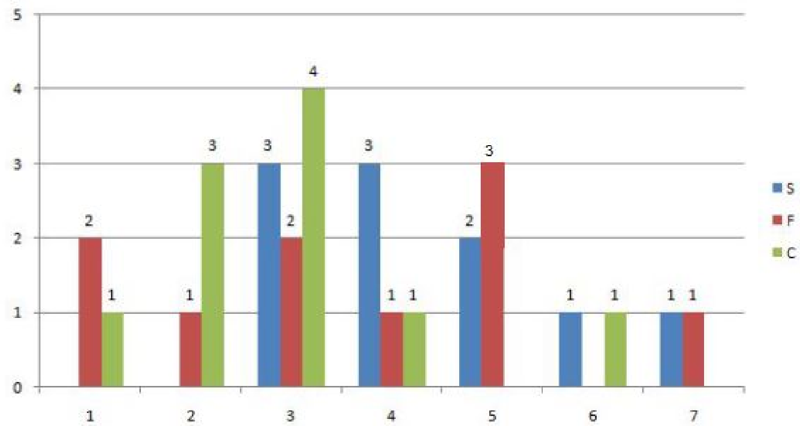
VIDEO SOLUTION

Instructions

Simple Happiness index (SHI) of a country is computed on the basis of three, parameters: social support (S),freedom to life choices (F) and corruption perception (C). Each of these three parameters is measured on a scale of 0 to 8 (integers only). A country is then categorised based on the total score obtained by summing the scores of all the three parameters, as shown in the following table:

Toatal Score	0-4	5-8	9-13	14-19	20-24
Category	Very Unhappy	Unhappy	Neutral	Happy	Very Happy

Following diagram depicts the frequency distribution of the scores in S, F and C of 10 countries - Amda, Benga, Calla, Delma, Eppa, Varsa, Wanna, Xanda,Yanga and Zooma:



Further, the following are known.
1. Amda and Calla jointly have the lowest total score, 7, with identical scores in all the three parameters.
2. Zooma has a total score of 17.
3. All the 3 countries, which are categorised as happy, have the highest score In exactly one parameter.

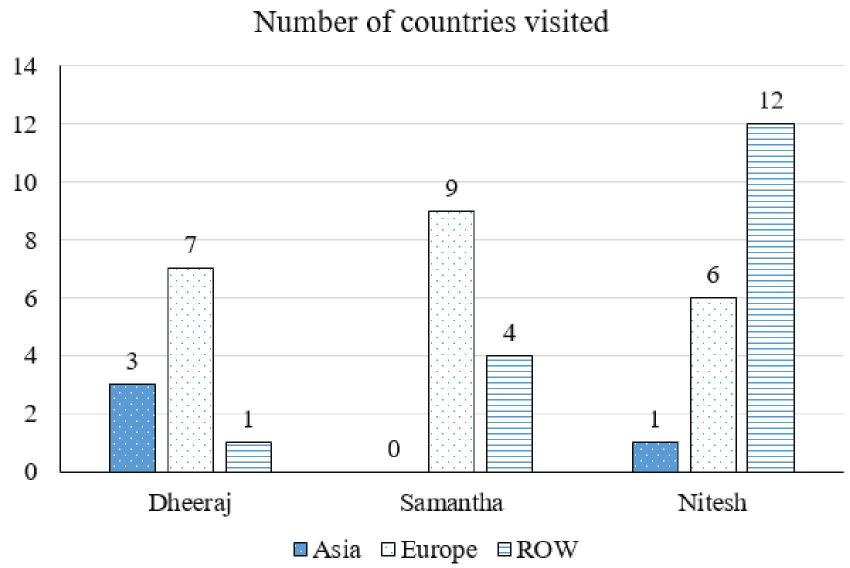
Question 67

What is Amda's score in F?

VIDEO SOLUTION

Instructions

The chart below provides complete information about the number of countries visited by Dheeraj, Samantha and Nitesh, in Asia, Europe and the rest of the world (ROW).

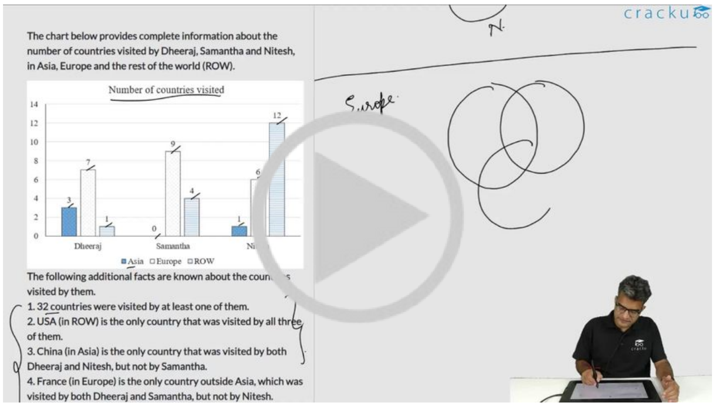


The following additional facts are known about the countries visited by them.

1. 32 countries were visited by at least one of them.
2. USA (in ROW) is the only country that was visited by all three of them.
3. China (in Asia) is the only country that was visited by both Dheeraj and Nitesh, but not by Samantha.
4. France (in Europe) is the only country outside Asia, which was visited by both Dheeraj and Samantha, but not by Nitesh.
5. Half of the countries visited by both Samantha and Nitesh are in Europe.

Question 68

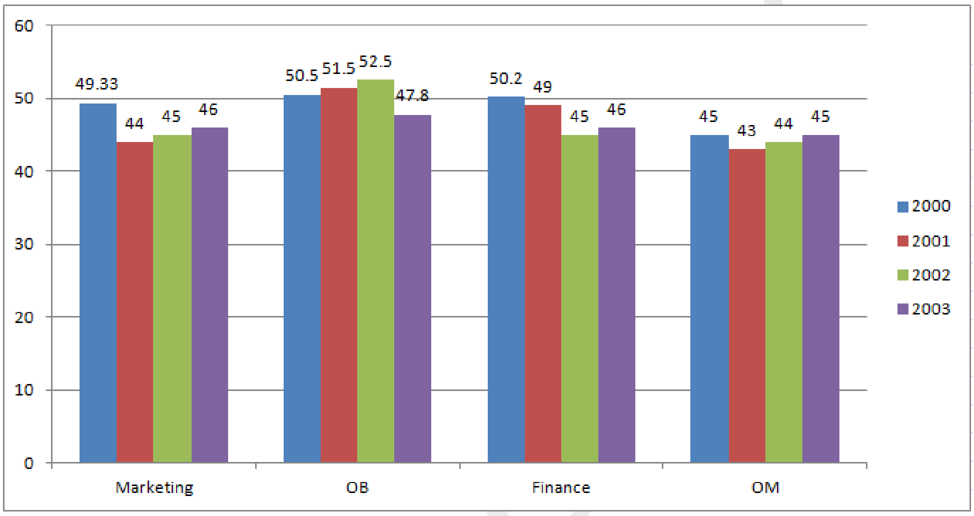
How many countries in Asia were visited by at least one of Dheeraj, Samantha and Nitesh?



VIDEO SOLUTION

Instructions

A management institute was established on January 1, 2000 with 3, 4, 5, and 6 faculty members in the Marketing, Organisational Behaviour (OB), Finance, and Operations Management (OM) areas respectively, to start with. No faculty member retired or joined the institute in the first three months of the year 2000. In the next four years, the institute recruited one faculty member in each of the four areas. All these new faculty members, who joined the institute subsequently over the years, were 25 years old at the time of their joining the institute. All of them joined the institute on April a) During these four years, one of the faculty members retired at the age of 60. The following diagram gives the area-wise average age (in terms of number of completed years) of faculty members as on April 1 of 2000, 2001, 2002, and 2003.



Question 69

What was the age of the new faculty member, who joined the OM area, as on April 1, 2003?

A 25

B 26

C 27

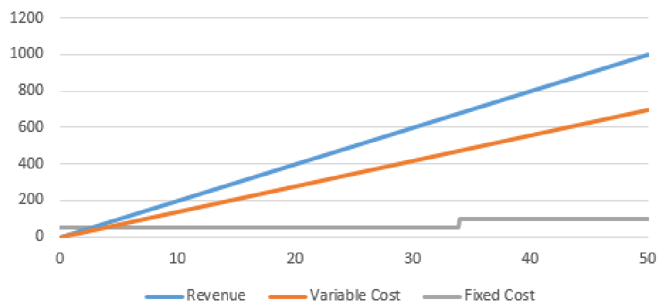
D 28

VIEW SOLUTION

Instructions

Answer the questions based on the following information.

Ghosh Babu has a manufacturing unit. The following graph gives the cost for the various number of units. Given: Profit = Revenue - Variable cost - Fixed cost. The fixed cost remains constant up to 34 units after which additional investment is to be done in fixed assets. In any case, production cannot exceed 50 units.



Note: The fixed cost for upto 34 units is 50 and the the fixed cost for more is 100.
The revenue from 50 units is 1000 and the variable cost from 50 units is 700

Question 70

If the fixed cost of production goes up by Rs. 40, then what is the minimum number of units that need to be manufactured to make sure that there is no loss?

A 10

B 19

C 15

D 20

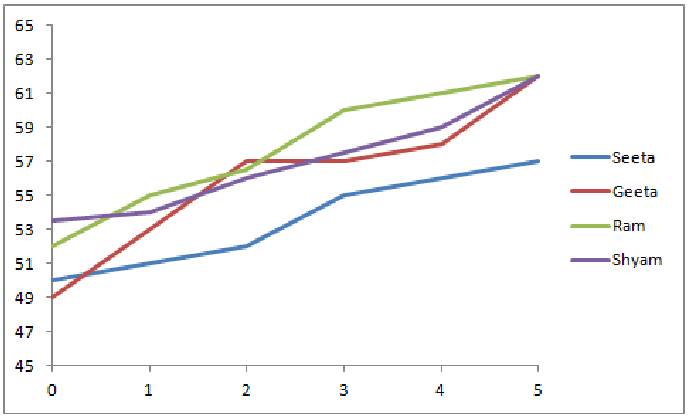
VIDEO SOLUTION

Instructions

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

The length of an infant is one of the measures of his/her development in the early stages of his/her life.

The figure below shows the growth chart of four infants in the first five months of life.



Question 71

The rate of growth during the third month was the lowest for

A Geeta

B Seeta

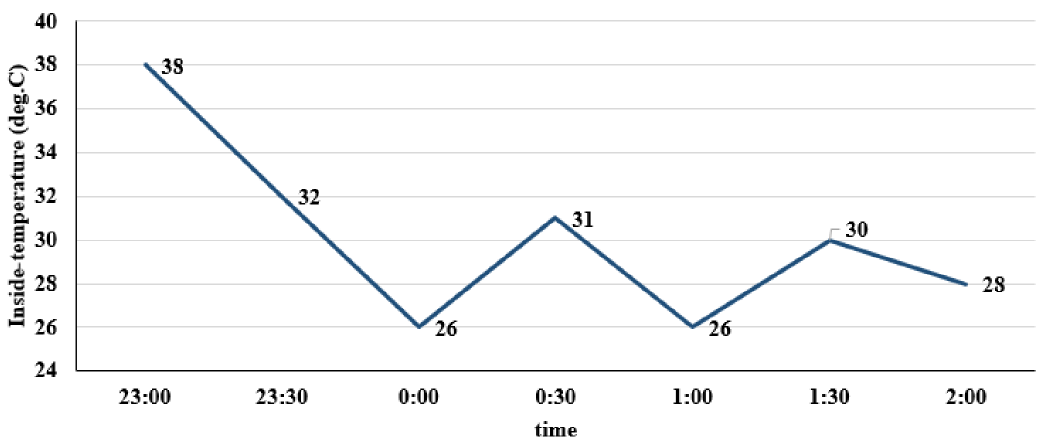
C Ram

D Shyam

VIEW SOLUTION

Instructions

The air-conditioner (AC) in a large room can be operated either in REGULAR mode or in POWER mode to reduce the temperature.
If the AC operates in REGULAR mode, then it brings down the temperature inside the room(called inside temperature) at a constant rate to the set temperature in 1 hour. If it operates in POWER mode, then this is achieved in 30 minutes.
If the AC is switched off, then the inside temperature rises at a constant rate so as to reach the temperature outside at the time of switching off in 1 hour.
The temperature outside has been falling at a constant rate from 7 pm onward until 3 am on a particular night.
The following graph shows the inside temperature between 11 pm (23:00)and 2 am (2:00) that night.



The following facts are known about the AC operation that night.
• The AC was turned on for the first time that night at 11 pm (23:00).
• The AC setting was changed (including turning it on/off, and/or setting different temperatures) only at the beginning of the hour or at 30 minutes after the hour.
• The AC was used in POWER mode for longer duration than in REGULAR mode during this 3-hour period.

Question 72

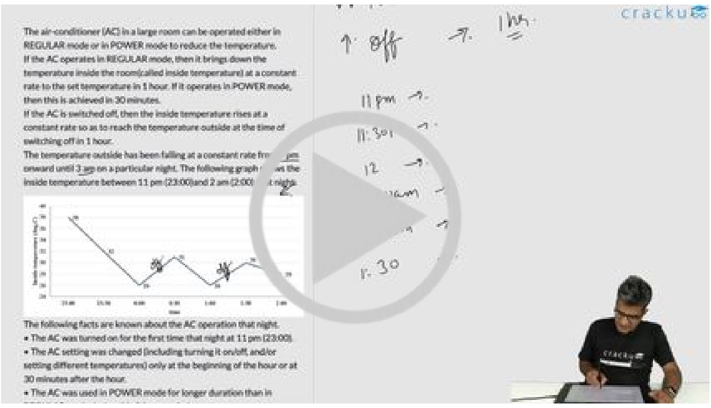
How many times the AC must have been turned off between 11:01 pm and 1:59 am?

A cannot be determined

B 2

C 0

D 1

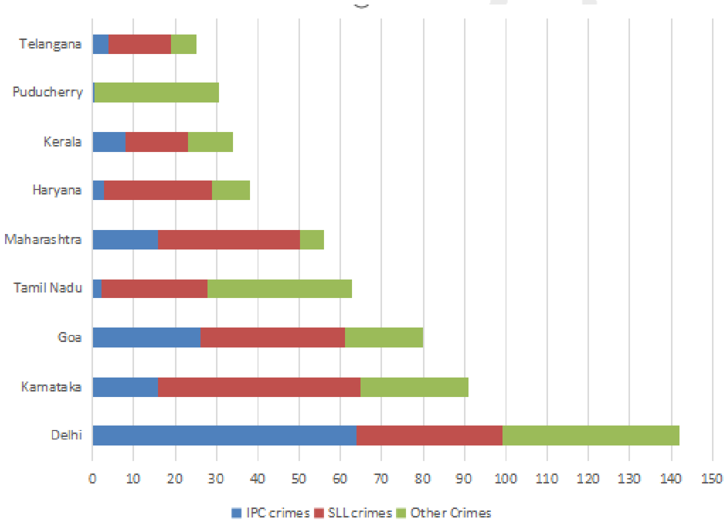


VIDEO SOLUTION

Instructions

The Ministry of Home Affairs is analysing crimes committed by foreigners in different states and union territories (UT) of India. All cases refer to the ones registered against foreigners in 2016.

The number of cases - classified into three categories: IPC crimes, SLL crimes and other crimes - for nine states/UTs are shown in the figure below. These nine belong to the top ten states/UTs in terms of the total number of cases registered. The remaining state (among top ten) is West Bengal, where all the 520 cases registered were SLL crimes.



The table below shows the ranks of the ten states/UTs mentioned above among ALL states/UTs of India in terms of the number of cases registered in each of the three category of crimes. A state/UT is given rank r for a category of crimes if there are (r-1) states/UTs having a larger number of cases registered in that category of crimes. For example, if two states have the same number of cases in a category, and exactly three other states/UTs have larger numbers of cases registered in the same category, then both the states are given rank 4 in that category. Missing ranks in the table are denoted by *.

	IPC crimes	SLL crimes	OtherCrimes
Delhi	*	*	*
Goa	*	4	*
Haryana	8	6	*
Karnataka	3	2	*
Kerala	*	9	*
Maharashtra	3	4	8
Puducherry	13	29	*
Tamil Nadu	11	7	*
Telangana	6	9	8
WestBengal	17	*	16

Question 73

What is the rank of Kerala in the 'IPC crimes' category?

VIDEO SOLUTION

Question 74

Which of the following is DEFINITELY true about the ranks of states/UT in the 'other crimes' category?
i) Tamil Nadu: 2
ii) Puducherry: 3

- **A** both i) and ii)
- **B** only ii)
- **C** neither i) , nor ii)
- **D** only i)

VIDEO SOLUTION

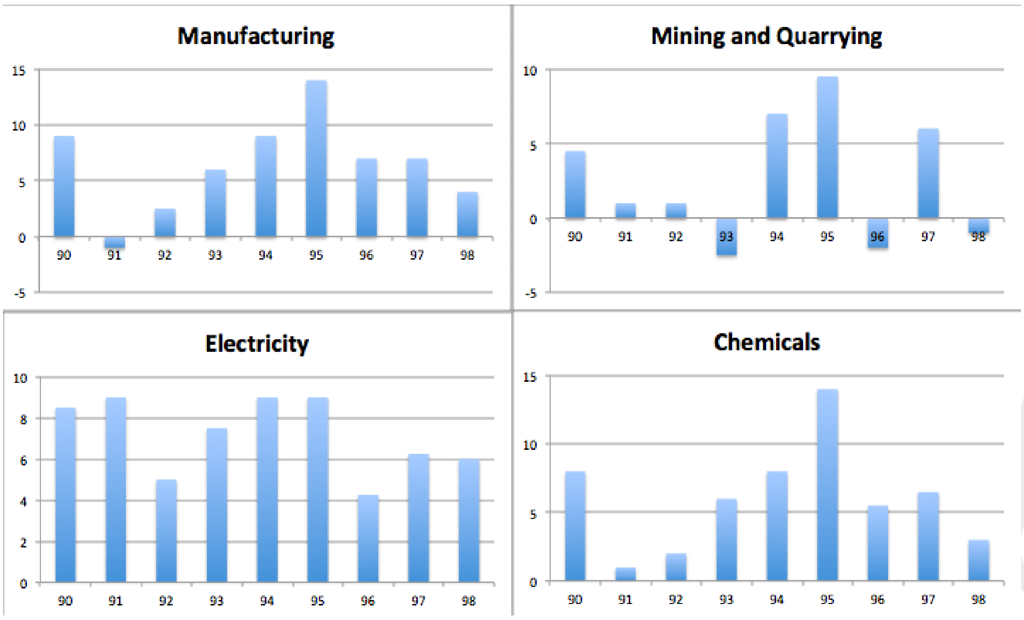
Question 75

What is the sum of the ranks of Delhi in the three categories of crimes?

VIDEO SOLUTION

Instructions

Answer these questions based on the data given below: The figures below present annual growth rate, expressed as the % change relative to the previous year, in four sectors of the economy of the Republic of Reposia during the 9 year period from 1990 to 1998. Assume that the index of production for each of the four sectors is set at 100 in 1989 Further, the four sectors: manufacturing, mining and quarrying, electricity, and chemicals, respectively, constituted 20%, 15%, 10%, 15 % of total industrial production 1989.



Question 76

When was the highest level of production in the manufacturing sector achieved during the nine-year period 1990-98?

A 1998

B 1995

C 1990

D Cannot be determined

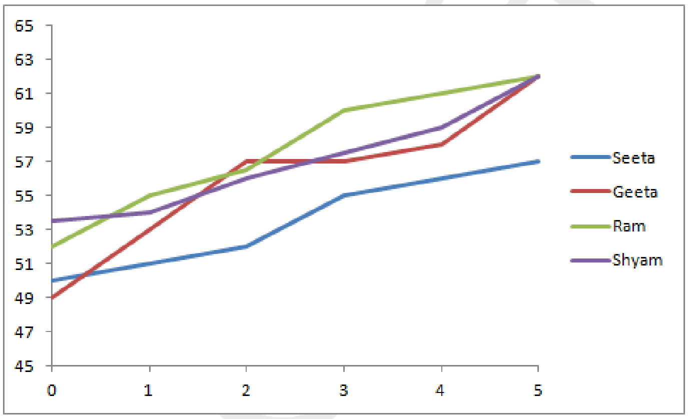
VIEW SOLUTION

Instructions

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

The length of an infant is one of the measures of his/her development in the early stages of his/her life.

The figure below shows the growth chart of four infants in the first five months of life.



Question 77

Who grew at the fastest rate in the first two months of life?

A Geeta

B Seeta

C Ram

D Shyam

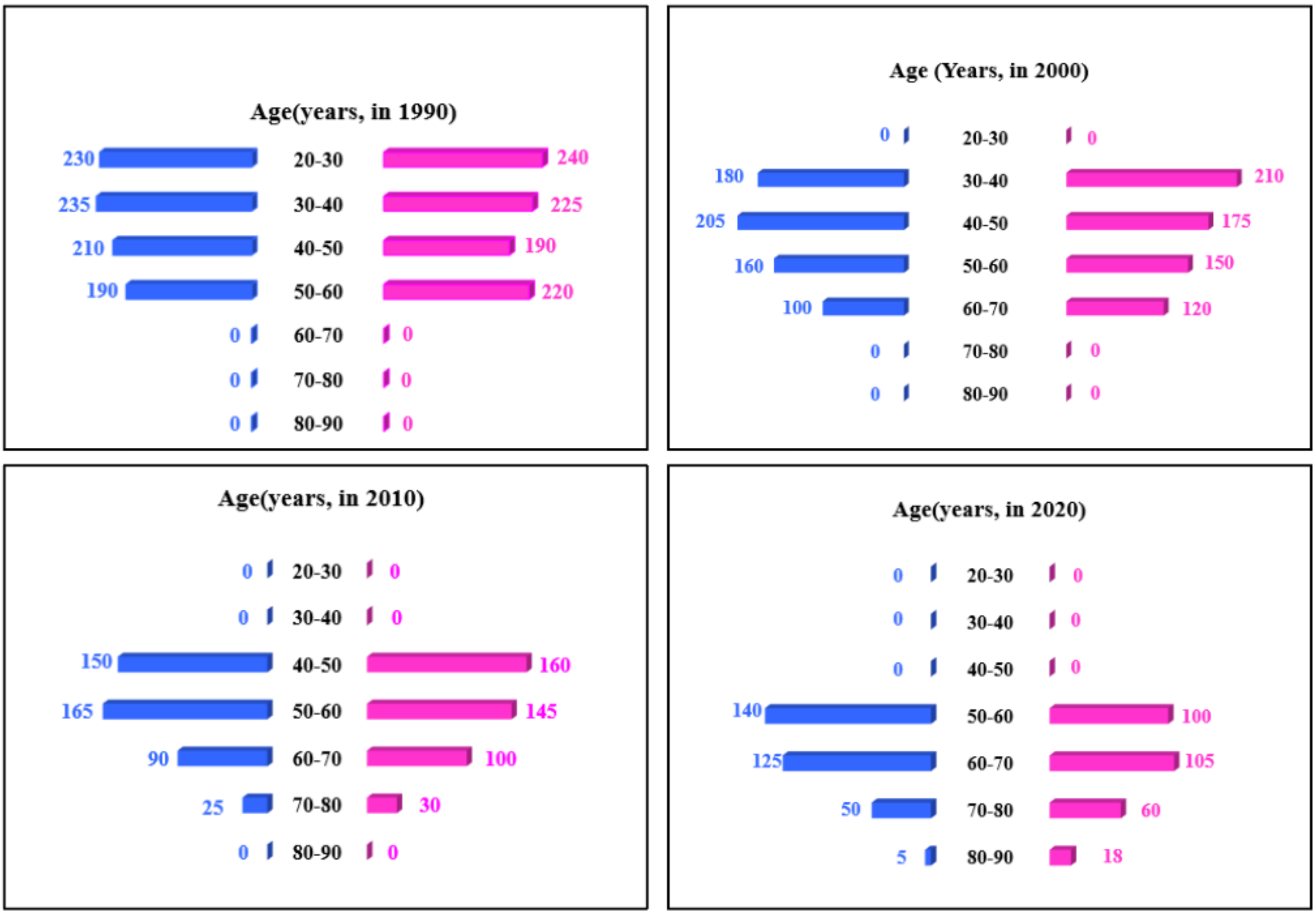
VIEW SOLUTION

Instructions

In the following, a year corresponds to 1st of January of that year.

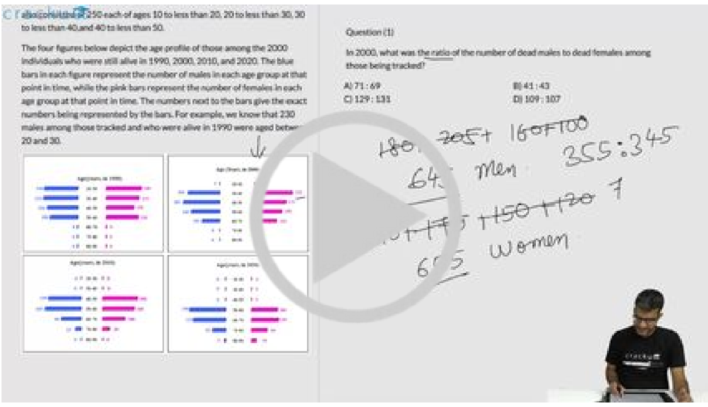
A study to determine the mortality rate for a disease began in 1980. The study chose 1000 males and 1000 females and followed them for forty years or until they died, whichever came first. The 1000 males chosen in 1980 consisted of 250 each of ages 10 to less than 20, 20 to less than 30, 30 to less than 40, and 40 to less than 50. The 1000 females chosen in 1980 also consisted of 250 each of ages 10 to less than 20, 20 to less than 30, 30 to less than 40,and 40 to less than 50.

The four figures below depict the age profile of those among the 2000 individuals who were still alive in 1990, 2000, 2010, and 2020. The blue bars in each figure represent the number of males in each age group at that point in time, while the pink bars represent the number of females in each age group at that point in time. The numbers next to the bars give the exact numbers being represented by the bars. For example, we know that 230 males among those tracked and who were alive in 1990 were aged between 20 and 30.



Question 78

How many of the males who were being tracked and who were between 20 and 30 years of age in 1980 died in the period 2000 to 2010?

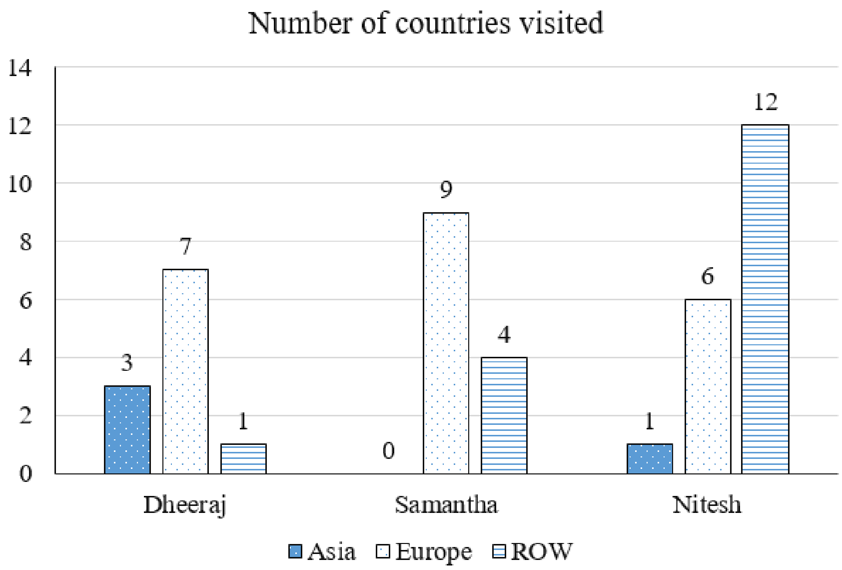


VIDEO SOLUTION

CAT Previous Papers PDF

Instructions

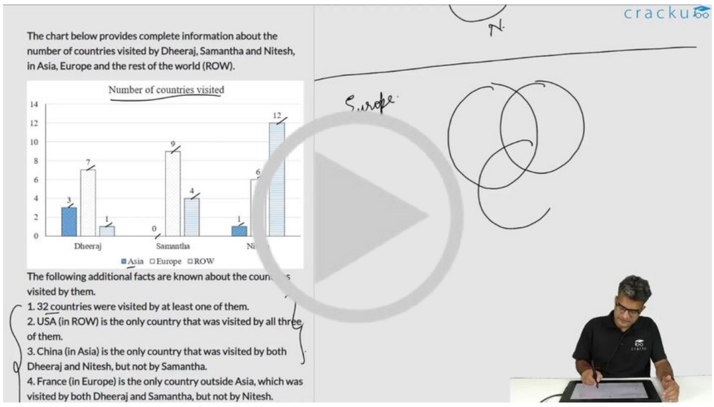
The chart below provides complete information about the number of countries visited by Dheeraj, Samantha and Nitesh, in Asia, Europe and the rest of the world (ROW).



The following additional facts are known about the countries visited by them.
1. 32 countries were visited by at least one of them.
2. USA (in ROW) is the only country that was visited by all three of them.
3. China (in Asia) is the only country that was visited by both Dheeraj and Nitesh, but not by Samantha.
4. France (in Europe) is the only country outside Asia, which was visited by both Dheeraj and Samantha, but not by Nitesh.
5. Half of the countries visited by both Samantha and Nitesh are in Europe.

Question 79

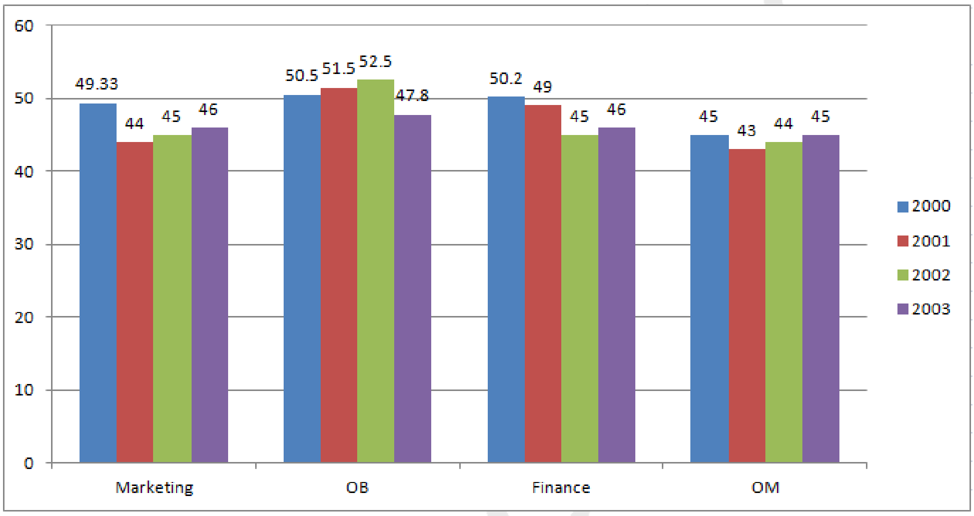
How many countries in Europe were visited only by Nitesh?



VIDEO SOLUTION

Instructions

A management institute was established on January 1, 2000 with 3, 4, 5, and 6 faculty members in the Marketing, Organisational Behaviour (OB), Finance, and Operations Management (OM) areas respectively, to start with. No faculty member retired or joined the institute in the first three months of the year 2000. In the next four years, the institute recruited one faculty member in each of the four areas. All these new faculty members, who joined the institute subsequently over the years, were 25 years old at the time of their joining the institute. All of them joined the institute on April a) During these four years, one of the faculty members retired at the age of 60. The following diagram gives the area-wise average age (in terms of number of completed years) of faculty members as on April 1 of 2000, 2001, 2002, and 2003.



Question 80

Professors Naresh and Devesh, two faculty members in the Marketing area, who have been with the Institute since its inception, share a birthday, which falls on 20th November. One was born in 1947 and the other one in 1950. On April 1 2005, what was the age of the third faculty member, who has been in the same area since inception?

A 47

B 50

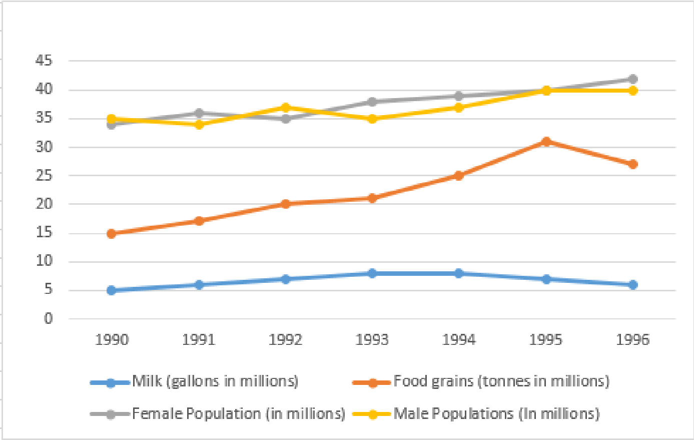
C 51

D 52

VIEW SOLUTION

Instructions

The graph given below shows the quantity of milk and food grains consumed annually along with female and male population (in millions). Use the data to answer the questions that follow.



Question 81

In which year was the difference between the percentage increase in the production of food grains and milk maximum?

A 1993

B 1994

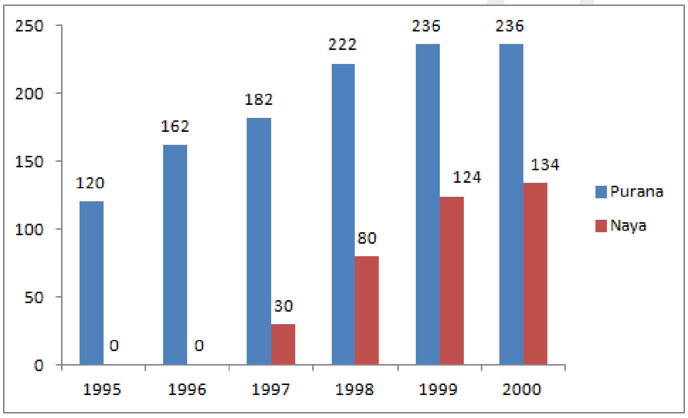
C 1995

D 1996

VIEW SOLUTION

Instructions

Directions for the following four questions: Answer the questions on the basis of the information given below.



Purana and Naya are two brands of kitchen mixer-grinders available in the local market. Purana is an old brand that was introduced in 1990, while Naya was introduced in 1997. For both these brands, 20% of the mixer-grinders bought in a particular year are disposed off as junk exactly two years later. It is knows that 10 Purana mixer-grinders were disposed off in 1997. The following figures show the number of Purana and Naya mixer-grinders in operation from 1995 to 2000, as at the end of the year.

Question 82

How many Purana mixer-grinders were purchased in 1999?

A 20

B 23

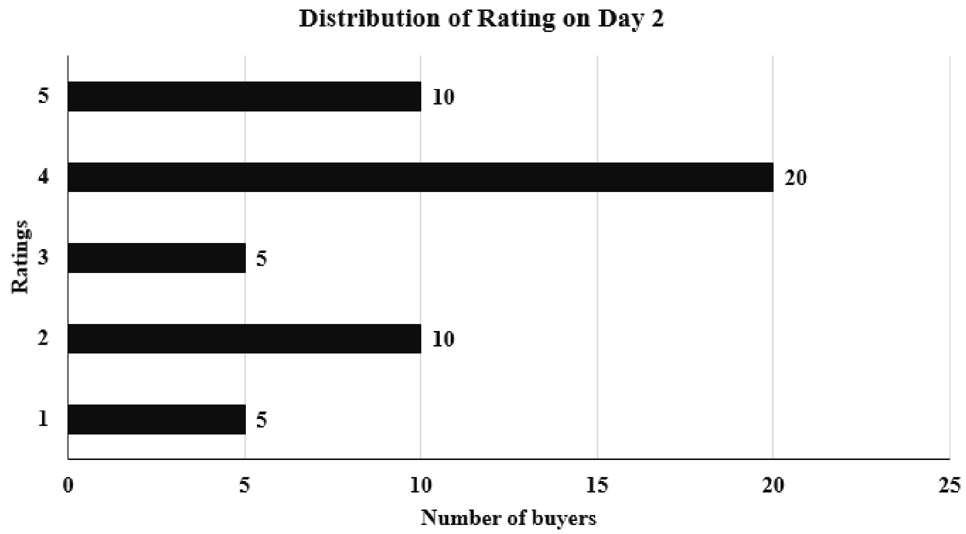
C 50

D Cannot be determined from the data

VIEW SOLUTION

Instructions

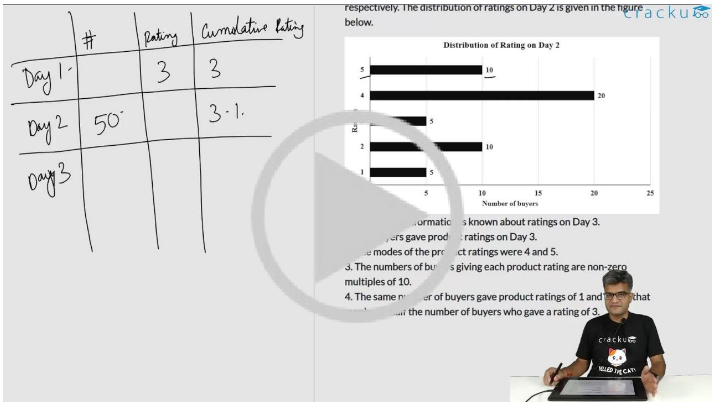
An online e-commerce firm receives daily integer product ratings from 1 through 5 given by buyers. The daily average is the average of the ratings given on that day. The cumulative average is the average of all ratings given on or before that day. The rating system began on Day 1, and the cumulative averages were 3 and 3.1 at the end of Day 1 and Day 2, respectively. The distribution of ratings on Day 2 is given in the figure below.



The following information is known about ratings on Day 3.
1. 100 buyers gave product ratings on Day 3.
2. The modes of the product ratings were 4 and 5.
3. The numbers of buyers giving each product rating are non-zero multiples of 10.
4. The same number of buyers gave product ratings of 1 and 2, and that number is half the number of buyers who gave a rating of 3.

Question 83

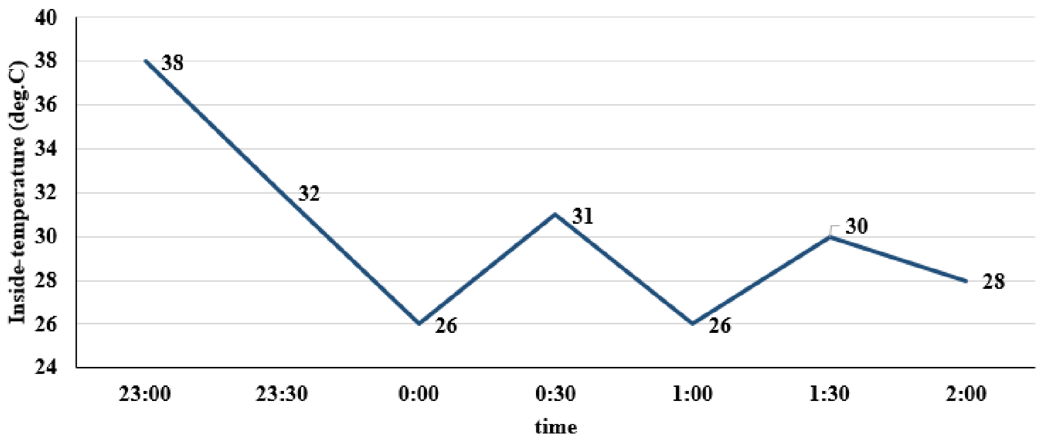
What is the median of all ratings given on Day 3?



VIDEO SOLUTION

Instructions

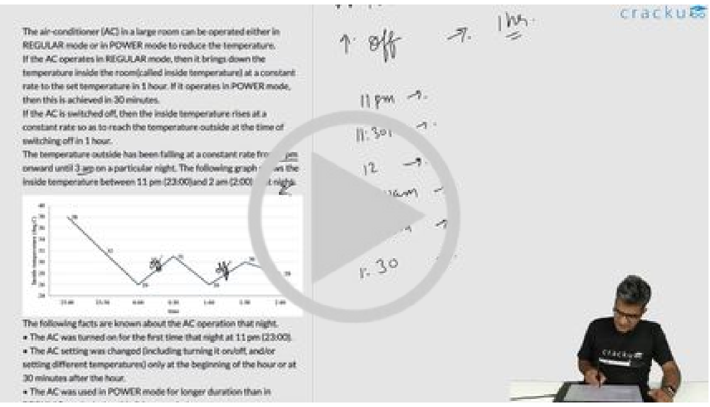
The air-conditioner (AC) in a large room can be operated either in REGULAR mode or in POWER mode to reduce the temperature.
If the AC operates in REGULAR mode, then it brings down the temperature inside the room(called inside temperature) at a constant rate to the set temperature in 1 hour. If it operates in POWER mode, then this is achieved in 30 minutes.
If the AC is switched off, then the inside temperature rises at a constant rate so as to reach the temperature outside at the time of switching off in 1 hour.
The temperature outside has been falling at a constant rate from 7 pm onward until 3 am on a particular night. The following graph shows the inside temperature between 11 pm (23:00)and 2 am (2:00) that night.



The following facts are known about the AC operation that night.
• The AC was turned on for the first time that night at 11 pm (23:00).
• The AC setting was changed (including turning it on/off, and/or setting different temperatures) only at the beginning of the hour or at 30 minutes after the hour.
• The AC was used in POWER mode for longer duration than in REGULAR mode during this 3-hour period.

Question 84

What was the maximum difference between temperature outside and inside temperature, in degree Celsius, between 11:01 pm and 1:59 am?



VIDEO SOLUTION

Free CAT Exam Preparation App

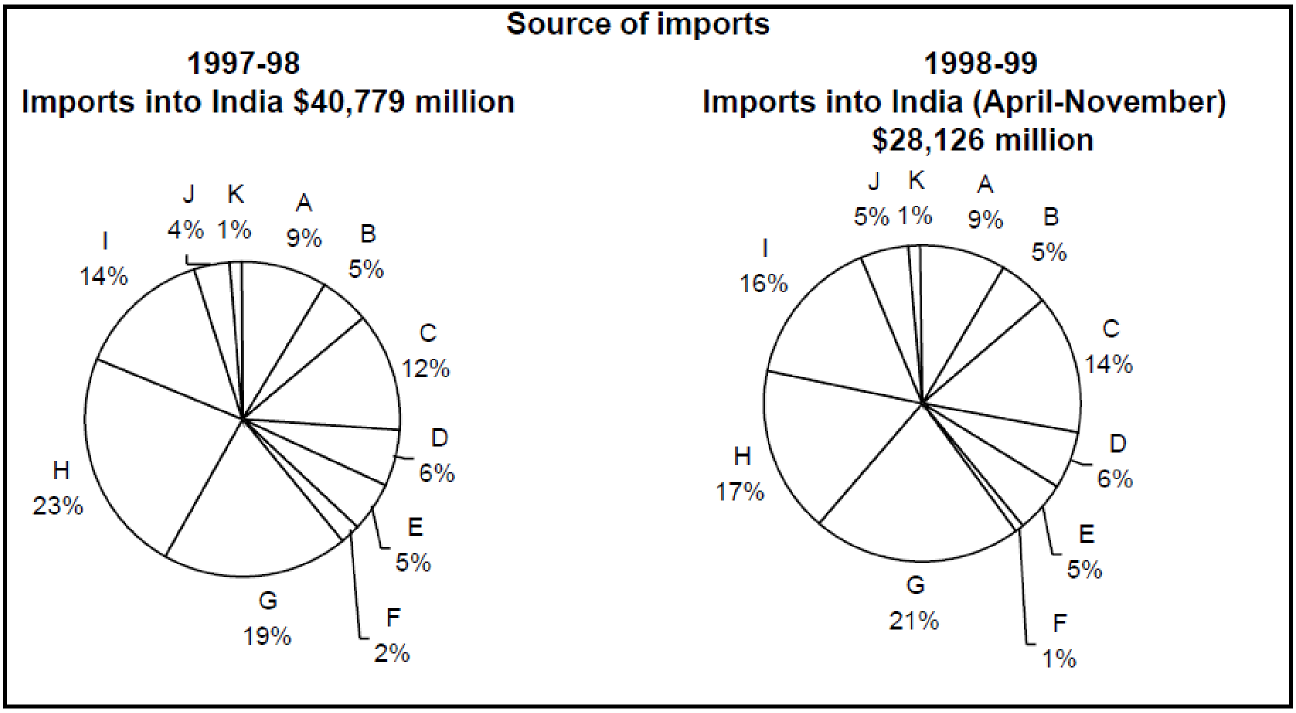
Instructions

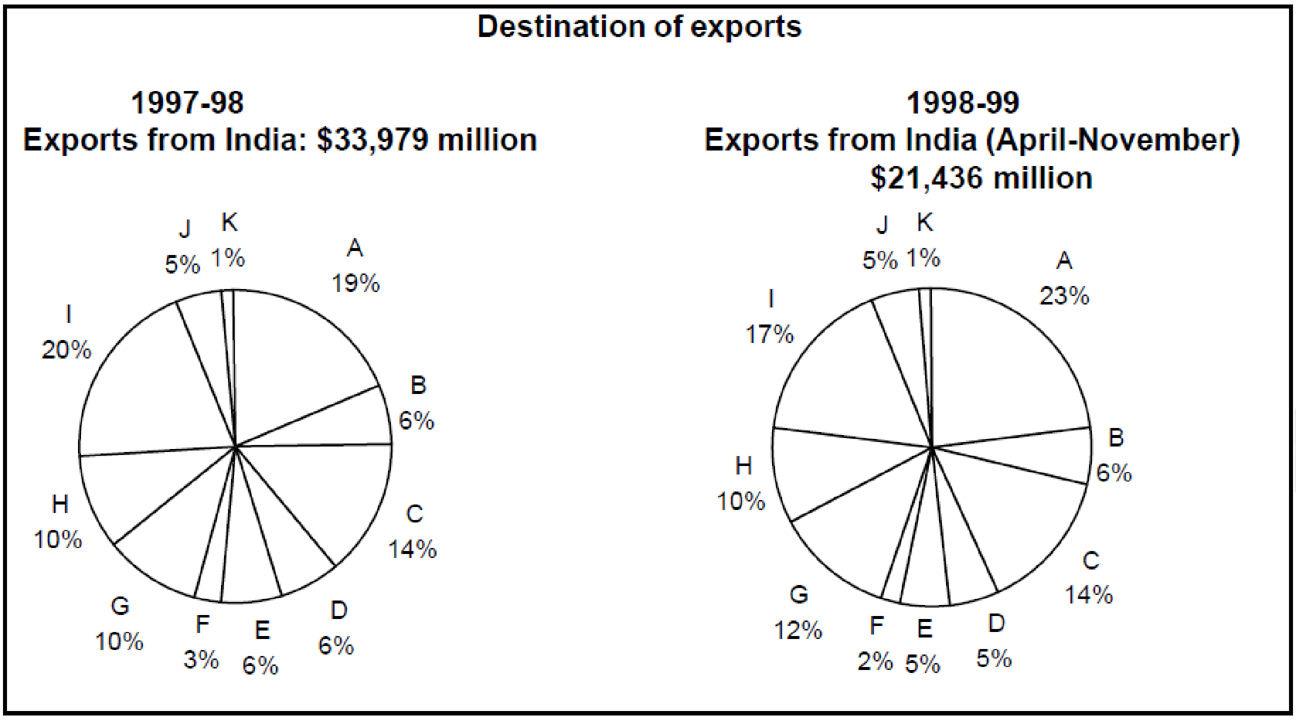
Consider the information provided in the figure below relating to India's foreign trade in 1997-98 and the first eight months of 1998-99.

Total trade with a region is defined as the sum of exports to and imports from that region.
Trade deficit is defined as the excess of imports over exports. Trade deficit may be negative.

A:USA. B:Germany C:Other EU. D:U.K. E:Japan F:Russia

G:Other East Europe H:OPEC   I:Asia J:Other LDCs K:Others





Question 85

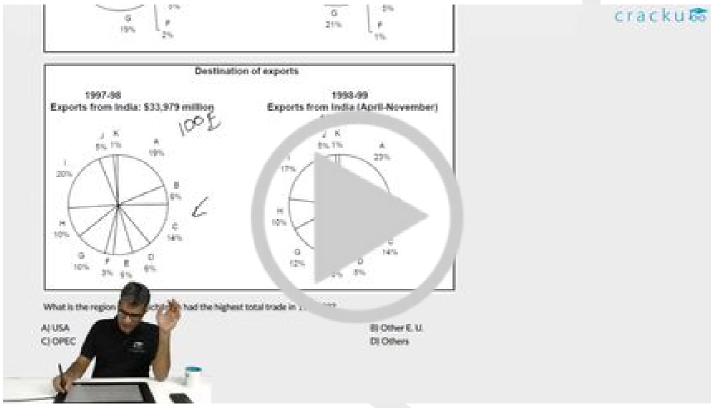
Assume that the average monthly exports from India and imports to India during the remaining four months of 1998-99 would be the same as that for the first eight months of the year. What is the percentage growth rate in India's total trade deficit between 1997-98 and 1998-99?

A 43.5

B 47.5

C 50

D 40

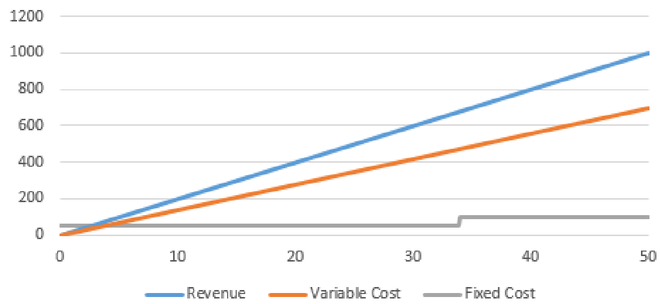


VIDEO SOLUTION

Instructions

Answer the questions based on the following information.

Ghosh Babu has a manufacturing unit. The following graph gives the cost for the various number of units. Given: Profit = Revenue - Variable cost - Fixed cost. The fixed cost remains constant up to 34 units after which additional investment is to be done in fixed assets. In any case, production cannot exceed 50 units.



Note: The fixed cost for upto 34 units is 50 and the the fixed cost for more is 100.
The revenue from 50 units is 1000 and the variable cost from 50 units is 700

Question 86

If at the most 40 units can be manufactured, then what is the number of units that can be manufactured to maximise profit per unit?

A 40

B 34

C 35

D 25

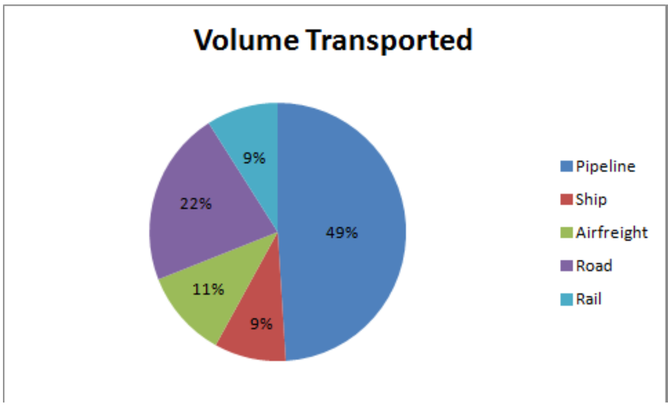
VIDEO SOLUTION

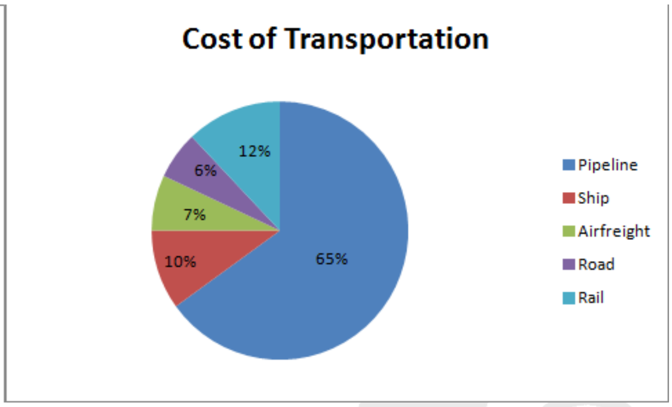
Instructions

Directions for the following three questions: Answer the questions based on the pie charts given below.

Chart 1 shows the distribution of 12 million tonnes of crude oil transported through different modes over a specific period of time.

Chart 2 shows the distribution of the cost of transporting this crude oil. The total cost was Rs. 30 million.





Question 87

If the costs per tonne of transport by ship, air and road are represented by P, Q and R respectively, which of the following is true?

A R > Q > P

B P > R > Q

C P > Q > R

D R > P > Q

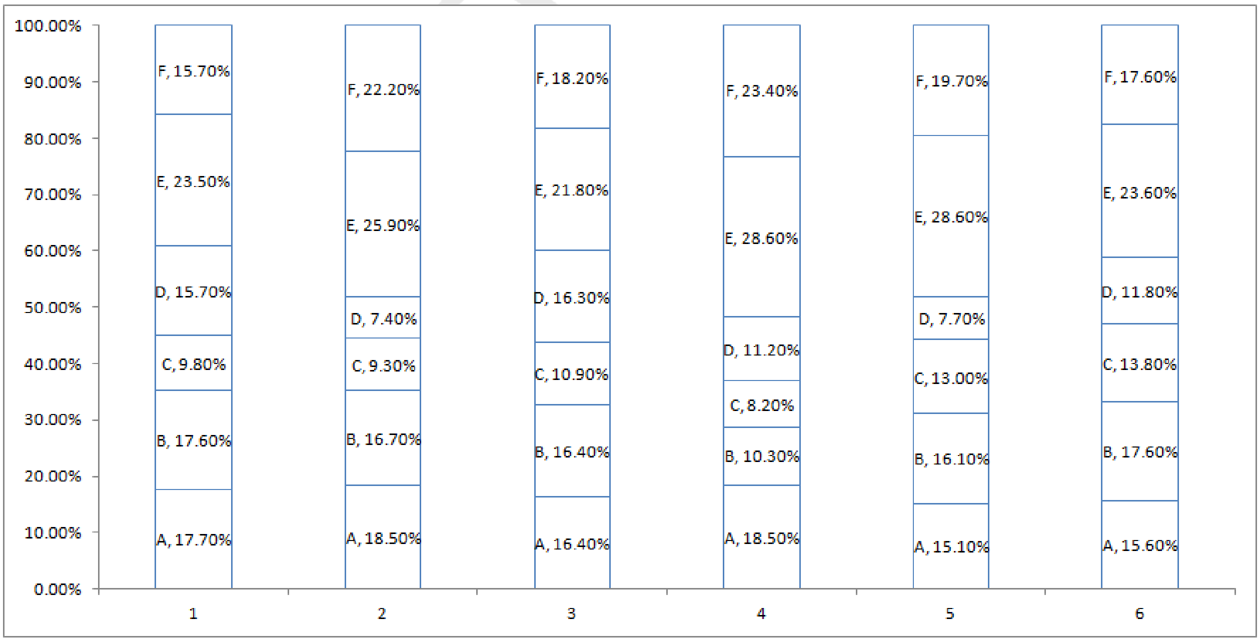
VIEW SOLUTION

Instructions

Directions for the following three questions: Answer these questions based on the data given below:

There are six companies, 1 through 6. All of these companies use six operations, A through F. The following graph shows the distribution of efforts put in by each company in these six operations.

The Y axis represents the % distribution of effort and the X axis represents the company



Question 88

Suppose effort allocation is inter-changed between operations B and C, then C and D, and then D and E. If companies are then ranked in ascending order of effort in E, what will be the rank of company 3?

A 2

B 3

C 4

D 5

VIEW SOLUTION

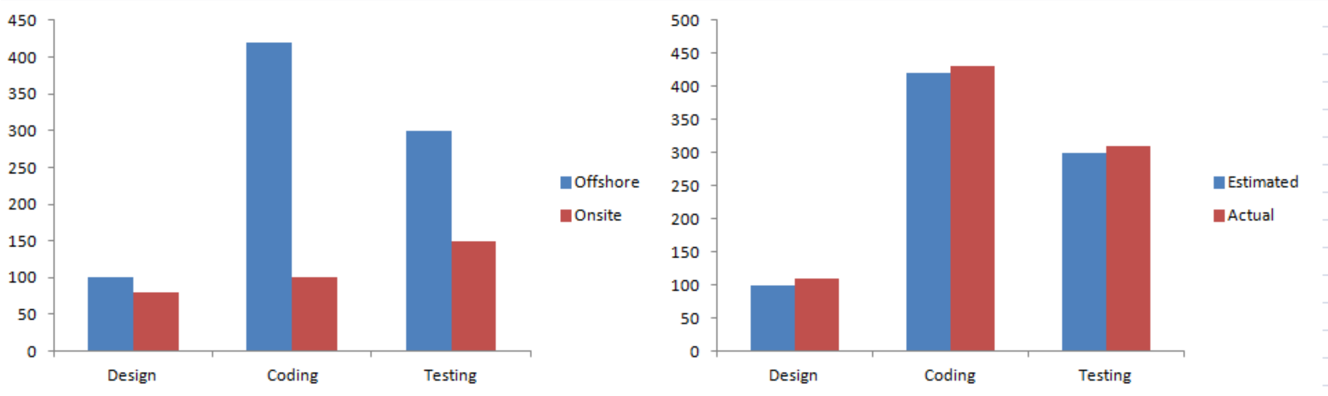
Instructions

Directions for the following five questions: Answer the questions based on the two graphs shown below.

Figure I (the chart on the left) shows the amount of work distribution, in man-hours, for a software company between offshore and onsite activities.

Figure 2 (the chart on the right) shows the estimated and actual work effort involved in the different offshore activities in the same company during the same period.

[Note: Onsite refers to work performed at the customer's premise and offshore refers to work performed at the developer's premise.]



Question 89

If 50% of the offshore work were to be carried out onsite, with the distribution of effort between the tasks remaining the same, the proportion of testing carried out offshore would be

A 40%

B 30%

C 50%

D 70%

VIEW SOLUTION

Question 90

The total effort in man-hours spent onsite is nearest to which of the following?

A The sum of the estimated and actual effort for offshore design

B The estimated man-hours of offshore coding

C The actual man-hours of offshore testing

D Half of the man-hours of estimated offshore coding

VIEW SOLUTION

CAT Formulas PDF [Download Now]

Question 91

Roughly, what percentage of the total work is carried out onsite?

A 40%

B 20 %

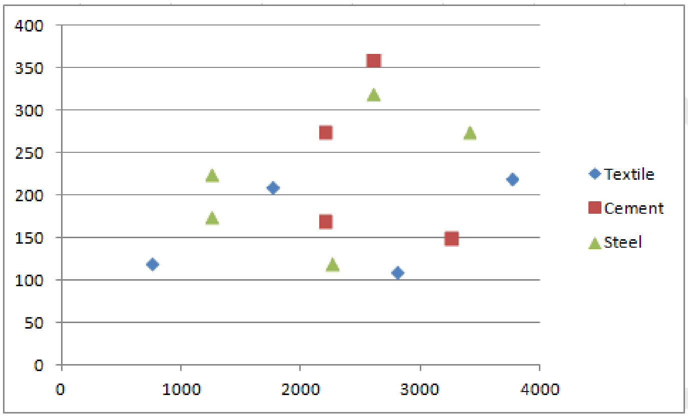
C 30 %

D 10 %

VIEW SOLUTION

Instructions

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.



Each point in the graph below shows the profit and turnover data for a company. Each company belongs to one of the three industries: textile, cement and steel.

Question 92

For how many steel companies with a turnover of more than 2000 is the profit less than 300?

A 0

B 1

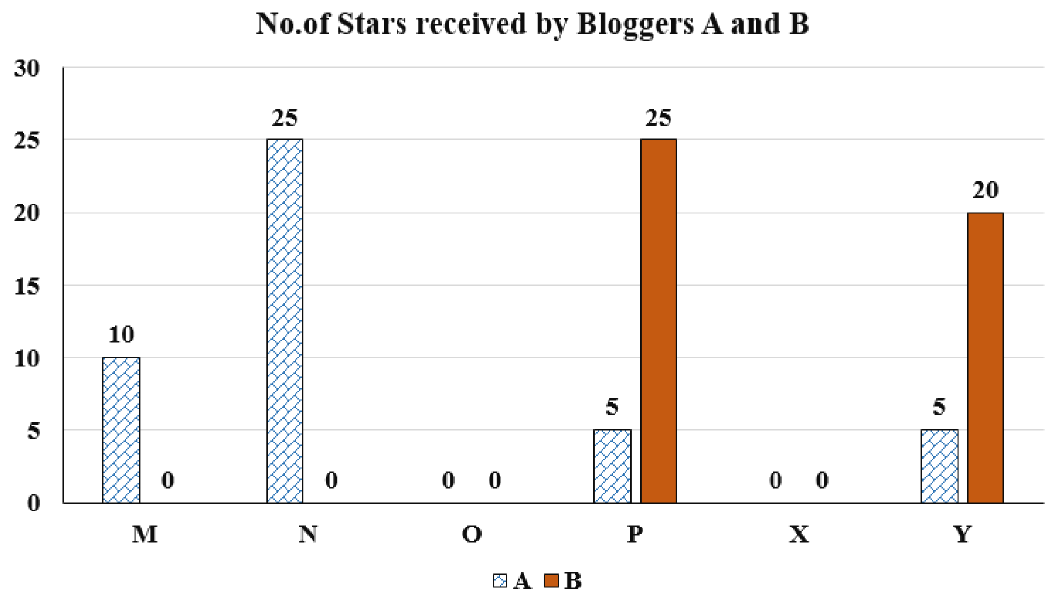
C 2

D 7

VIEW SOLUTION

Instructions

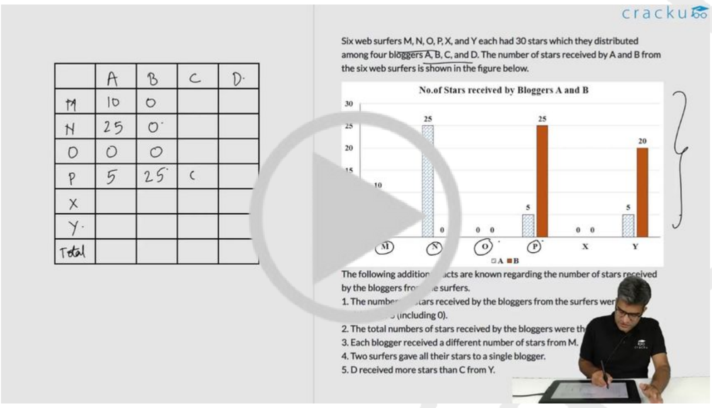
Six web surfers M, N, O, P, X, and Y each had 30 stars which they distributed among four bloggers A, B, C, and D. The number of stars received by A and B from the six web surfers is shown in the figure below.



The following additional facts are known regarding the number of stars received by the bloggers from the surfers.
1. The numbers of stars received by the bloggers from the surfers were all multiples of 5 (including 0).
2. The total numbers of stars received by the bloggers were the same.
3. Each blogger received a different number of stars from M.
4. Two surfers gave all their stars to a single blogger.
5. D received more stars than C from Y.

Question 93

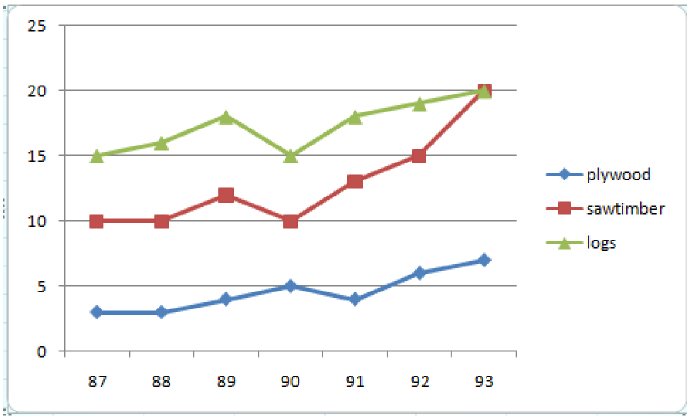
How many surfers distributed their stars among exactly 2 bloggers?



VIDEO SOLUTION

Instructions

Answer the questions based on the following information. In the following chart, the price of logs shown is per cubic metre that of plywood and saw timber is per tonne.



Question 94

If the volume sales of plywood, saw timber and logs were 40%, 30% and 30% respectively, then what was the average realisation in 1993 per cubic metre of sales? (Weight of one cubic metre of saw dust and plywood both = 800 kg)

A 18

B 15

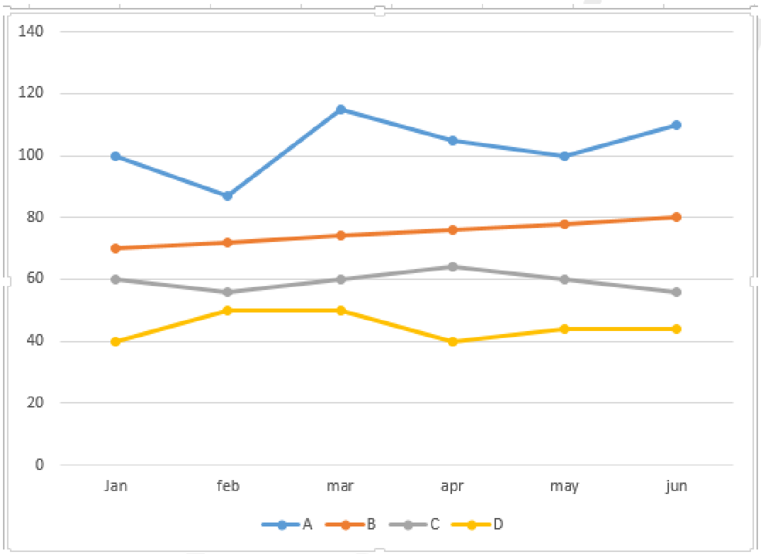
C 16

D 13

VIEW SOLUTION

Instructions

The graph below shows the end of the month market values of 4 shares for the period from January to June. Answer the following questions based on this graph.



Question 95

In which month was the greatest percentage increase in market value for any share recorded?

A February

B March

C April

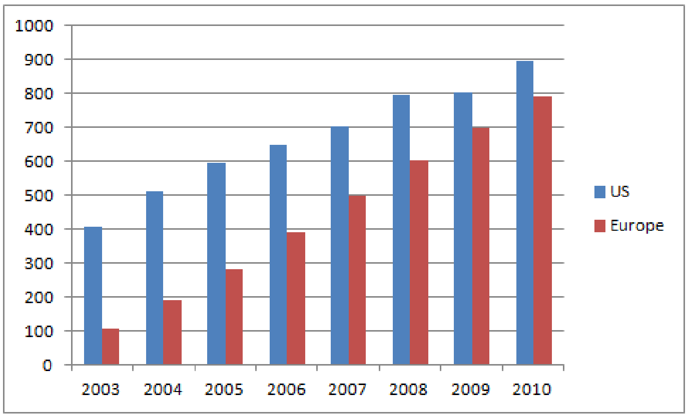
D January

VIEW SOLUTION

Instructions

Directions for the following four questions: Answer the following questions based on the information given below: The bar chart below shows the revenue received in million US Dollars (USD), from subscribers to a particular Internet service. The data covers the period 2003 to 2007 for the United States (US) and Europe. The bar chart also shows the estimated revenues from subscription to this service for the period 2008 to 2010.

The Y axis represents the subscription revenue in Million (USD) and the X axis represents the years.



Question 96

In 2003, sixty percent of subscribers in Europe were men. Given that women subscribers increase at the rate of 10 percent annum and men at the rate of 5 percent per annum, what is the approximate percentage growth of subscribers between 2003 and 2010 in Europe? The subscription prices are volatile and may change each year.

A 62

B 15

C 78

D 84

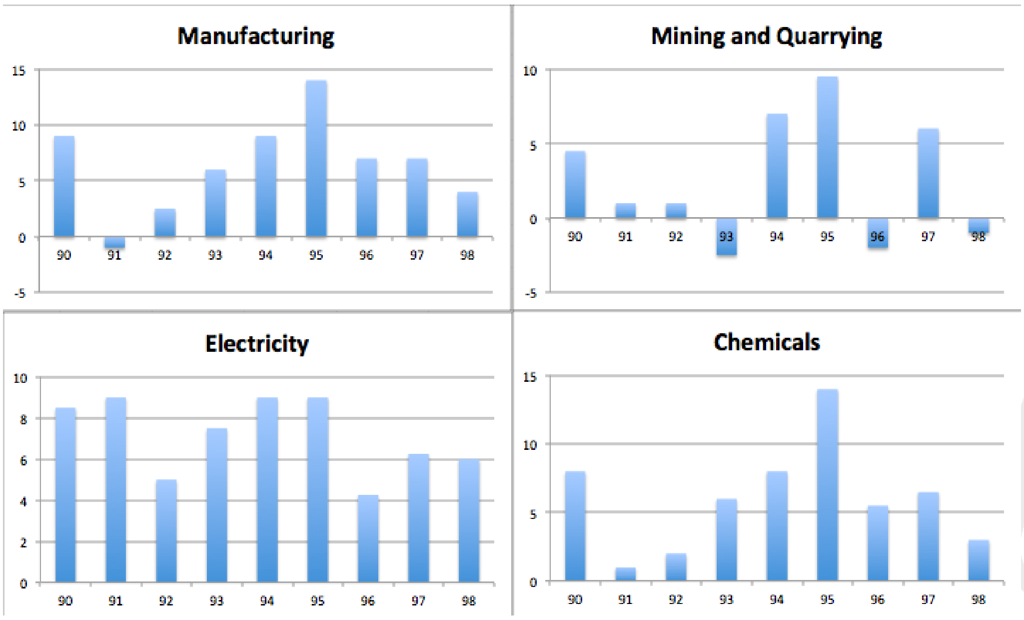
E 50

VIEW SOLUTION

Top-500 Free CAT Questions (With Solutions)

Instructions

Answer these questions based on the data given below: The figures below present annual growth rate, expressed as the % change relative to the previous year, in four sectors of the economy of the Republic of Reposia during the 9 year period from 1990 to 1998. Assume that the index of production for each of the four sectors is set at 100 in 1989 Further, the four sectors: manufacturing, mining and quarrying, electricity, and chemicals, respectively, constituted 20%, 15%, 10%, 15 % of total industrial production 1989.



Question 97

The overall growth rate in 1991 of the four sectors together is approximately:

A 10%

B 1%

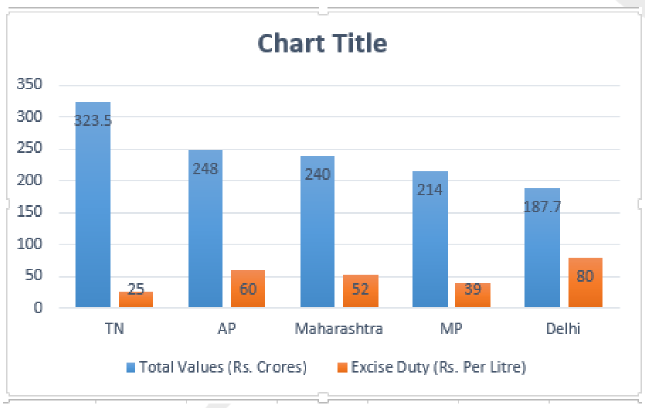
C 2.5%

D 1.5%

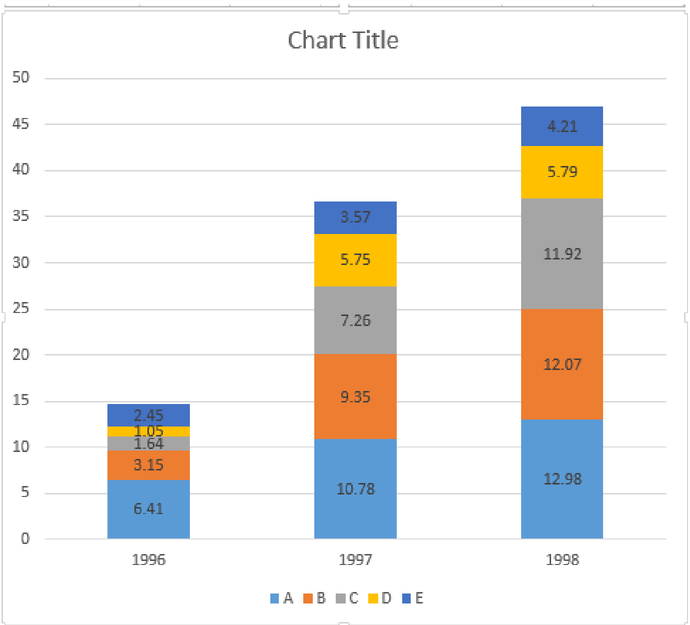
VIEW SOLUTION

Instructions

The following graph shows the value of liquor supplied by the 5 states in 1996 and the excise duty rates in each state.



Amount of liquor supplied in Tamil Nadu Distilleries A, B, C, D, E (from bottom to top) in lakh litres.



Question 98

If Excise duty is levied before the goods leave the factory (on the value of the liquor), then which of the following choices shows distilleries in ascending order of the excise duty paid by them for the year 1996? (Assume the total liquor in TN is supplied by only these 5 distilleries).

A ECABD

B ADEBC

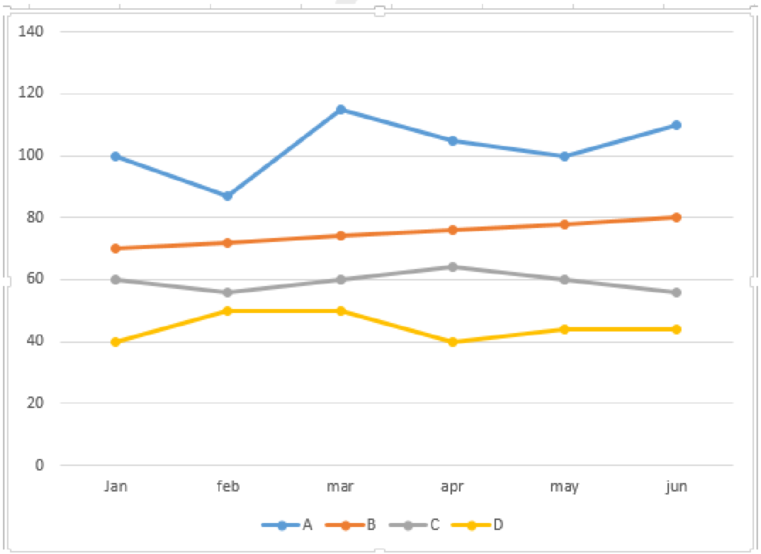
C DCEBA

D Cannot be determined.

VIEW SOLUTION

Instructions

The graph below shows the end of the month market values of 4 shares for the period from January to June. Answer the following questions based on this graph.



Question 99

Ram gives Shyam a share of C and a share of D and takes a share of A in return. What is the maximum loss suffered by Ram from this transaction?

A 15

B 10

C 5

D 0

VIEW SOLUTION

Free CAT Study Material

Question 100

Ram gives Shyam a share of C and a share of D and takes a share of A in return. At which month-end would the Ram's loss from this transaction, be the most?

A June

B March

C April

D February

VIEW SOLUTION

Question 101

In which month was the greatest absolute change in market value for any share recorded?

A March

B April

C May

D June

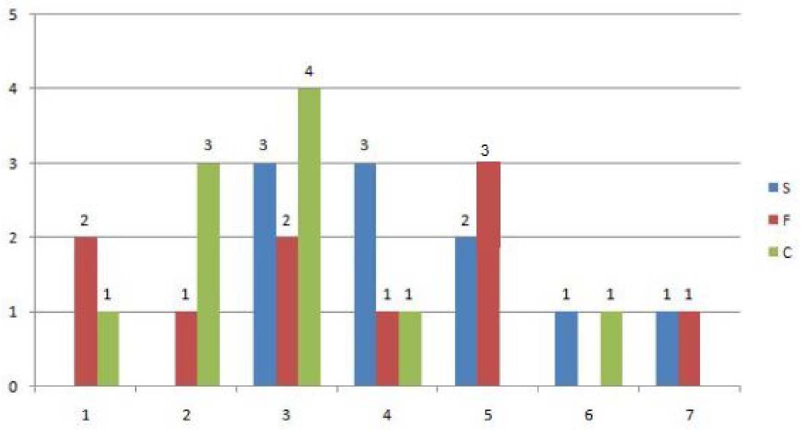
VIEW SOLUTION

Instructions

Simple Happiness index (SHI) of a country is computed on the basis of three, parameters: social support (S),freedom to life choices (F) and corruption perception (C). Each of these three parameters is measured on a scale of 0 to 8 (integers only). A country is then categorised based on the total score obtained by summing the scores of all the three parameters, as shown in the following table:

Toatal Score	0-4	5-8	9-13	14-19	20-24
Category	Very Unhappy	Unhappy	Neutral	Happy	Very Happy

Following diagram depicts the frequency distribution of the scores in S, F and C of 10 countries - Amda, Benga, Calla, Delma, Eppa, Varsa, Wanna, Xanda,Yanga and Zooma:



Further, the following are known.
1. Amda and Calla jointly have the lowest total score, 7, with identical scores in all the three parameters.
2. Zooma has a total score of 17.
3. All the 3 countries, which are categorised as happy, have the highest score In exactly one parameter.

Question 102

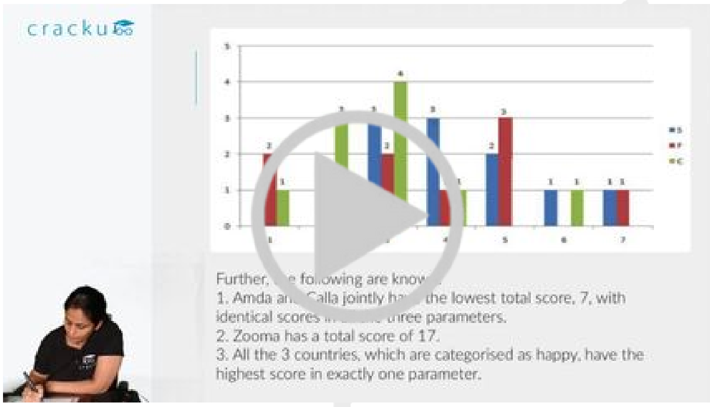
Benga and Delma, two countries categorized as happy, are tied with the same total score. What is the maximum score they can have?

A 14

B 15

C 16

D 17



VIDEO SOLUTION

Question 103

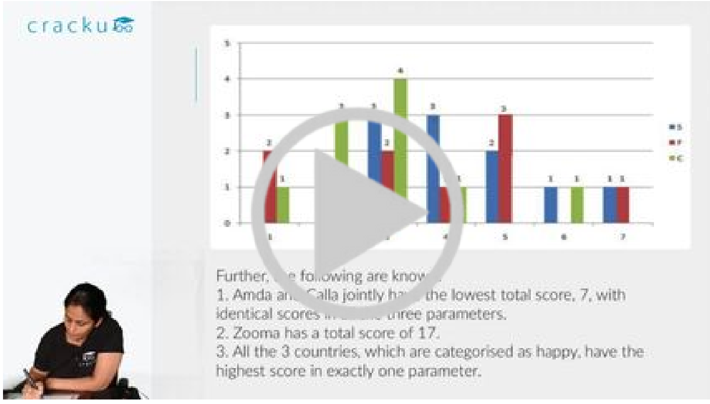
If Benga scores 16 and Delma scores 15, then what is the maximum number of countries with a score of 13?

A 0

B 1

C 2

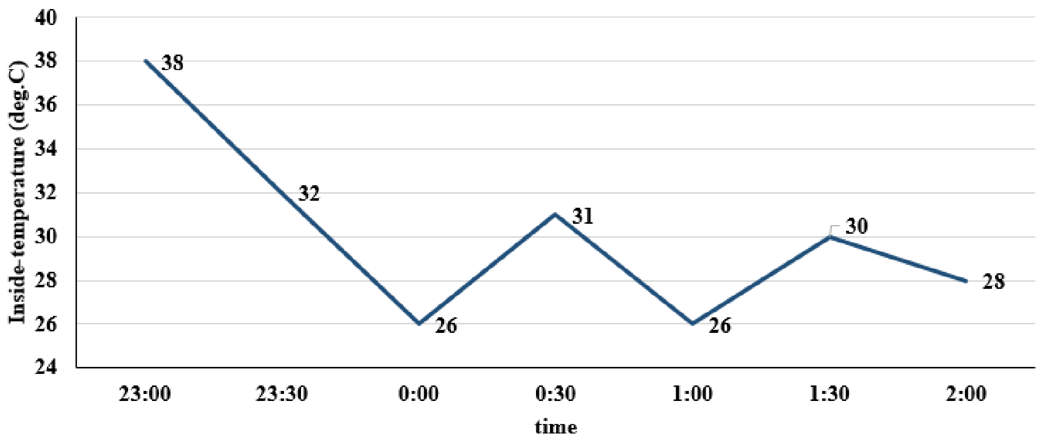
D 3



VIDEO SOLUTION

Instructions

The air-conditioner (AC) in a large room can be operated either in REGULAR mode or in POWER mode to reduce the temperature.
If the AC operates in REGULAR mode, then it brings down the temperature inside the room(called inside temperature) at a constant rate to the set temperature in 1 hour. If it operates in POWER mode, then this is achieved in 30 minutes.
If the AC is switched off, then the inside temperature rises at a constant rate so as to reach the temperature outside at the time of switching off in 1 hour.
The temperature outside has been falling at a constant rate from 7 pm onward until 3 am on a particular night.
The following graph shows the inside temperature between 11 pm (23:00)and 2 am (2:00) that night.



The following facts are known about the AC operation that night.
• The AC was turned on for the first time that night at 11 pm (23:00).
• The AC setting was changed (including turning it on/off, and/or setting different temperatures) only at the beginning of the hour or at 30 minutes after the hour.
• The AC was used in POWER mode for longer duration than in REGULAR mode during this 3-hour period.

Question 104

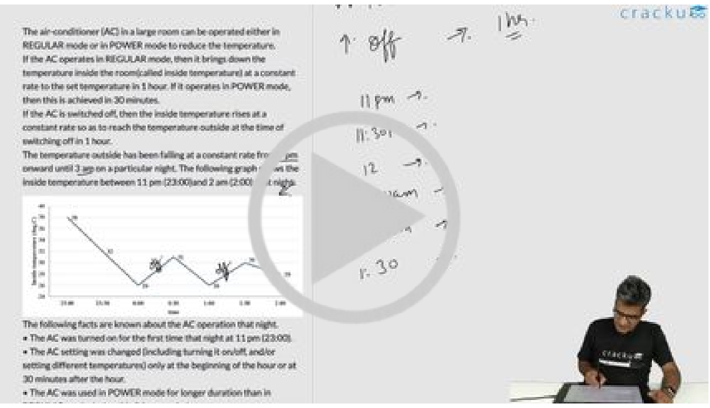
What best can be concluded about the number of times the AC must have either been turned on or the AC temperature setting been altered between 11:01 pm and 1:59 am?

A More than 3

B Either 2 or 3

C Exactly 2

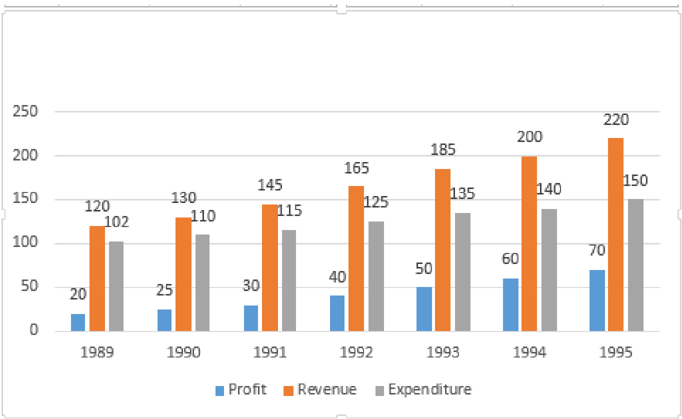
D Exactly 3



VIDEO SOLUTION

Instructions

Answer the questions based on the following information.



Question 105

If the profit in 1996 shows the annual rate of growth as it had shown in 1995 over the previous year, then what approximately will be the profit in 1996?

A Rs. 72 lakh

B Rs. 82 lakh

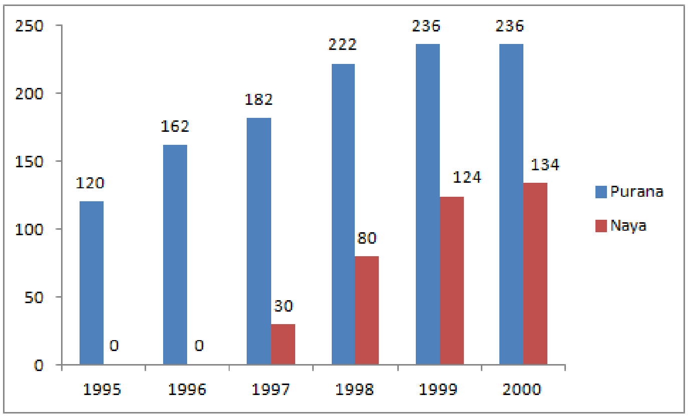
C Rs. 93 lakh

D Rs. 78 lakh

VIEW SOLUTION

Instructions

Directions for the following four questions: Answer the questions on the basis of the information given below.



Purana and Naya are two brands of kitchen mixer-grinders available in the local market. Purana is an old brand that was introduced in 1990, while Naya was introduced in 1997. For both these brands, 20% of the mixer-grinders bought in a particular year are disposed off as junk exactly two years later. It is knows that 10 Purana mixer-grinders were disposed off in 1997. The following figures show the number of Purana and Naya mixer-grinders in operation from 1995 to 2000, as at the end of the year.

Question 106

How many Naya mixer-grinders were disposed off by the end of 2000?

A 10

B 16

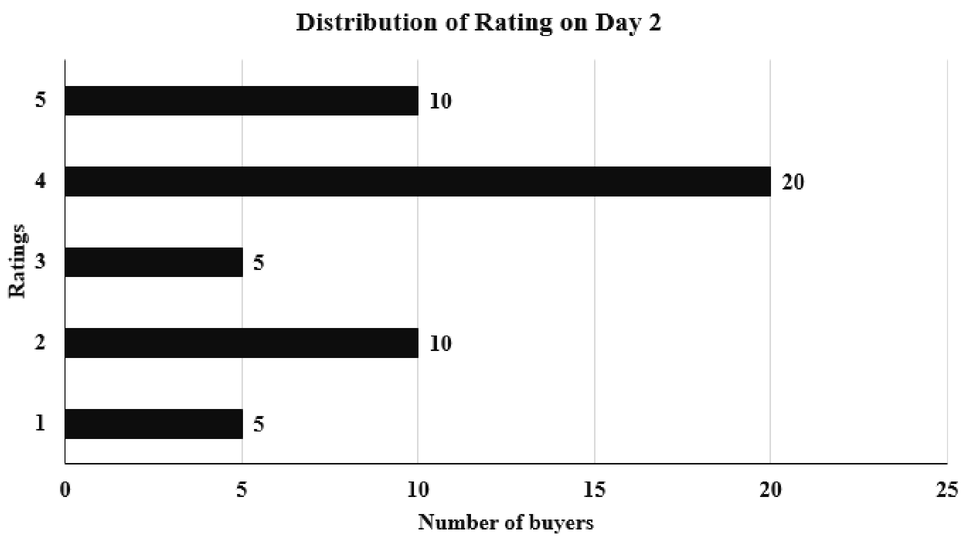
C 22

D Cannot be determined from the data

VIEW SOLUTION

Instructions

An online e-commerce firm receives daily integer product ratings from 1 through 5 given by buyers. The daily average is the average of the ratings given on that day. The cumulative average is the average of all ratings given on or before that day. The rating system began on Day 1, and the cumulative averages were 3 and 3.1 at the end of Day 1 and Day 2, respectively. The distribution of ratings on Day 2 is given in the figure below.



The following information is known about ratings on Day 3.
1. 100 buyers gave product ratings on Day 3.
2. The modes of the product ratings were 4 and 5.

3. The numbers of buyers giving each product rating are non-zero multiples of 10.
4. The same number of buyers gave product ratings of 1 and 2, and that number is half the number of buyers who gave a rating of 3.

Question 107

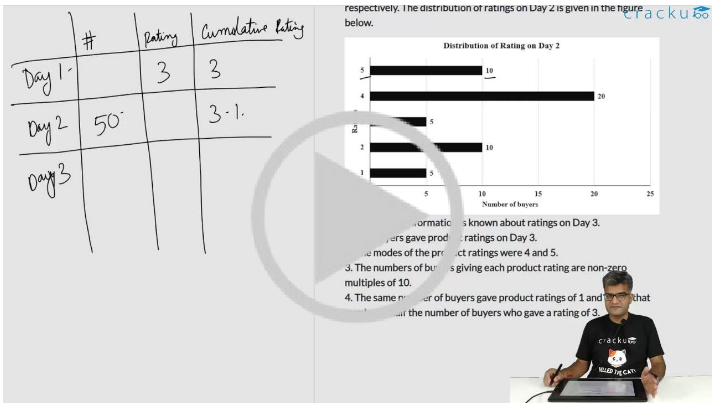
What is the daily average rating of Day 3?

A 3.6

B 3.0

C 3.2

D 3.5

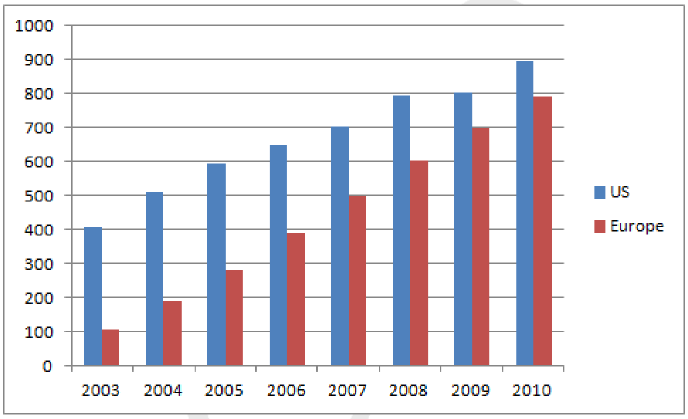


VIDEO SOLUTION

Instructions

Directions for the following four questions: Answer the following questions based on the information given below: The bar chart below shows the revenue received in million US Dollars (USD), from subscribers to a particular Internet service. The data covers the period 2003 to 2007 for the United States (US) and Europe. The bar chart also shows the estimated revenues from subscription to this service for the period 2008 to 2010.

The Y axis represents the subscription revenue in Million (USD) and the X axis represents the years.



Question 108

Consider the annual percent change in the gap between subscription revenues in the US and Europe. What is the year in which the absolute value of this change is the highest?

A 03 - 04

B 05 - 06

C 06 - 07

D 08 - 09

E 09 - 10

VIEW SOLUTION

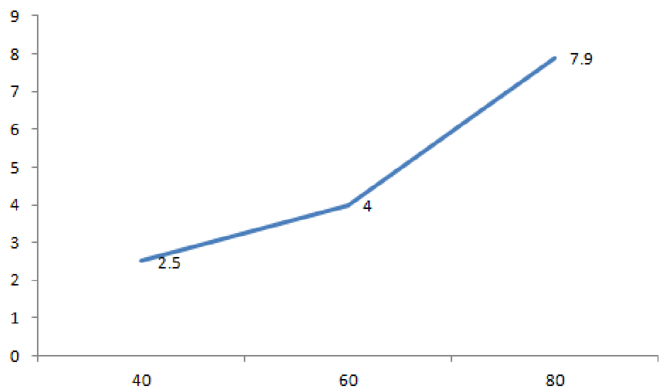
Free Videos for CAT Preparation

Instructions

Directions for the following two questions: Answer the questions based on the following information.

The petrol consumption rate of a new model car 'Palto' depends on its speed and may be described by the graph below.

The axis represents the speed and the Y axis represents the Fuel Consumption (Liters per hour)



Question 109

Manasa would like to minimize the fuel consumption for the trip by driving at the appropriate speed. How should she change the speed?

[CAT 2001]

A Increase the speed

B Decrease the speed

C Maintain the speed at 60 km/hr

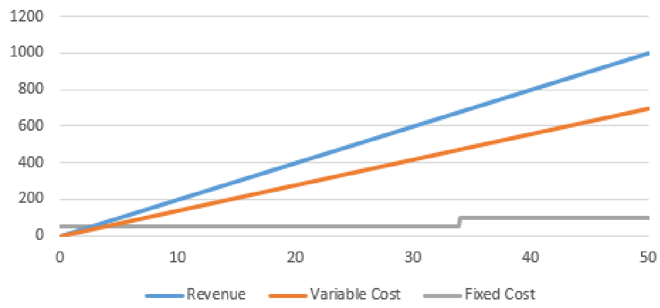
D Cannot be determined

VIEW SOLUTION

Instructions

Answer the questions based on the following information.

Ghosh Babu has a manufacturing unit. The following graph gives the cost for the various number of units. Given: Profit = Revenue - Variable cost - Fixed cost. The fixed cost remains constant up to 34 units after which additional investment is to be done in fixed assets. In any case, production cannot exceed 50 units.



Note: The fixed cost for upto 34 units is 50 and the the fixed cost for more is 100.
The revenue from 50 units is 1000 and the variable cost from 50 units is 700

Question 110

If the production cannot exceed 45 units, then what is the number of units that can maximise profit per unit?

A 40

B 34

C 45

D 35

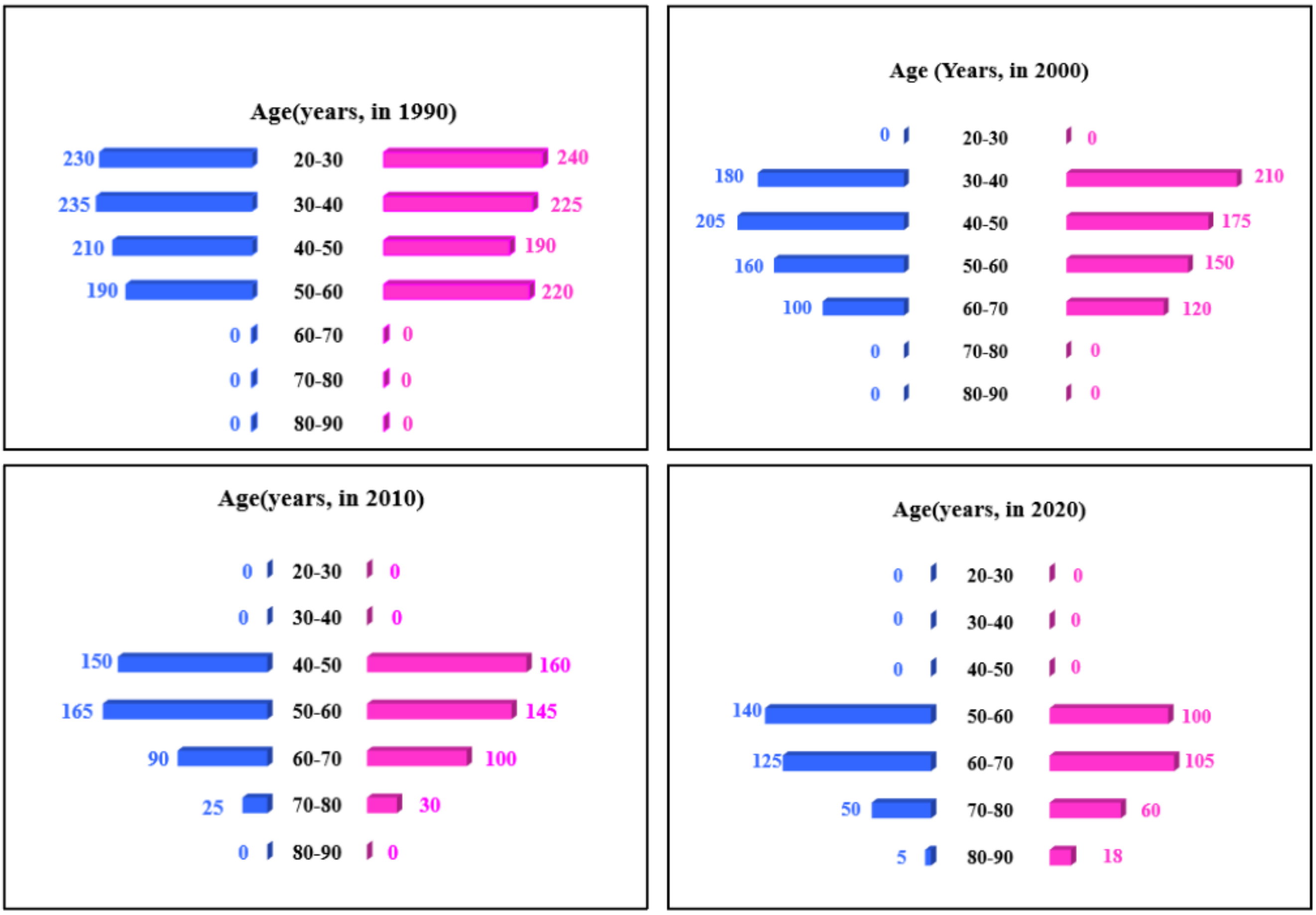
VIDEO SOLUTION

Instructions

In the following, a year corresponds to 1st of January of that year.

A study to determine the mortality rate for a disease began in 1980. The study chose 1000 males and 1000 females and followed them for forty years or until they died, whichever came first. The 1000 males chosen in 1980 consisted of 250 each of ages 10 to less than 20, 20 to less than 30, 30 to less than 40, and 40 to less than 50. The 1000 females chosen in 1980 also consisted of 250 each of ages 10 to less than 20, 20 to less than 30, 30 to less than 40,and 40 to less than 50.

The four figures below depict the age profile of those among the 2000 individuals who were still alive in 1990, 2000, 2010, and 2020. The blue bars in each figure represent the number of males in each age group at that point in time, while the pink bars represent the number of females in each age group at that point in time. The numbers next to the bars give the exact numbers being represented by the bars. For example, we know that 230 males among those tracked and who were alive in 1990 were aged between 20 and 30.



Question 111

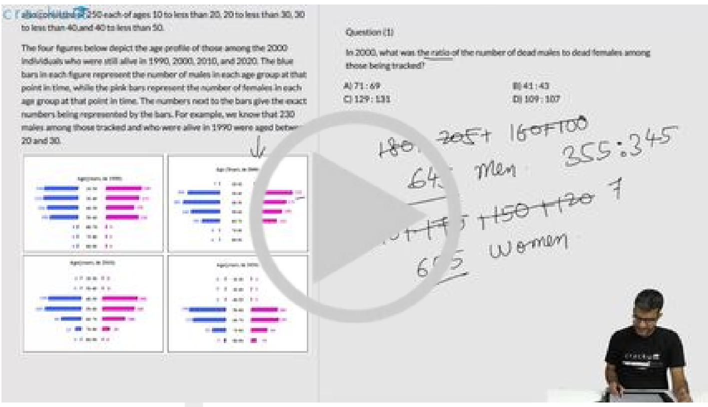
In 2000, what was the ratio of the number of dead males to dead females among those being tracked?

A 71 : 69

B 41 : 43

C 129 : 131

D 109 : 107



VIDEO SOLUTION

CAT Percentile Predictor

Question 112

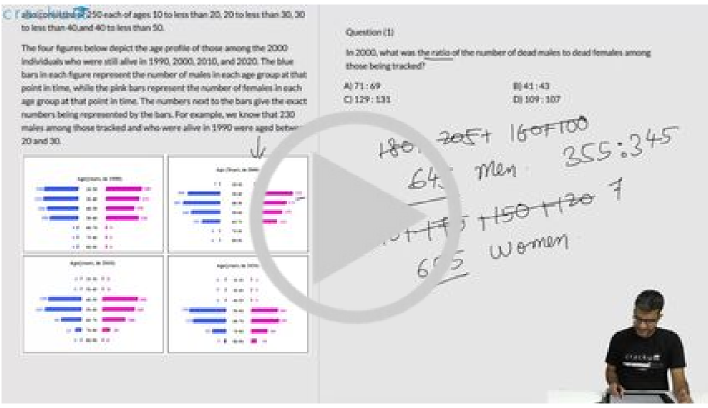
How many individuals who were being tracked and who were less than 30 years of age in 1980 survived until 2020?

A 240

B 580

C 470

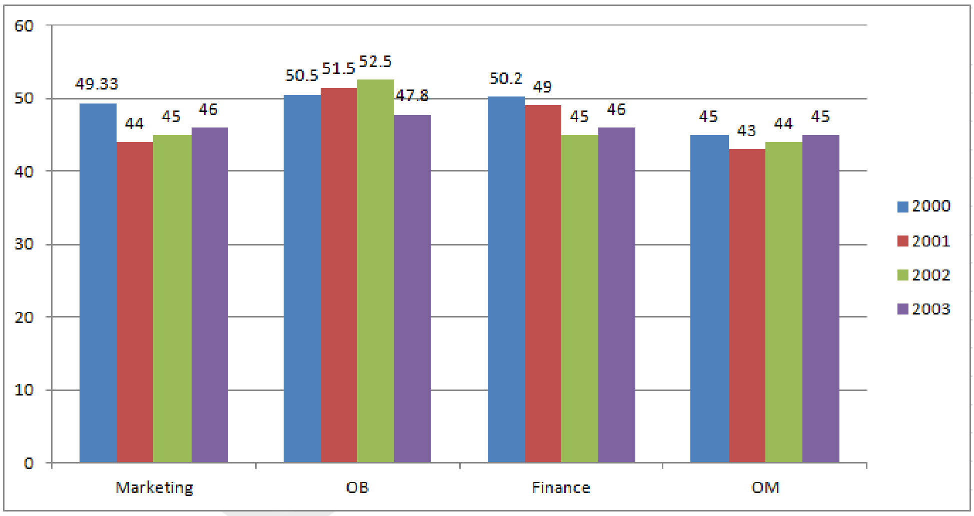
D 230



VIDEO SOLUTION

Instructions

A management institute was established on January 1, 2000 with 3, 4, 5, and 6 faculty members in the Marketing, Organisational Behaviour (OB), Finance, and Operations Management (OM) areas respectively, to start with. No faculty member retired or joined the institute in the first three months of the year 2000. In the next four years, the institute recruited one faculty member in each of the four areas. All these new faculty members, who joined the institute subsequently over the years, were 25 years old at the time of their joining the institute. All of them joined the institute on April a) During these four years, one of the faculty members retired at the age of 60. The following diagram gives the area-wise average age (in terms of number of completed years) of faculty members as on April 1 of 2000, 2001, 2002, and 2003.



Question 113

In which year did the new faculty member join the Finance area?

A 2000

B 2001

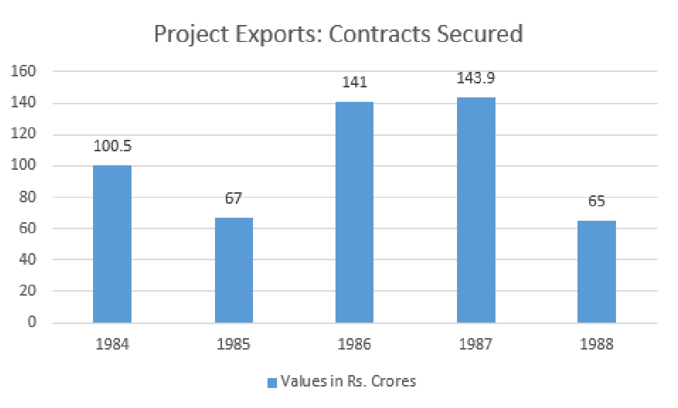
C 2002

D 2003

VIEW SOLUTION

Instructions

Refer to the following Bar-chart and answer the questions that follow :



Question 114

What is the average value of the contract secured during the years shown in the diagram?

A Rs. 103.48 crore

B Rs. 105 crore

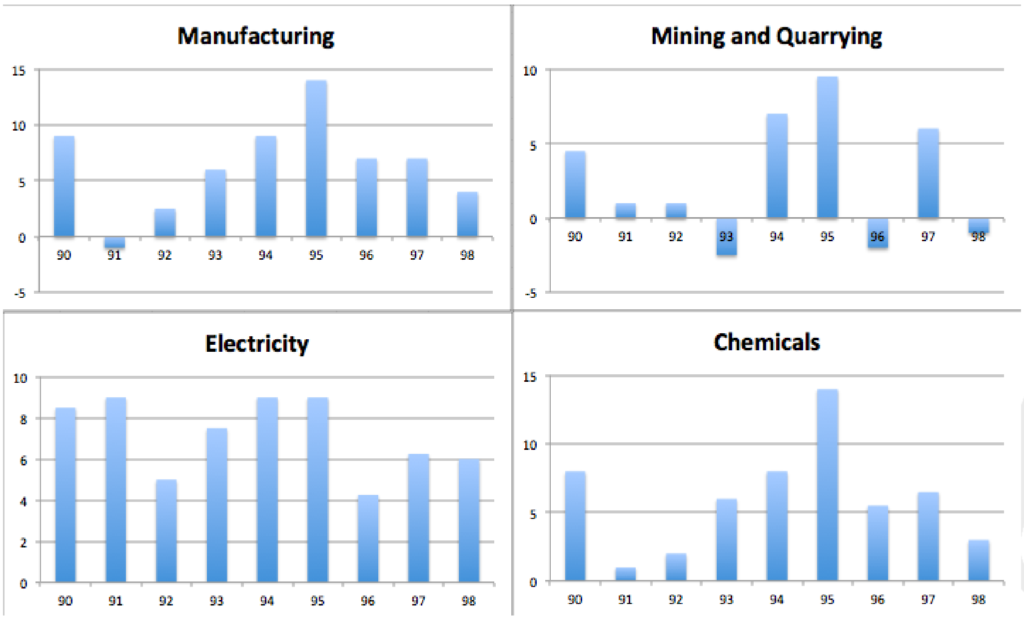
C Rs. 100 crore

D Rs.125.2 crore

VIEW SOLUTION

Instructions

Answer these questions based on the data given below: The figures below present annual growth rate, expressed as the % change relative to the previous year, in four sectors of the economy of the Republic of Reposia during the 9 year period from 1990 to 1998. Assume that the index of production for each of the four sectors is set at 100 in 1989 Further, the four sectors: manufacturing, mining and quarrying, electricity, and chemicals, respectively, constituted 20%, 15%, 10%, 15 % of total industrial production 1989.



Question 115

When was the lowest level of production of the mining and quarrying sector achieved during the nine year period 1990-1998?

A 1996

B 1993

C 1990

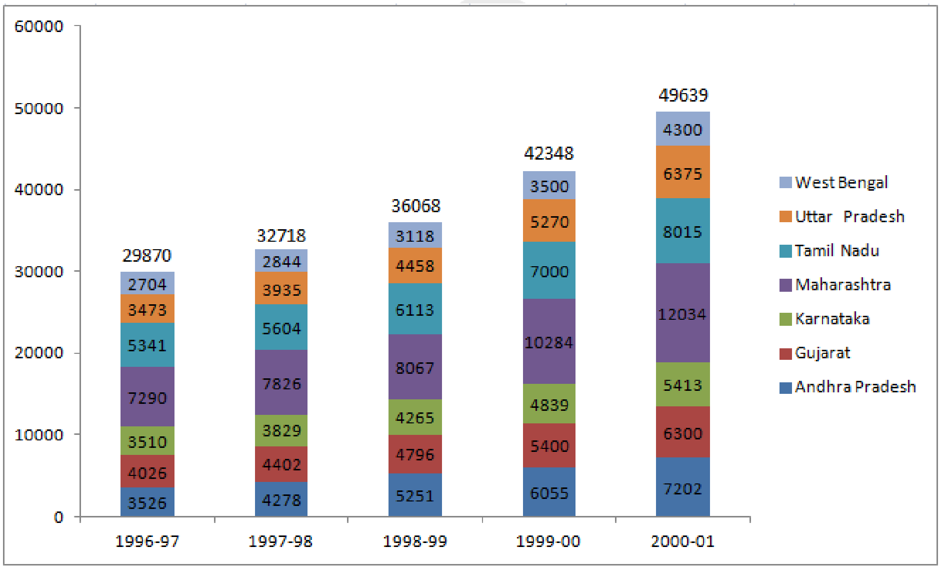
D Cannot be determined

VIEW SOLUTION

Instructions

The chart given below indicates the annual sales tax revenue collections (in rupees in crores) of seven states from 1997 to 2001.

The values given at the top of each bar represents the total collections in that year



Question 116

Which of the following states has changed its relative ranking most number of times when you rank the states in terms of the descending volume of sales tax collections each year?

A Andhra Pradesh

B Uttar Pradesh

C Karnataka

D Tamil Nadu

VIEW SOLUTION

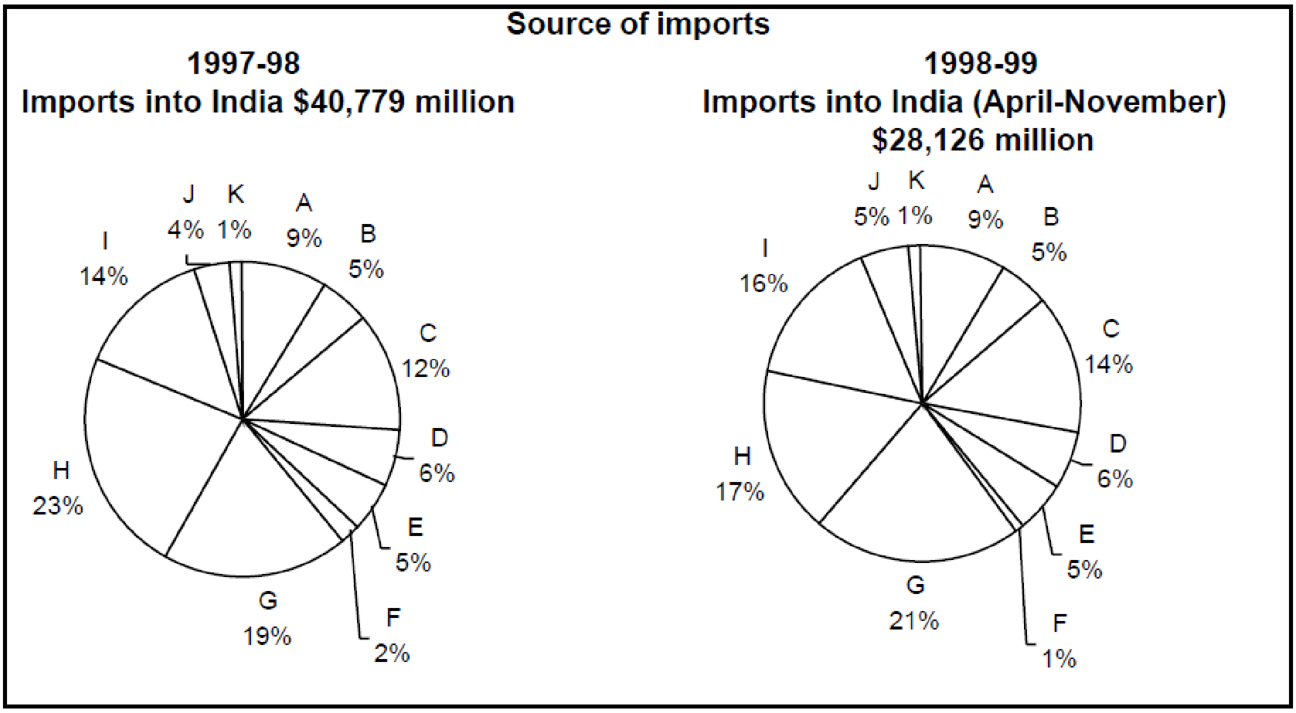
Instructions

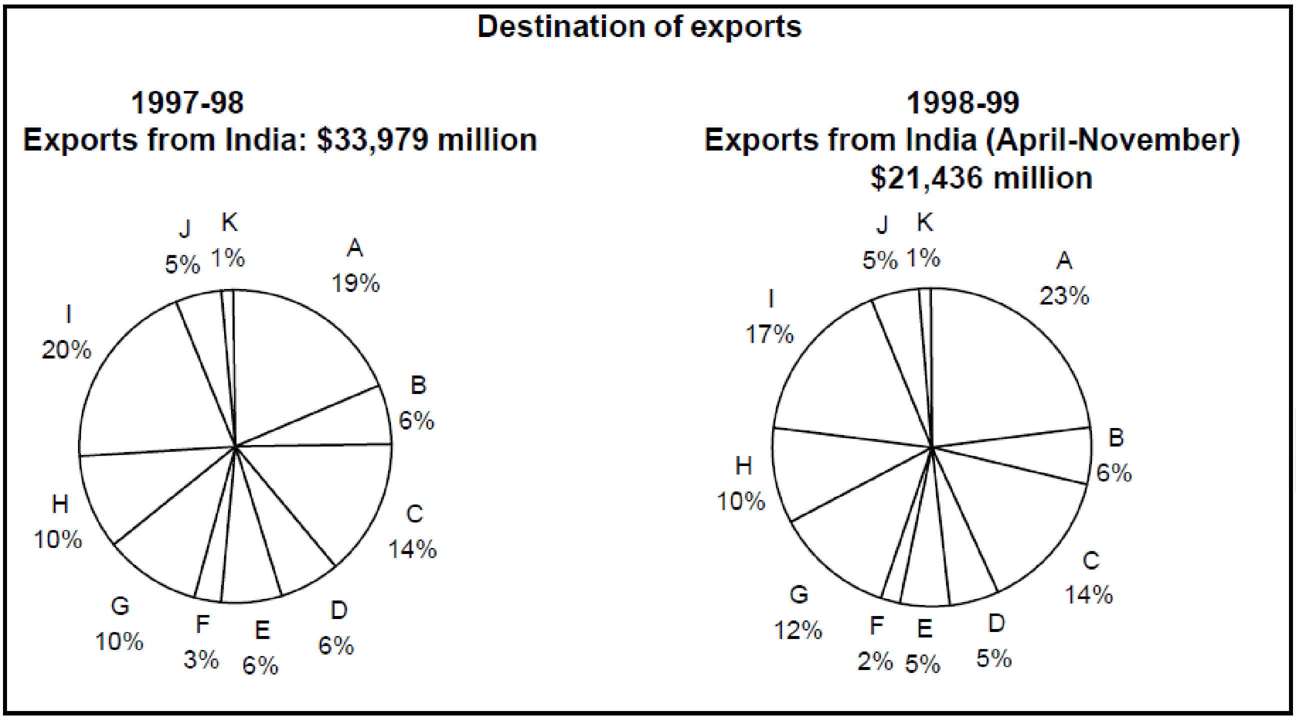
Consider the information provided in the figure below relating to India's foreign trade in 1997-98 and the first eight months of 1998-99.

Total trade with a region is defined as the sum of exports to and imports from that region.
Trade deficit is defined as the excess of imports over exports. Trade deficit may be negative.

A:USA. B:Germany C:Other EU. D:U.K. E:Japan F:Russia

G:Other East Europe H:OPEC I:Asia J:Other LDCs K:Others





Question 117

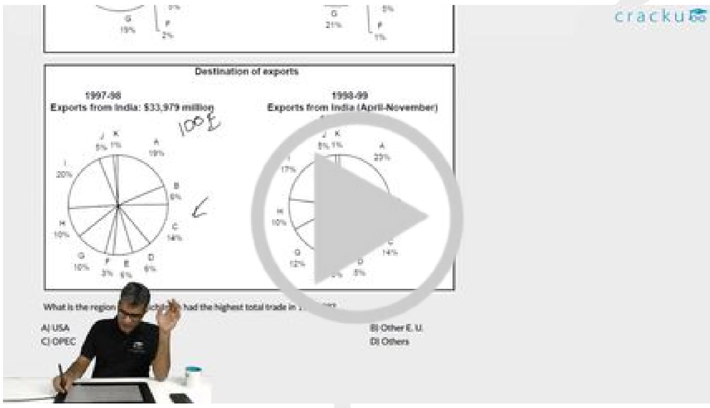
What is the region with which India had the highest total trade in 1997-98?

A USA

B Other E. U.

C OPEC

D Others

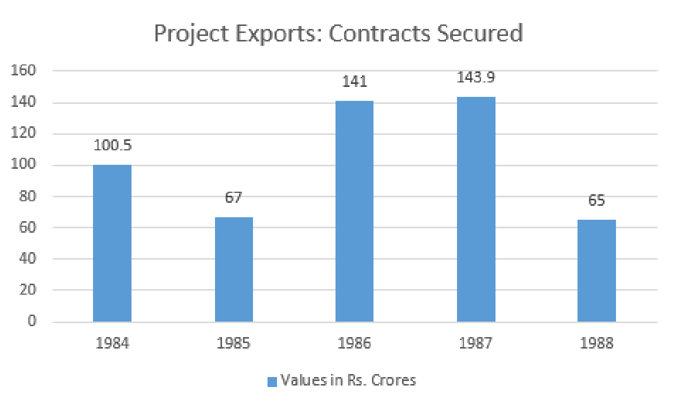


VIDEO SOLUTION

Data Interpretation for CAT Questions (download pdf)

Instructions

Refer to the following Bar-chart and answer the questions that follow :



Question 118

Which is the year in which the highest percentage decline is seen in the value of contract secured compared to the preceding year?

A 1985

B 1988

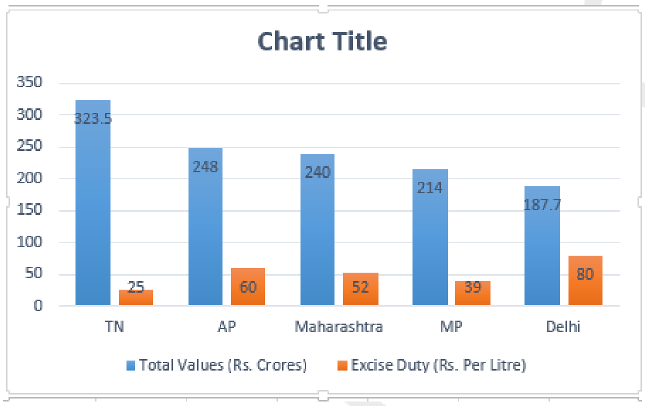
C 1984

D 1986

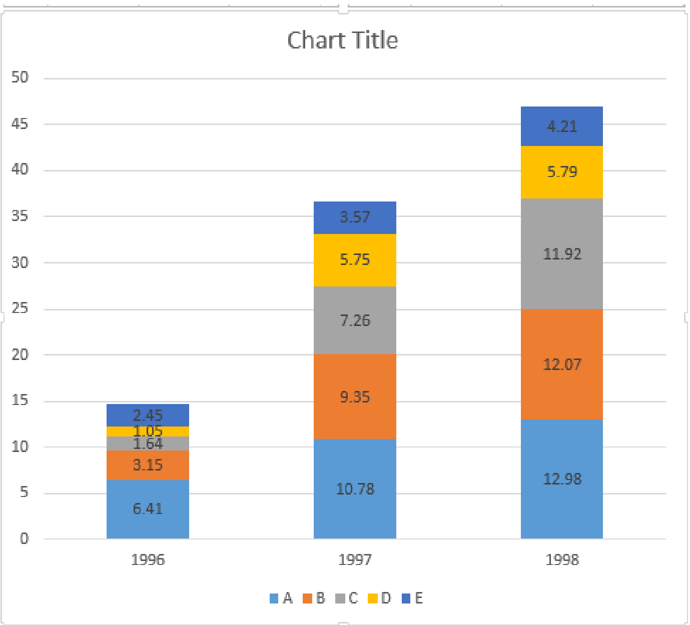
VIEW SOLUTION

Instructions

The following graph shows the value of liquor supplied by the 5 states in 1996 and the excise duty rates in each state.



Amount of liquor supplied in Tamil Nadu Distilleries A, B, C, D, E (from bottom to top) in lakh litres.



Question 119

If the Tamil Nadu distillery, with the least average simple annual growth in amount of liquor supplied in the given period had shown the same rate of growth as the one which grew fastest, what would that company's supply have been in 1998, in lakh liters?

A 13

B 15.11

C 130

D Cannot be determined.

VIEW SOLUTION

Question 120

Which of the five states manufactured liquor at the lowest cost?

A Tamil Nadu

B Delhi

C The states which has the lowest value for (wholesale price-Excise duty) per litre

D Cannot be determined.

VIEW SOLUTION

Question 121

What is the lowest percentage difference in the excise duty rates for any two states?

A 12

B 15

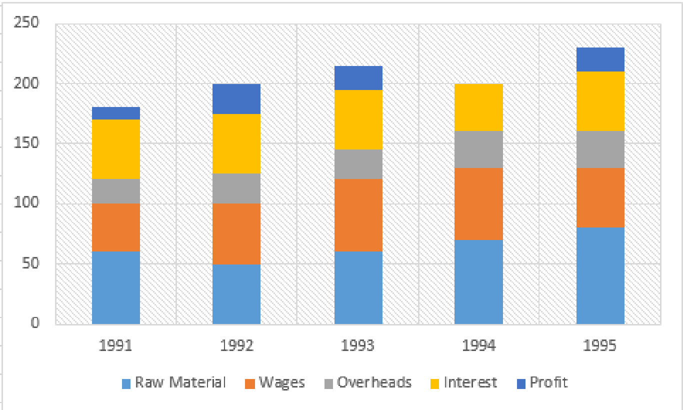
C 20

D 25

VIEW SOLUTION

Instructions

The graph given below gives the yearly details of money invested in producing a certain product over the years 1991 to 1995. It also gives the profit (in ‘000 rupees).



Question 122

If the interest component is not included in the total cost calculation, which year would show the maximum profit per unit cost?

A 1991

B 1992

C 1993

D 1995

VIEW SOLUTION

Question 123

What percentage of the costs did the profits form over the period?

A 3%

B 5%

C 8%

D 11%

VIEW SOLUTION

Question 124

In which year were the overheads, as a percentage of the raw material, maximum?

A 1995

B 1994

C 1992

D 1993

VIEW SOLUTION

Question 125

Which component of the cost production has remained more or less constant over the period?

A Interest

B Overheads

C Wages

D Raw material

VIEW SOLUTION

Question 126

In which period was the decrease in profit maximum (as a percentage)?

A 1991-92

B 1992-93

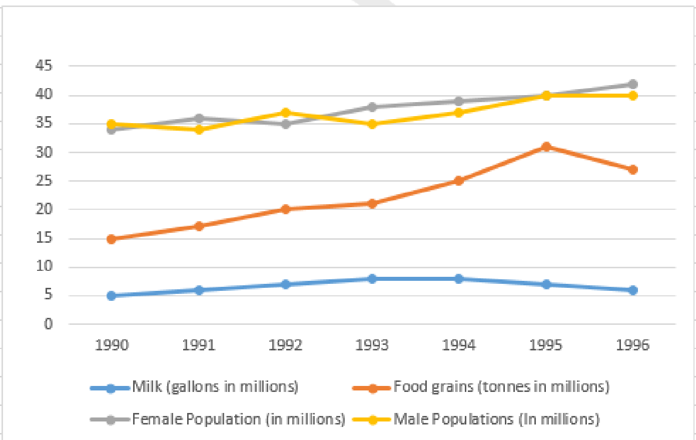
C 1993-94

D 1994-95

VIEW SOLUTION

Instructions

The graph given below shows the quantity of milk and food grains consumed annually along with female and male population (in millions). Use the data to answer the questions that follow.



Question 127

Referring to the previous question, in which year was the per capita consumption of this nutrient highest?

A 1993

B 1994

C 1995

D 1996

VIEW SOLUTION

Question 128

If one gallon milk contains 120 g of a particular nutrient and one tonne of foodgrains contains 80 g of the same nutrient, in which year was the availability of this nutrient maximum?

A 1993

B 1994

C 1995

D 1996

VIEW SOLUTION

Question 129

If milk contains 320 calories and foodgrains contain 160 calories, in which year was the per capita consumption of calories highest?

A 1993

B 1994

C 1995

D 1996

VIEW SOLUTION

Instructions

Answer the questions based on the following information. The first table gives the percentage of students in MBA class, who sought employment in the areas of finance, marketing and software. The second table gives the average starting salaries of the students per month, (rupees in thousands) in these areas. The third table gives the number of students who passed out in each year.

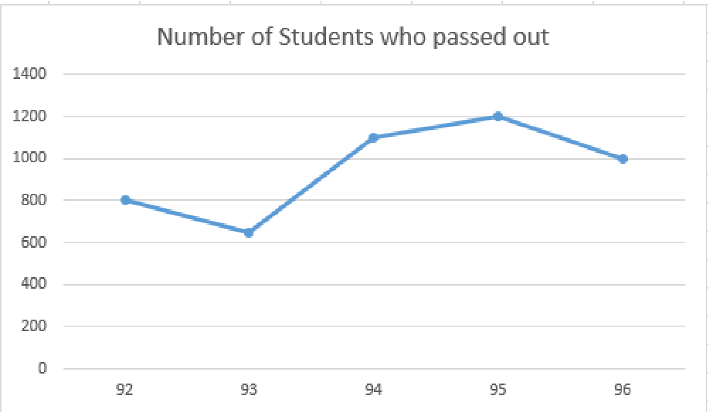
Table 1:

	Finance	Marketing	Software	Others
1992	22	36	19	23
1993	17	48	23	12
1994	23	43	21	13
1995	19	37	16	28
1996	32	32	20	16

Table 2:

	Finance	Marketing	Software
1992	5450	5170	5290
1993	6380	6390	6440
1994	7550	7630	7050
1995	8920	8960	7760
1996	9810	10220	8640

Table 3:



Question 130

In 1994, students seeking jobs in finance earned ___ more than those opting for software (per annum).

A Rs. 43 lakh

B Rs. 33.8 lakh

C Rs. 28.4 lakh

D Rs. 38.8 lakh

VIEW SOLUTION

Question 131

What is the average monthly salary offered to a management graduate in 1993?

A Rs. 6,403

B Rs. 6,330

C Rs. 6,333

D Cannot be determined

VIEW SOLUTION

Question 132

What is the percentage increase in the average salary of finance from 1992 to 1996?

A 60%

B 32%

C 96%

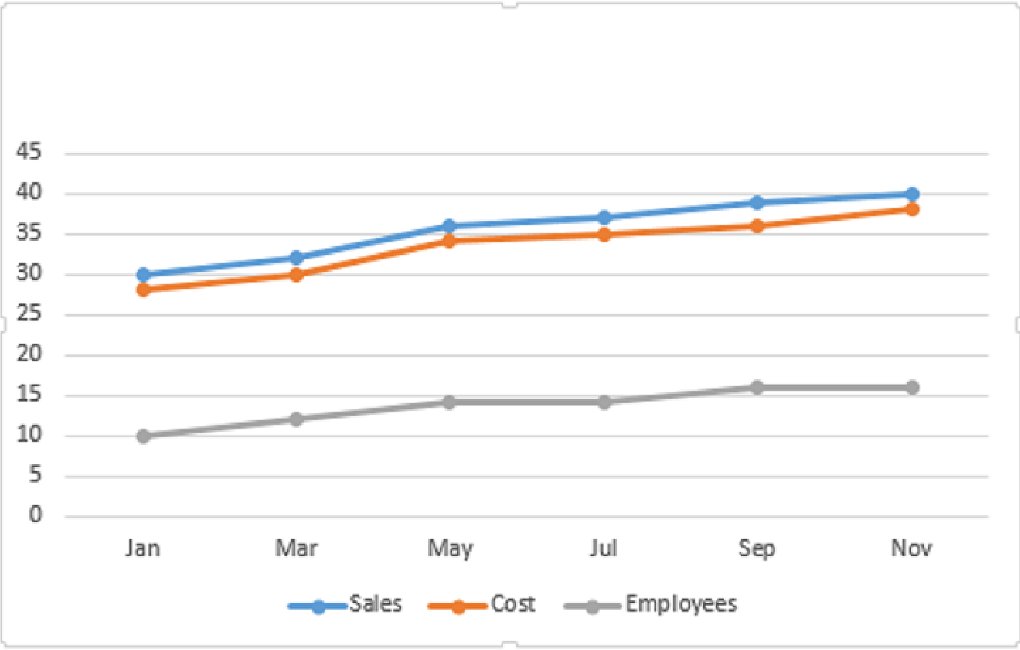
D 80%

VIEW SOLUTION

Enroll for Excellent CAT/MBA courses

Instructions

Answer the questions based on the following graph.



Employess in thousand Sales - cost = profit
(all values are integers)

Question 133

Assuming that no employees left the job, how many more people did the company take on in the given period?

A 4,600

B 5,000

C 5,800

D 6,400

VIEW SOLUTION

Question 134

In which month is the percentage increase in sales two months before, the highest?

A March

B September

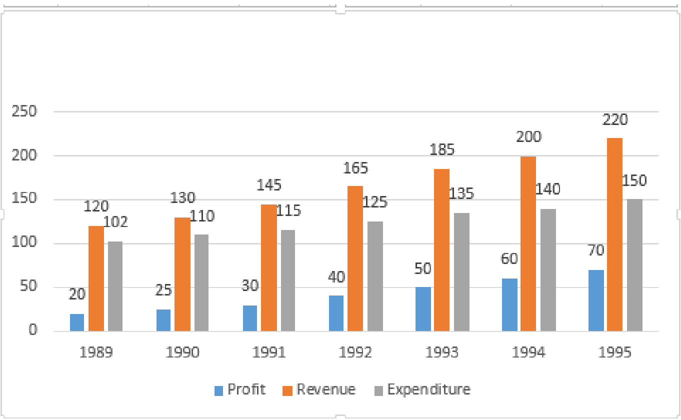
C July

D May

VIEW SOLUTION

Instructions

Answer the questions based on the following information.



Question 135

In which year was the growth in expenditure maximum as compared to the previous year?

A 1993

B 1995

C 1991

D 1992

VIEW SOLUTION

Cracku CAT Success Stories

Question 136

Which year showed the greatest percentage increase in profit as compared to the previous year?

A 1993

B 1994

C 1990

D 1992

VIEW SOLUTION

Question 137

The expenditure for the 7 years together form what per cent of the revenues during the same period?

A 75%

B 67%

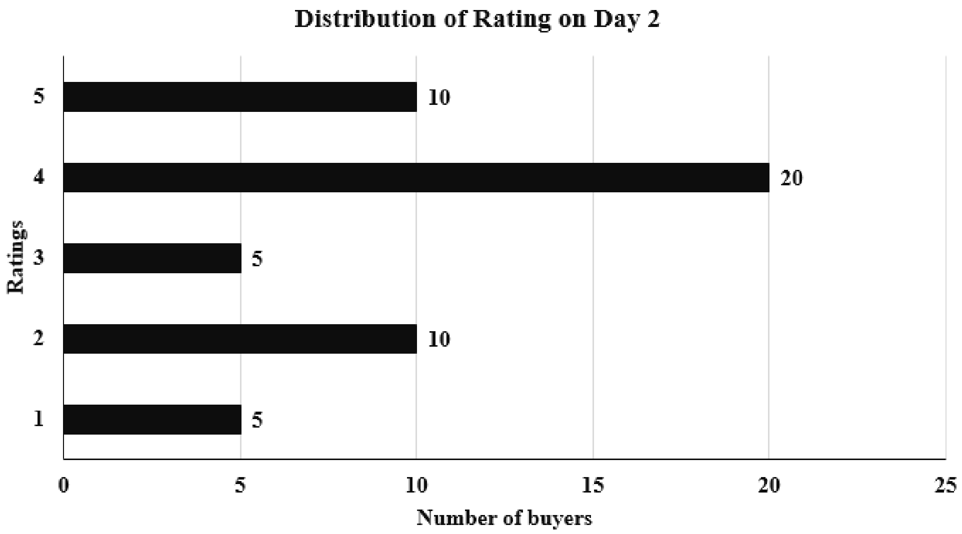
C 62%

D 83%

VIEW SOLUTION

Instructions

An online e-commerce firm receives daily integer product ratings from 1 through 5 given by buyers. The daily average is the average of the ratings given on that day. The cumulative average is the average of all ratings given on or before that day. The rating system began on Day 1, and the cumulative averages were 3 and 3.1 at the end of Day 1 and Day 2, respectively. The distribution of ratings on Day 2 is given in the figure below.



The following information is known about ratings on Day 3.
1. 100 buyers gave product ratings on Day 3.
2. The modes of the product ratings were 4 and 5.
3. The numbers of buyers giving each product rating are non-zero multiples of 10.
4. The same number of buyers gave product ratings of 1 and 2, and that number is half the number of buyers who gave a rating of 3.

Question 138

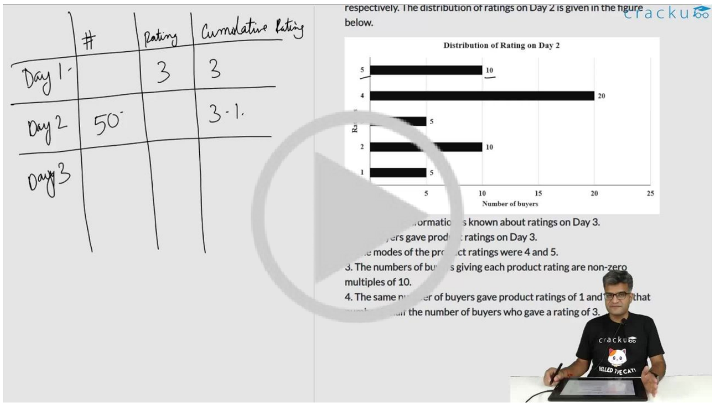
Which of the following is true about the cumulative average ratings of Day 2 and Day 3?

A The cumulative average of Day 3 increased by less than 5% from Day 2.

B The cumulative average of Day 3 decreased from Day 2.

C The cumulative average of Day 3 increased by a percentage between 5% and 8% from Day 2.

D The cumulative average of Day 3 increased by more than 8% from Day 2.

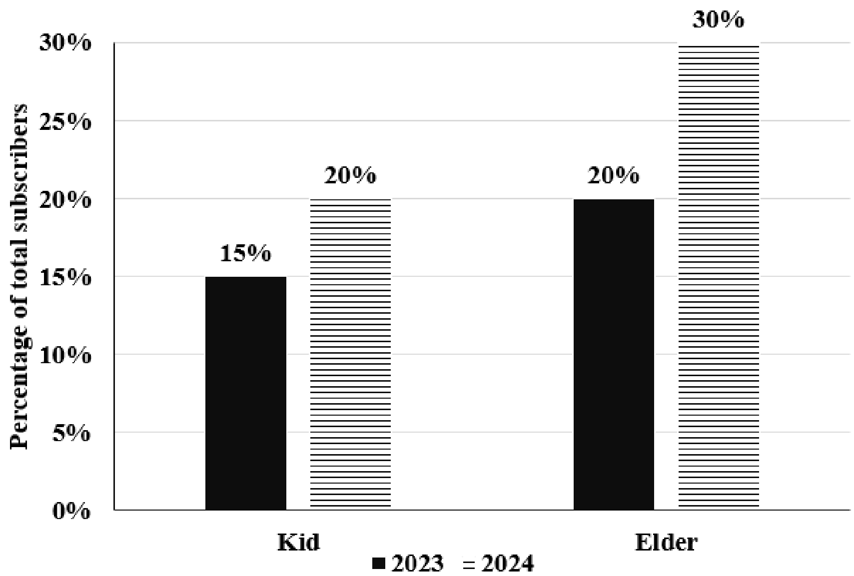


VIDEO SOLUTION

Subscribe to MBA Exams Youtube Channel

Instructions

Comprehension:
Over the top (OTT) subscribers of a platform are segregated into three categories: i) Kid, ii)Elder, and iii) Others. Some of the subscribers used one app and the others used multiple apps to access the platform. The figure below shows the percentage of the total number of subscribers in 2023 and 2024 who belong to the 'Kid' and 'Elder' categories.



The following additional facts are known about the numbers of subscribers.
1. The total number of subscribers increased by 10% from 2023 to 2024.
2. In 2024, 1/2 of the subscribers from the 'Kid' category and 2/3 of the subscribers from the 'Elder' category subscribers use one app.
3. In 2023, the number of subscribers from the 'Kid' category who used multiple apps was the same as the number of subscribers from the 'Elder' category who used one app.
4. 10,000 subscribers from the 'Kid' category used one app and 15,000 subscribers from the 'Elder' category used multiple apps in 2023.

Question 139

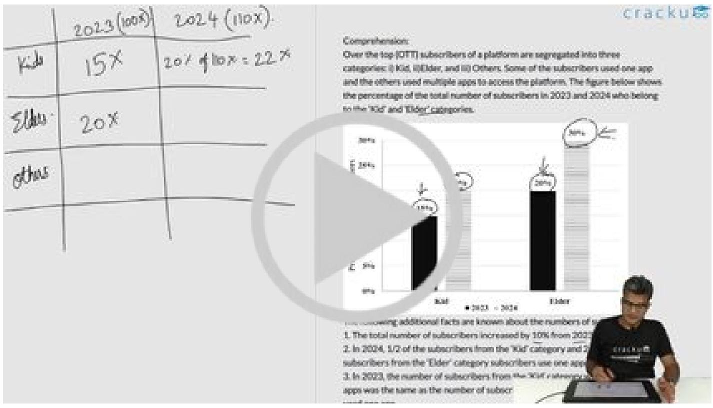
How many subscribers belonged to the 'Others' category in 2024?

A Cannot be determined

B 65000

C 55000

D 45000



VIDEO SOLUTION

Question 140

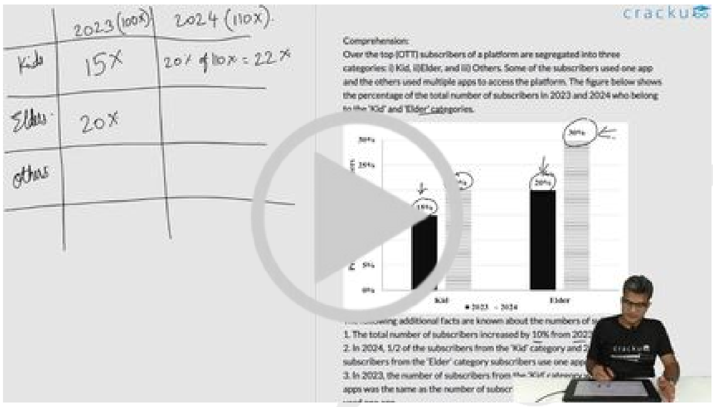
What percentage of subscribers in the ‘Kid’ category used multiple apps in 2023?

A 33.33%

B 5.00%

C 25.50%

D 50.00%



VIDEO SOLUTION

Question 141

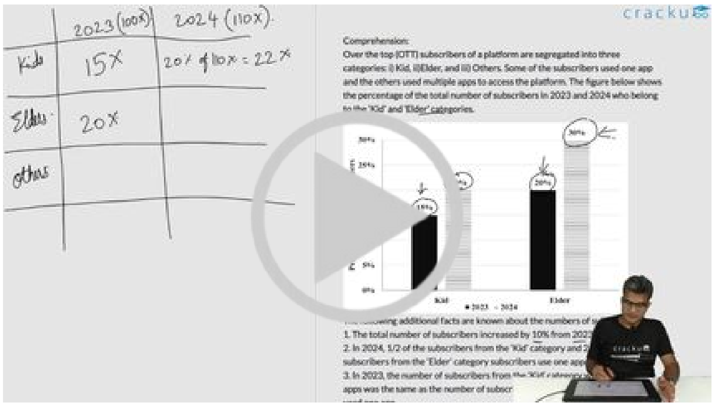
What was the percentage increase in the number of subscribers in the ‘Elder’ categoryfrom 2023 to 2024?

A 60%

B 50%

C 65%

D 40%



VIDEO SOLUTION

Question 142

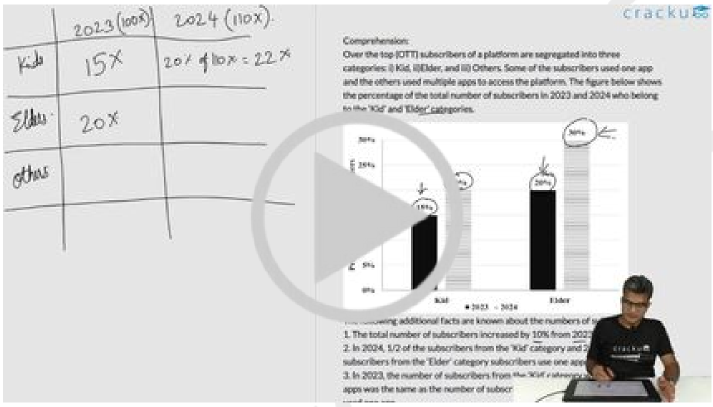
What could be the minimum percentage of subscribers who used multiple apps in 2024?

A 16.5%

B 20.0%

C 10.0%

D 22.00%

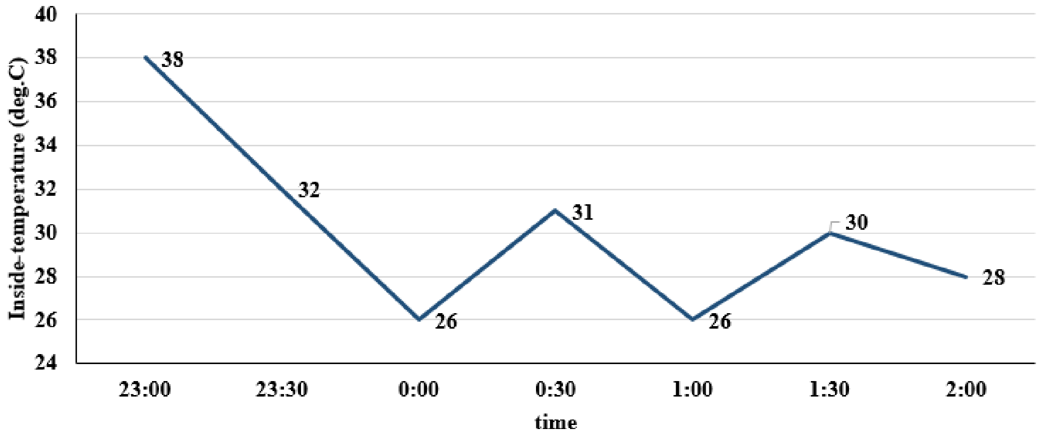


VIDEO SOLUTION

Instructions

The air-conditioner (AC) in a large room can be operated either in REGULAR mode or in POWER mode to reduce the temperature.
If the AC operates in REGULAR mode, then it brings down the temperature inside the room(called inside temperature) at a constant rate to the set temperature in 1 hour. If it operates in POWER mode, then this is achieved in 30 minutes.
If the AC is switched off, then the inside temperature rises at a constant rate so as to reach the temperature

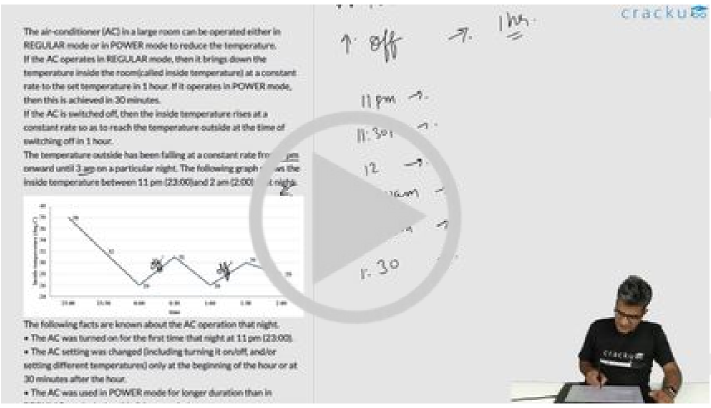
outside at the time of switching off in 1 hour.
The temperature outside has been falling at a constant rate from 7 pm onward until 3 am on a particular night.
The following graph shows the inside temperature between 11 pm (23:00)and 2 am (2:00) that night.



The following facts are known about the AC operation that night.
• The AC was turned on for the first time that night at 11 pm (23:00).
• The AC setting was changed (including turning it on/off, and/or setting different temperatures) only at the beginning of the hour or at 30 minutes after the hour.
• The AC was used in POWER mode for longer duration than in REGULAR mode during this 3-hour period.

Question 143

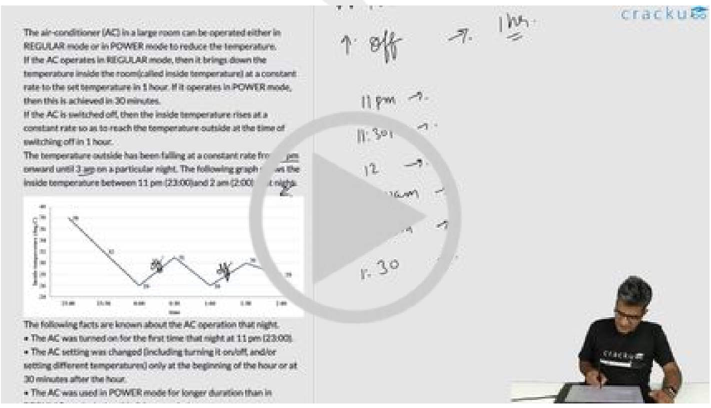
What was the temperature outside, in degree Celsius, at 1 am?



VIDEO SOLUTION

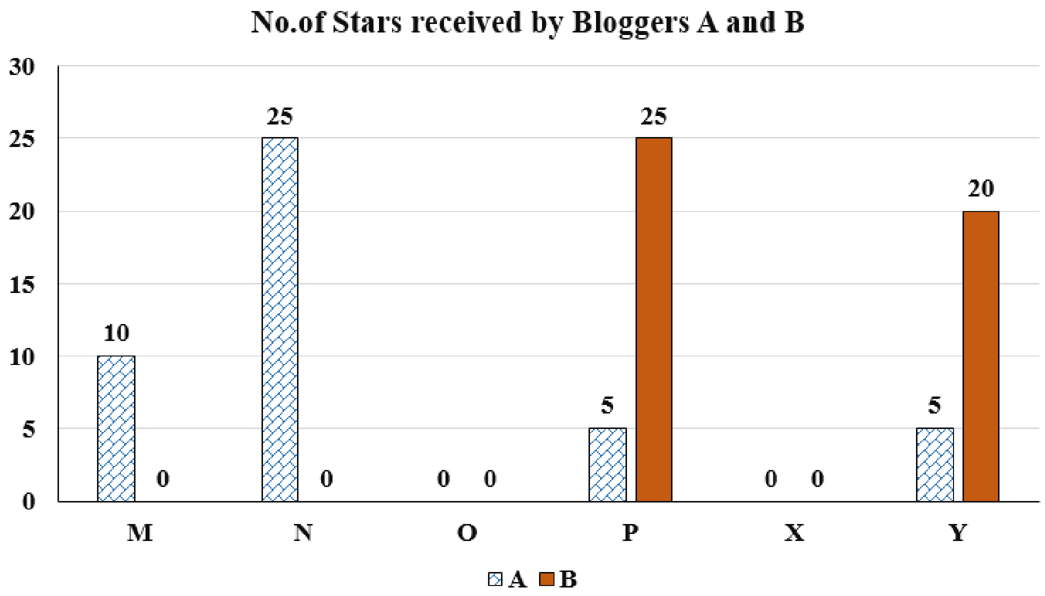
Question 144

What was the temperature outside, in degree Celsius, at 9 pm?



Instructions

Six web surfers M, N, O, P, X, and Y each had 30 stars which they distributed among four bloggers A, B, C, and D. The number of stars received by A and B from the six web surfers is shown in the figure below.



The following additional facts are known regarding the number of stars received by the bloggers from the surfers.

1. The numbers of stars received by the bloggers from the surfers were all multiples of 5 (including 0).
2. The total numbers of stars received by the bloggers were the same.
3. Each blogger received a different number of stars from M.
4. Two surfers gave all their stars to a single blogger.
5. D received more stars than C from Y.

Question 145

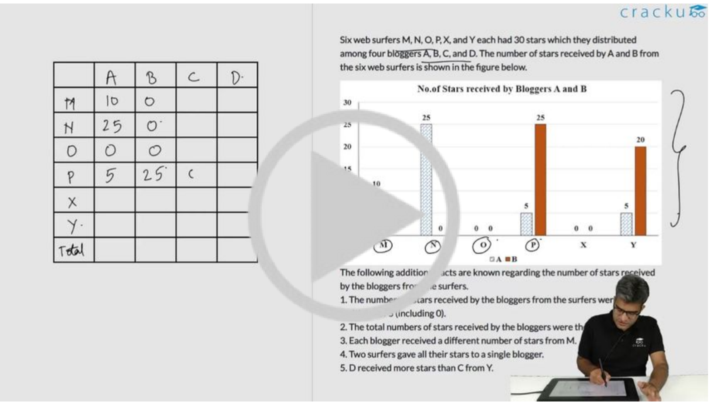
What was the number of stars received by D from Y?

A 5

B 10

C Cant be determined

D 0



VIDEO SOLUTION

Question 146

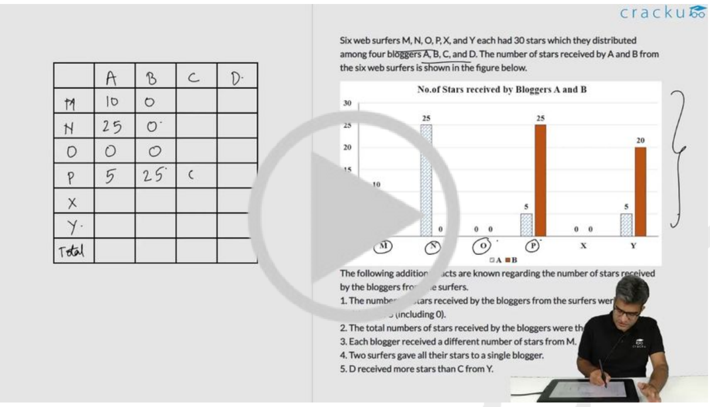
Which of the following can be determined with certainty?
I. The number of stars received by C from M
II. The number of stars received by D from O

A Neither I or II

B Only I

C Only II

D Both I and II



VIDEO SOLUTION

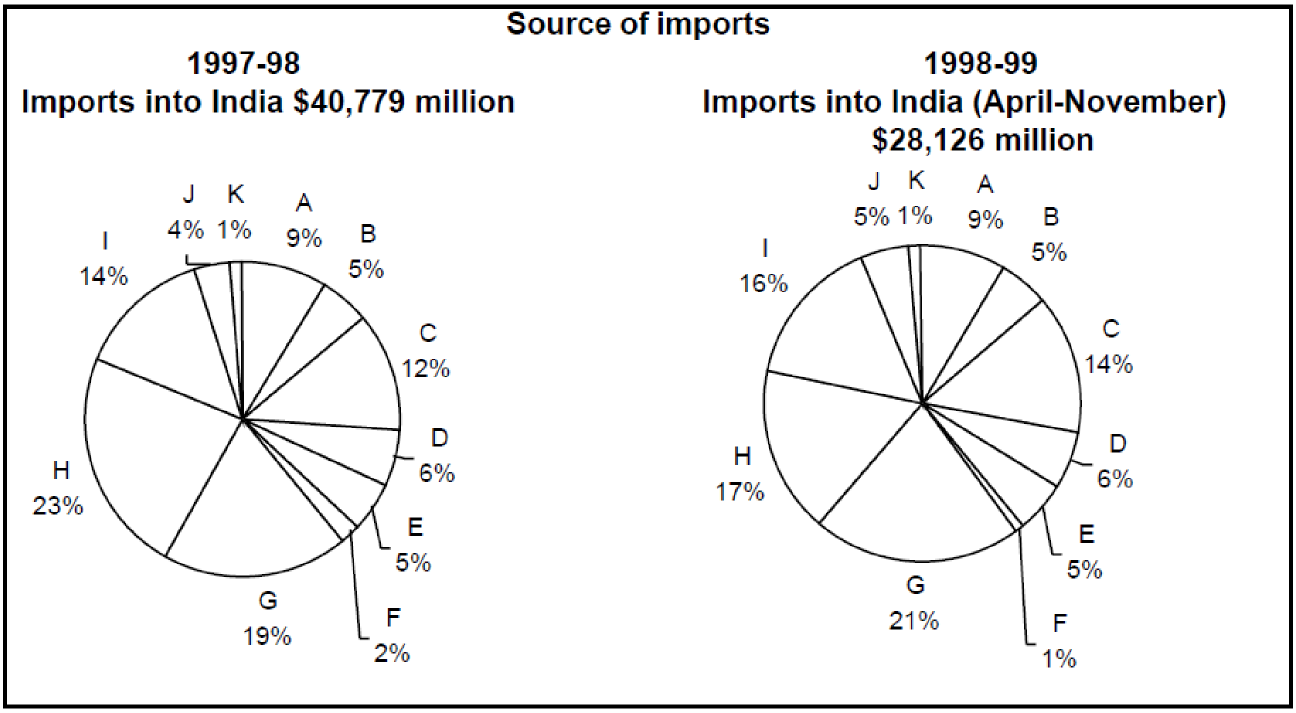
Instructions

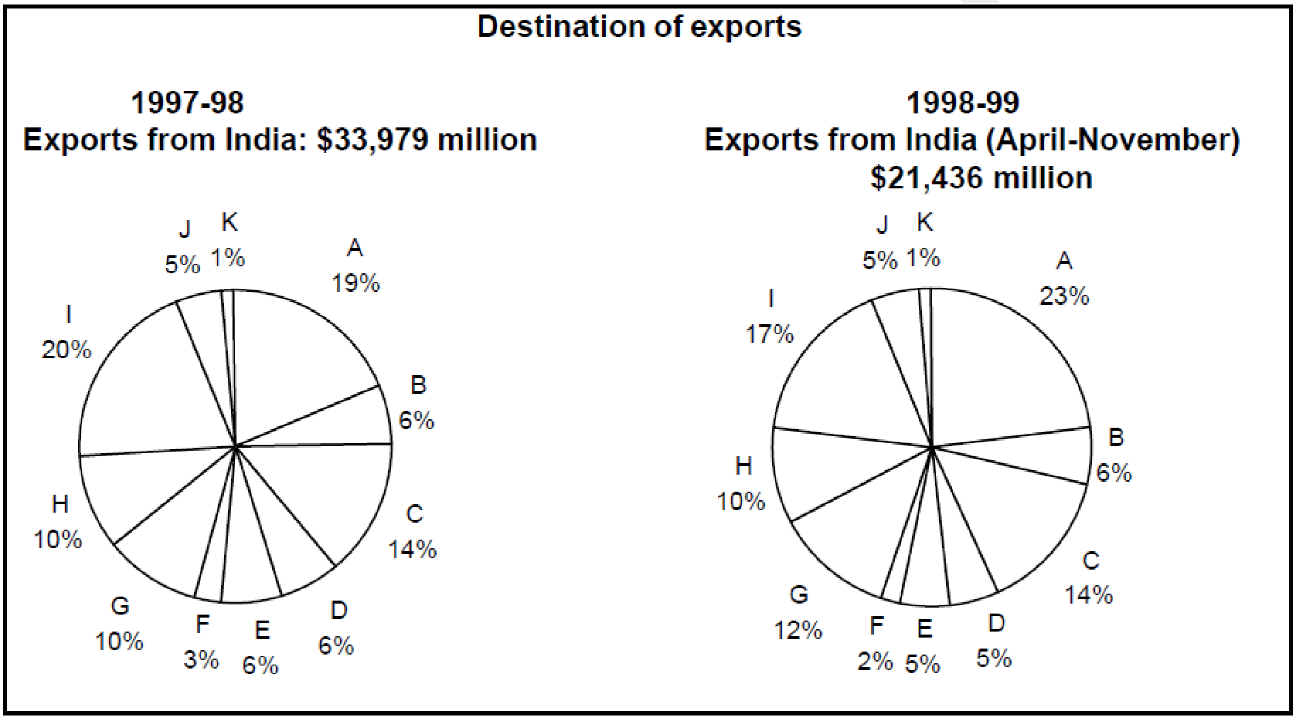
Consider the information provided in the figure below relating to India's foreign trade in 1997-98 and the first eight months of 1998-99.

Total trade with a region is defined as the sum of exports to and imports from that region.
Trade deficit is defined as the excess of imports over exports. Trade deficit may be negative.

A:USA. B:Germany C:Other EU. D:U.K. E:Japan F:Russia

G:Other East Europe H:OPEC I:Asia J:Other LDCs K:Others





Question 147

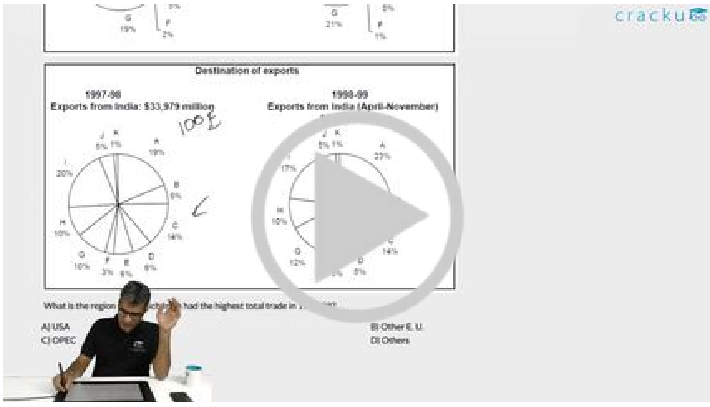
Assume that the average monthly exports from India and imports to India during the remaining four months of 1998-99 would be the same as that for the first eight months of the year. What is the region to which India’s exports registered the highest percentage growth between1997-98 and 1998-99?

A Other East European Countries

B USA

C Asia

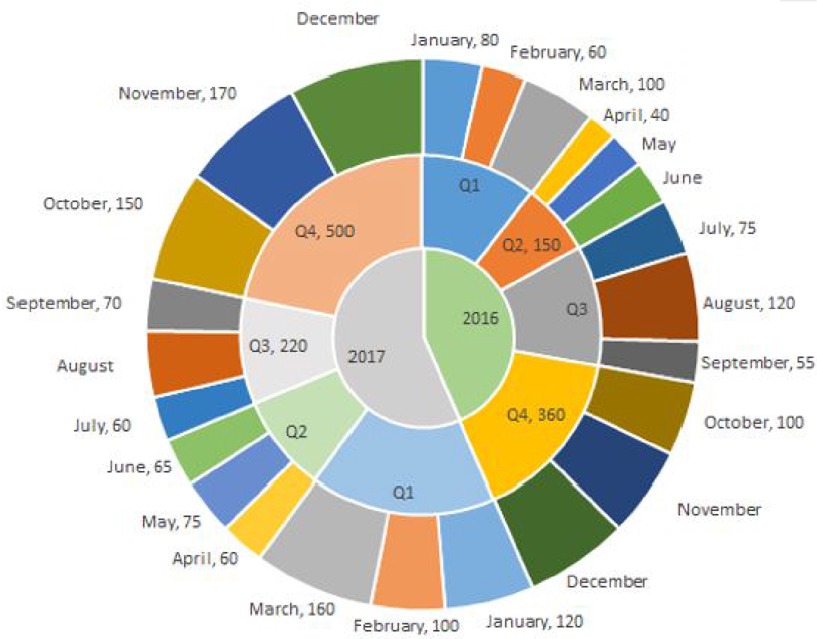
D Exports have declined, no growth



VIDEO SOLUTION

Instructions

The multi-layered pie-chart below shows the sales of LED television sets for a big retail electronics outlet during 2016 and 2017. The outer layer shows the monthly sales during this period, with each label showing the month followed by sales figure of that month. For some months, the sales figures are not given in the chart. The middle-layer shows quarterwise aggregate sales figures (in some cases, aggregate quarter-wise sales numbers are not given next to the quarter). The innermost layer shows annual sales. It is known that the sales figures during the three months of the second quarter (April, May, June) of 2016 form an arithmetic progression, as do the three monthly sales figures in the fourth quarter (October, November, December) of that year.



Question 148

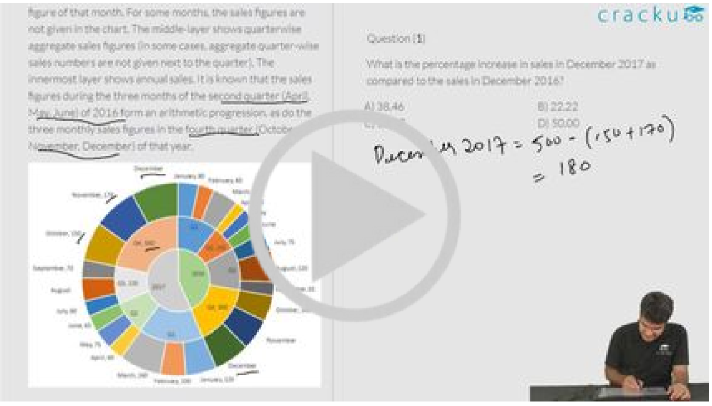
In which quarter of 2017 was the percentage increase in sales from the same quarter of 2016 the highest?

A Q2

B Q1

C Q4

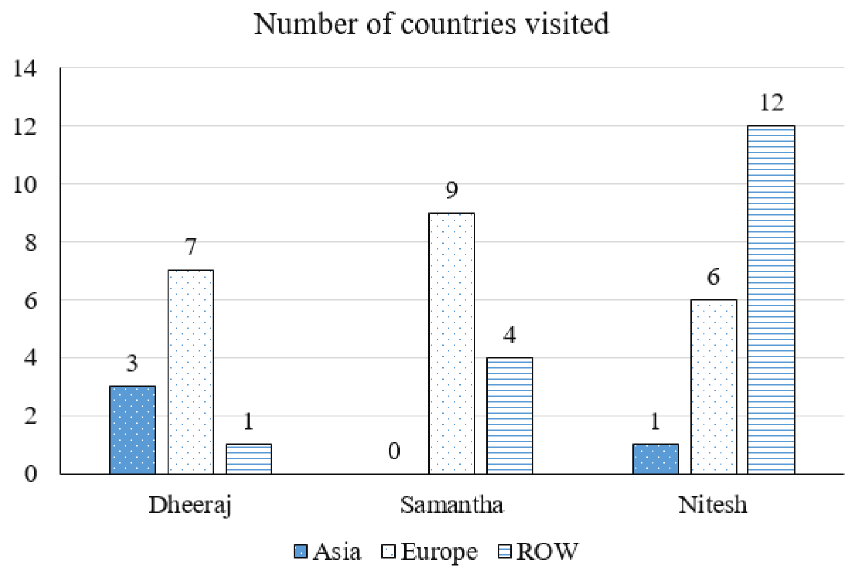
D Q3



VIDEO SOLUTION

Instructions

The chart below provides complete information about the number of countries visited by Dheeraj, Samantha and Nitesh, in Asia, Europe and the rest of the world (ROW).



The following additional facts are known about the countries visited by them.
1. 32 countries were visited by at least one of them.
2. USA (in ROW) is the only country that was visited by all three of them.
3. China (in Asia) is the only country that was visited by both Dheeraj and Nitesh, but not by Samantha.
4. France (in Europe) is the only country outside Asia, which was visited by both Dheeraj and Samantha, but not by Nitesh.
5. Half of the countries visited by both Samantha and Nitesh are in Europe.

Question 149

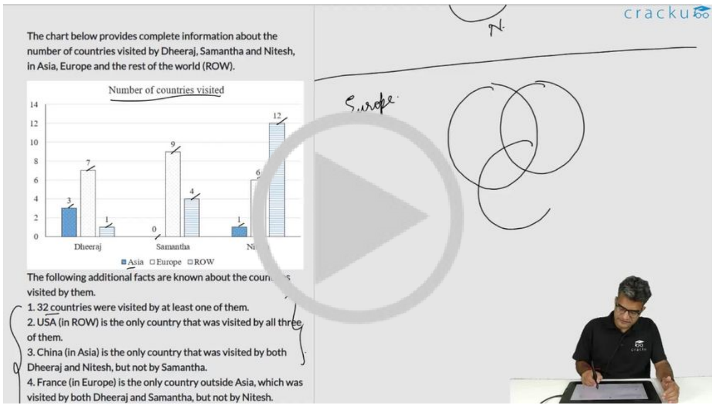
How many countries in Europe were visited by exactly one of Dheeraj, Samantha and Nitesh?

A 10

B 5

C 14

D 12

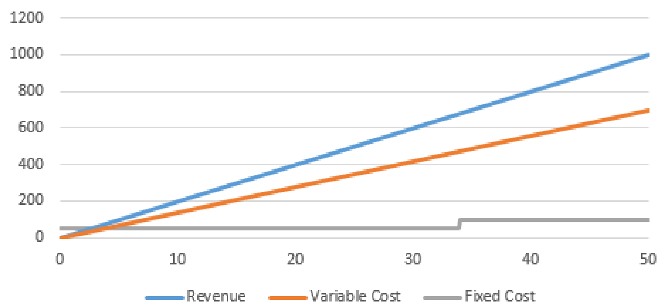


VIDEO SOLUTION

Instructions

Answer the questions based on the following information.

Ghosh Babu has a manufacturing unit. The following graph gives the cost for the various number of units. Given: Profit = Revenue - Variable cost - Fixed cost. The fixed cost remains constant up to 34 units after which additional investment is to be done in fixed assets. In any case, production cannot exceed 50 units.



Note: The fixed cost for upto 34 units is 50 and the the fixed cost for more is 100.
The revenue from 50 units is 1000 and the variable cost from 50 units is 700

Question 150

What is the least number of units that should be manufactured such that the profit was at least Rs. 50?

A 17

B 34

C 45

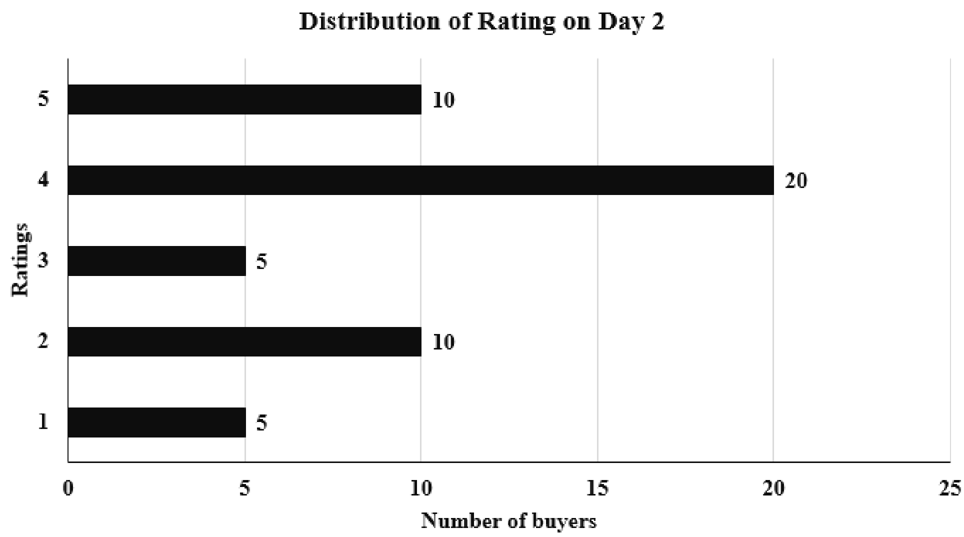
D 30

VIDEO SOLUTION

CAT Previous Papers PDF

Instructions

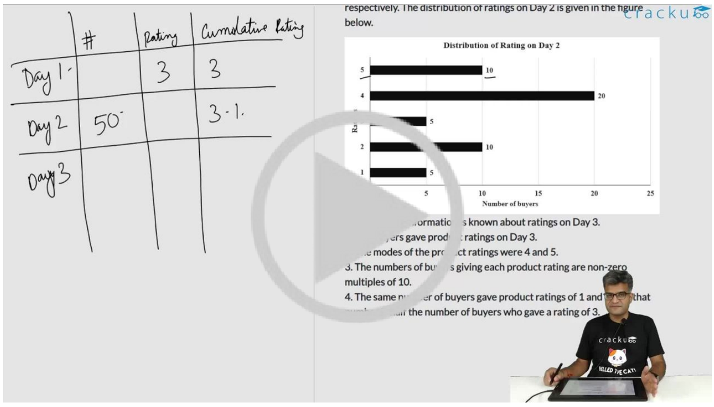
An online e-commerce firm receives daily integer product ratings from 1 through 5 given by buyers. The daily average is the average of the ratings given on that day. The cumulative average is the average of all ratings given on or before that day. The rating system began on Day 1, and the cumulative averages were 3 and 3.1 at the end of Day 1 and Day 2, respectively. The distribution of ratings on Day 2 is given in the figure below.



The following information is known about ratings on Day 3.
1. 100 buyers gave product ratings on Day 3.
2. The modes of the product ratings were 4 and 5.
3. The numbers of buyers giving each product rating are non-zero multiples of 10.
4. The same number of buyers gave product ratings of 1 and 2, and that number is half the number of buyers who gave a rating of 3.

Question 151

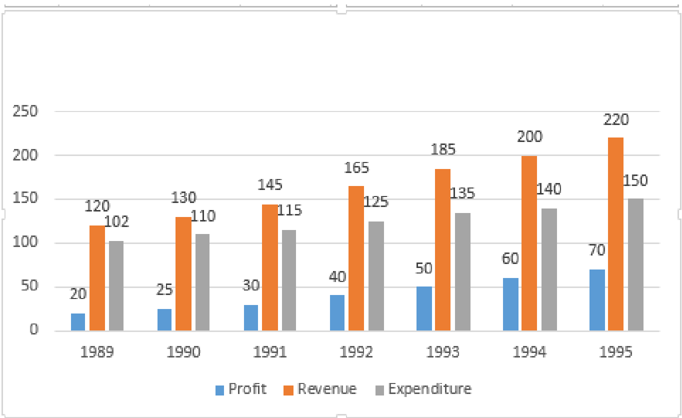
How many buyers gave ratings on Day 1?



VIDEO SOLUTION

Instructions

Answer the questions based on the following information.



Question 152

The average revenue collected in the given 7 years is approximately

A Rs. 164 lakh

B Rs. 166 lakh

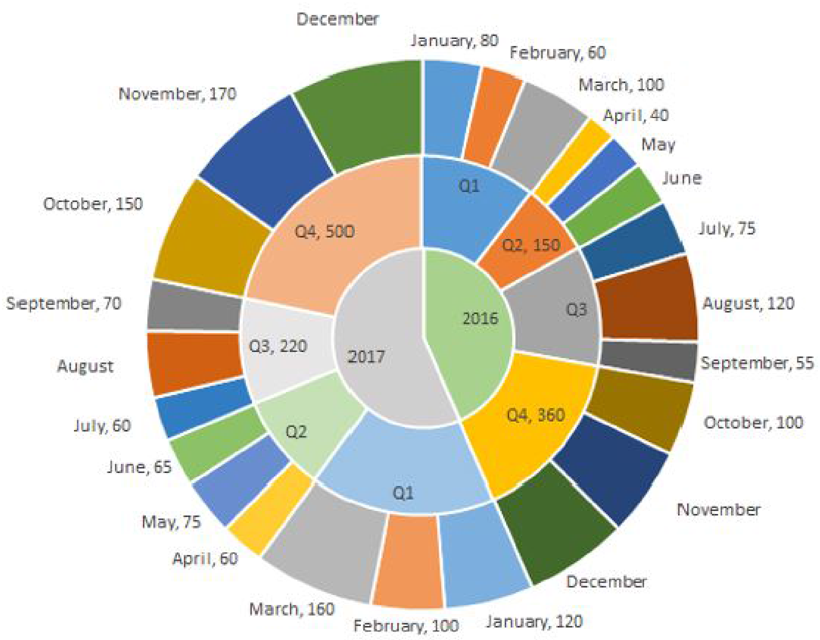
C Rs. 171 lakh

D Rs. 175 lakh

VIEW SOLUTION

Instructions

The multi-layered pie-chart below shows the sales of LED television sets for a big retail electronics outlet during 2016 and 2017. The outer layer shows the monthly sales during this period, with each label showing the month followed by sales figure of that month. For some months, the sales figures are not given in the chart. The middle-layer shows quarterwise aggregate sales figures (in some cases, aggregate quarter-wise sales numbers are not given next to the quarter). The innermost layer shows annual sales. It is known that the sales figures during the three months of the second quarter (April, May, June) of 2016 form an arithmetic progression, as do the three monthly sales figures in the fourth quarter (October, November, December) of that year.



Question 153

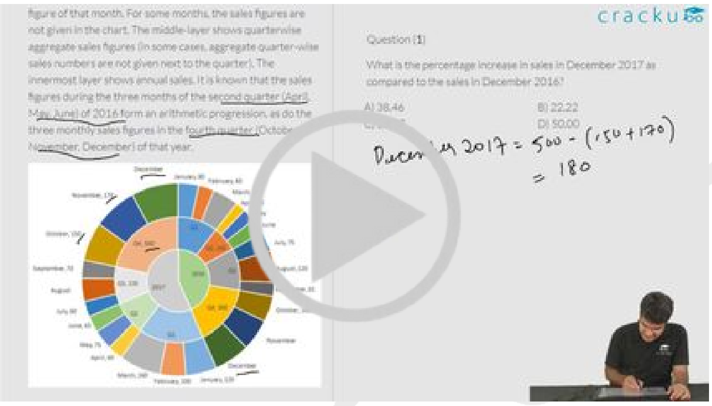
What is the percentage increase in sales in December 2017 as compared to the sales in December 2016?

A 38.46

B 22.22

C 28.57

D 50.00



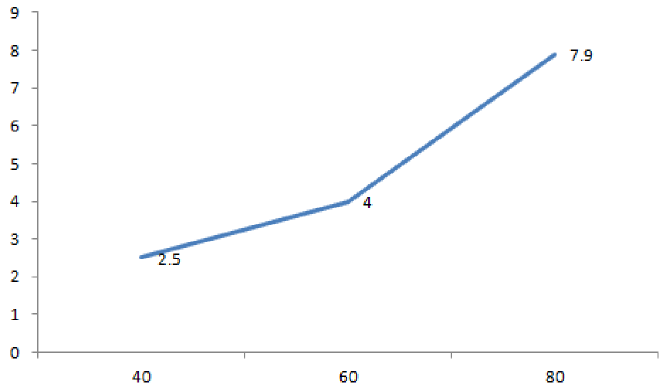
VIDEO SOLUTION

Instructions

Directions for the following two questions: Answer the questions based on the following information.

The petrol consumption rate of a new model car 'Palto' depends on its speed and may be described by the graph below.

The axis represents the speed and the Y axis represents the Fuel Consumption (Liters per hour)



Question 154

Manasa makes a 200 km trip from Mumbai to Pune at a steady speed of 60 km/hr. What is the volume of petrol consumed for the journey?

[CAT 2001]

A 12.5 L

B 13.33 L

C 16 L

D 19.75 L

VIEW SOLUTION

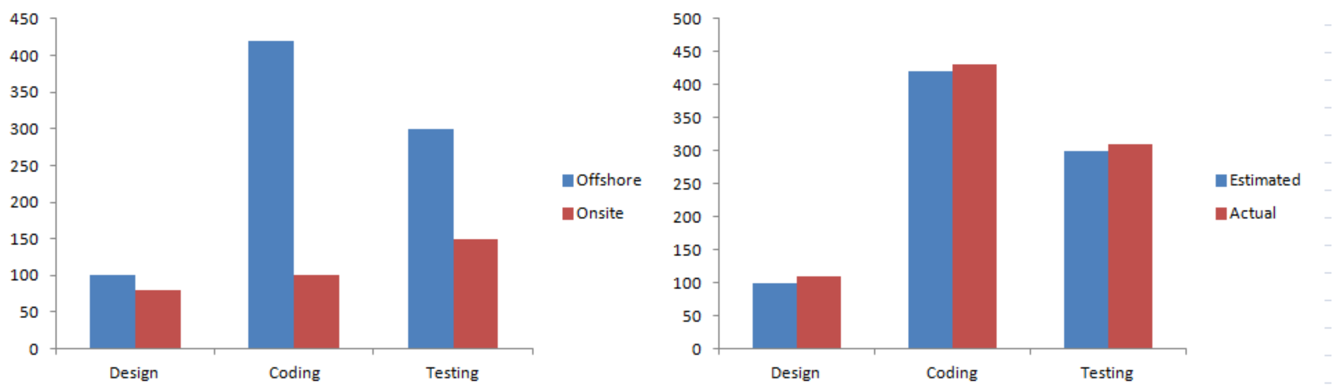
Instructions

Directions for the following five questions: Answer the questions based on the two graphs shown below.

Figure I (the chart on the left) shows the amount of work distribution, in man-hours, for a software company between offshore and onsite activities.

Figure 2 (the chart on the right) shows the estimated and actual work effort involved in the different offshore activities in the same company during the same period.

[Note: Onsite refers to work performed at the customer's premise and offshore refers to work performed at the developer's premise.]



Question 155

If the total working hours were 100, which of the following tasks will account for approximately 50 hr?

A Coding

B Design

C Offshore testing

D Offshore testing plus design

VIEW SOLUTION

Question 156

If 50% of the offshore work were to be carried out onsite, with the distribution of effort between the tasks remaining the same, which of the following is true of all work carried out onsite?

A The amount of coding done is greater than that of testing

B The amount of coding done onsite is less than that of design done onsite

C The amount of design carried out onsite is greater than that of testing

D The amount of testing carried out offshore is greater than that of total design

VIEW SOLUTION

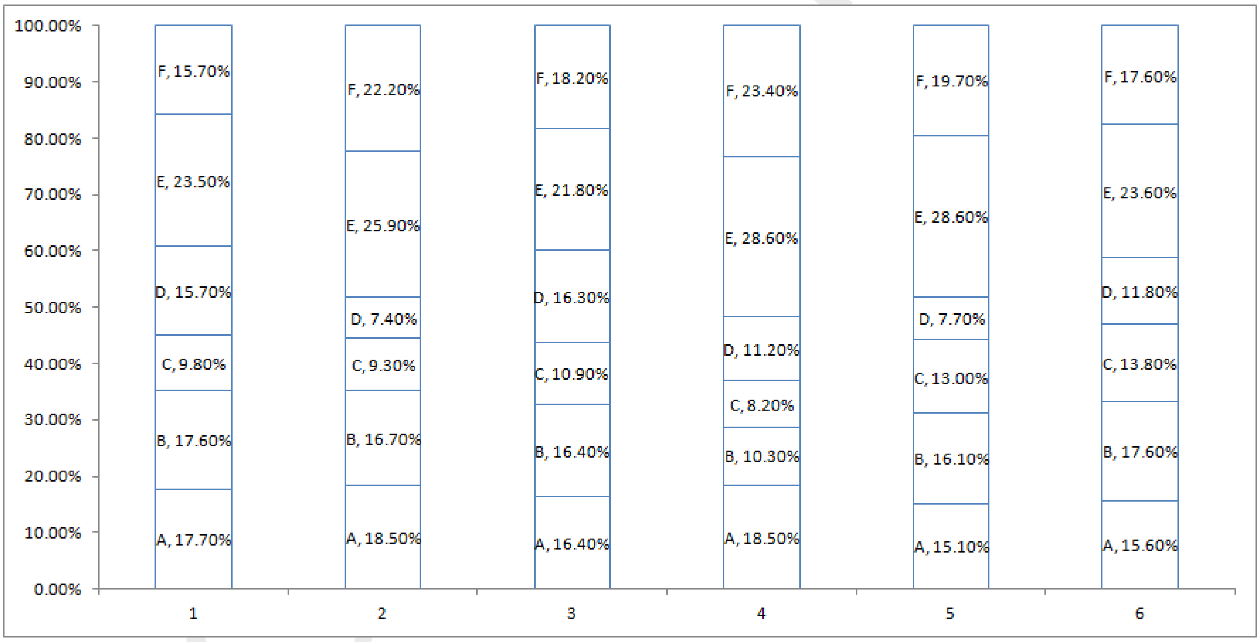
Free CAT Exam Preparation App

Instructions

Directions for the following three questions: Answer these questions based on the data given below:

There are six companies, 1 through 6. All of these companies use six operations, A through F. The following graph shows the distribution of efforts put in by each company in these six operations.

The Y axis represents the % distribution of effort and the X axis represents the company



Question 157

Suppose the companies find that they can remove operations B, C and D and redistribute the effort released equally among the remaining operations. Then which operation will show the maximum share across all companies and all operations?

A Operation E in company 1

B Operation E in company 4

C Operation F in company 5

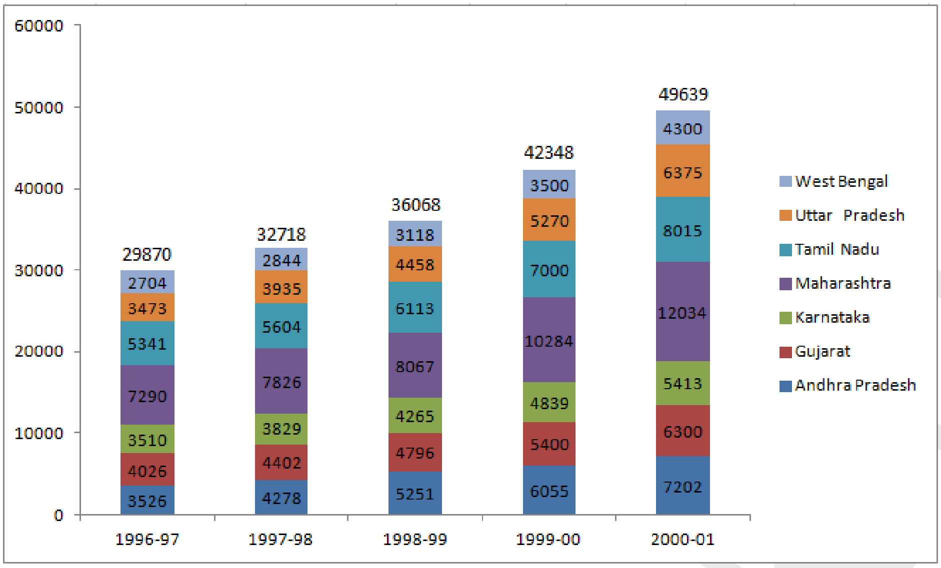
D Operation E in company 5

VIEW SOLUTION

Instructions

The chart given below indicates the annual sales tax revenue collections (in rupees in crores) of seven states from 1997 to 2001.

The values given at the top of each bar represents the total collections in that year



Question 158

Which state below has been maintaining a constant rank over the years in terms of its contribution to total tax collections?

A Andhra Pradesh

B Karnataka

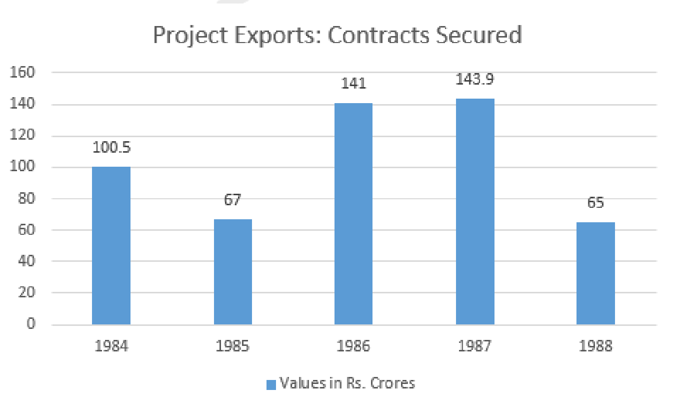
C Tamil Nadu

D Uttar Pradesh

VIEW SOLUTION

Instructions

Refer to the following Bar-chart and answer the questions that follow :



Question 159

Compared to the performance in 1985 (i.e. taking it as the base), what can you say about the performances in the years `84, `85, `86, `87, `88 respectively, in percentage terms?

A 150, 100, 211, 216, 97

B 100, 67, 141,144,65

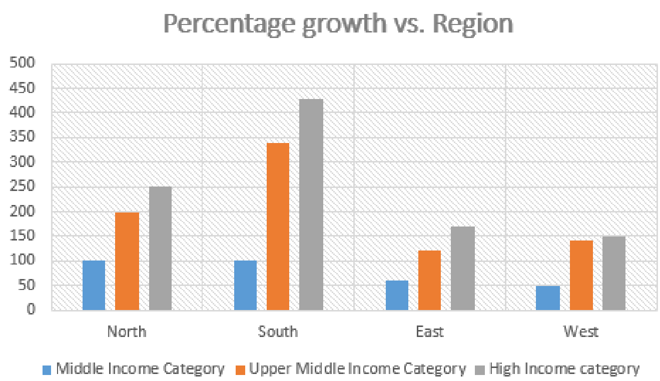
C 150, 100, 200, 215, 100

D 120, 100, 220, 230, 68

VIEW SOLUTION

Instructions

Answer the questions based on the following information. The following bar chart gives the growth percentage in the number of households in the middle, upper-middle and high-income categories in the four regions for the period between 1987-88 and 1994-95.



	Number of households in 1987-88	Average household income in 1987-88	Growth in average household income (1994-95 over 1987-88)
Middle Income	40	Rs. 30,000	50%
Upper-middle	10	Rs. 50,000	60%
High income	5	Rs. 75,000	90%

(Number of households in thousands)

Question 160

Which region showed the highest percentage growth in number of households in all the income categories for the period?

A North

B South

C West

D None of these

VIEW SOLUTION

Question 161

The ratio of total income for the high-income category to the upper-middle class increased by how much percentage in the given period (approximately) ?

A 20%

B 36%

C 25%

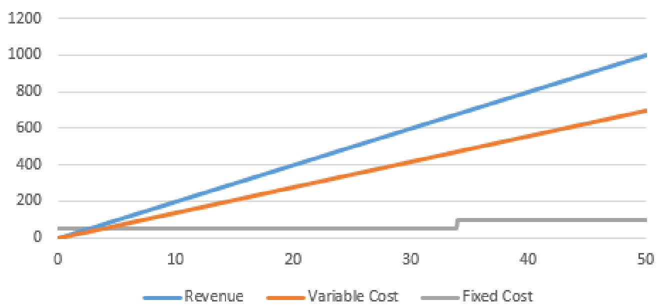
D Cannot be determined

VIEW SOLUTION

Instructions

Answer the questions based on the following information.

Ghosh Babu has a manufacturing unit. The following graph gives the cost for the various number of units. Given: Profit = Revenue - Variable cost - Fixed cost. The fixed cost remains constant up to 34 units after which additional investment is to be done in fixed assets. In any case, production cannot exceed 50 units.



Note: The fixed cost for upto 34 units is 50 and the the fixed cost for more is 100.
The revenue from 50 units is 1000 and the variable cost from 50 units is 700

Question 162

What is the minimum number of units that need to be produced to make sure that there was no loss?

A 5

B 9

C 20

D Indeterminable

VIDEO SOLUTION

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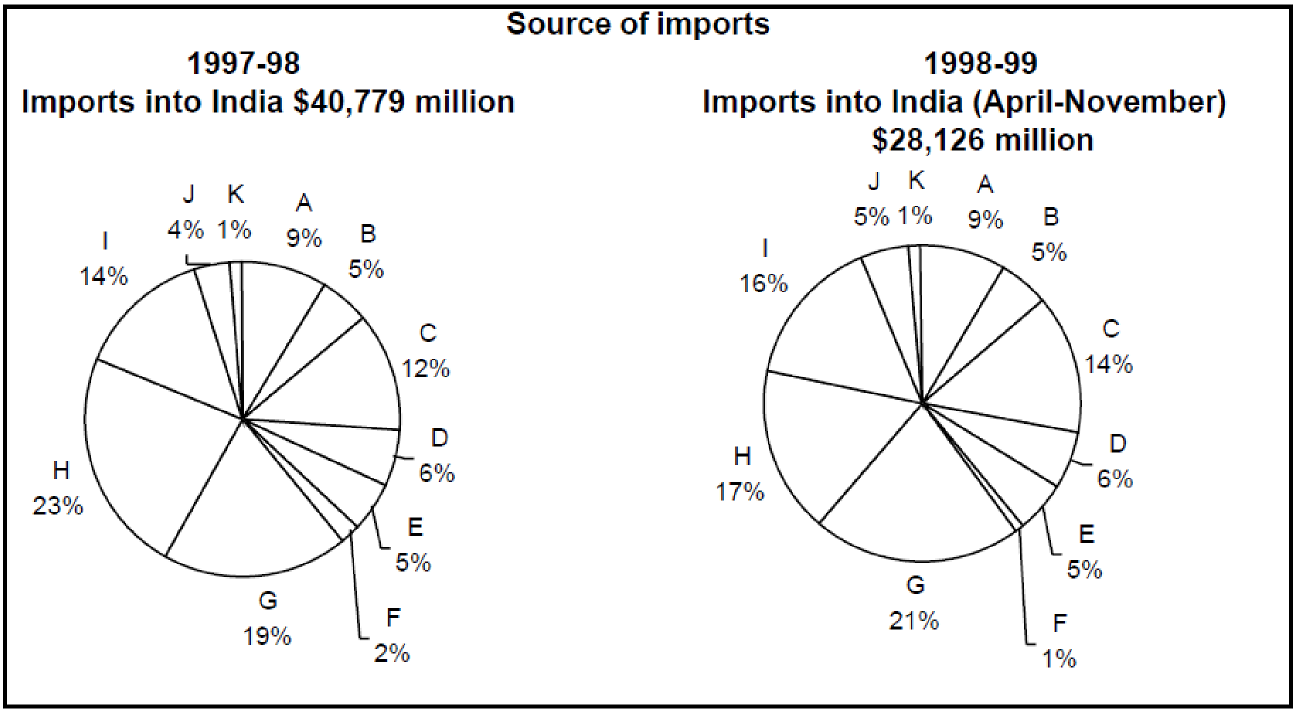
Instructions

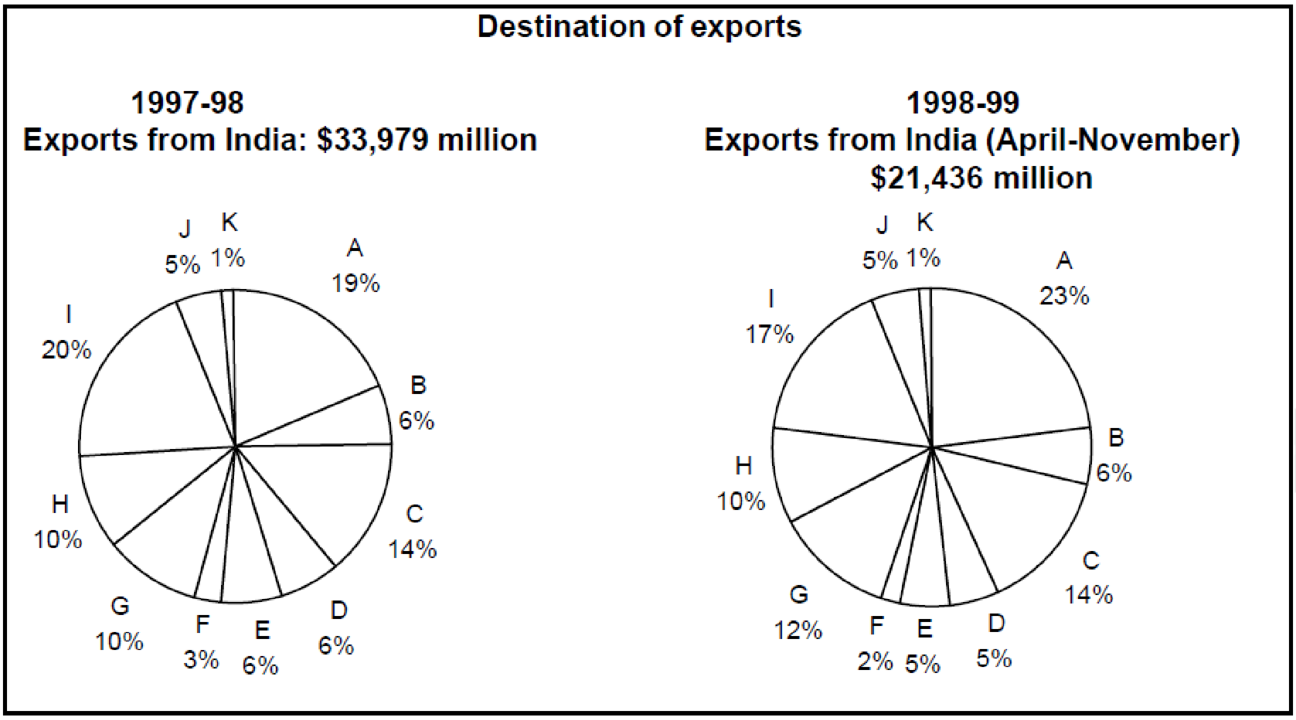
Consider the information provided in the figure below relating to India's foreign trade in 1997-98 and the first eight months of 1998-99.

Total trade with a region is defined as the sum of exports to and imports from that region.
Trade deficit is defined as the excess of imports over exports. Trade deficit may be negative.

A:USA. B:Germany C:Other EU. D:U.K. E:Japan F:Russia

G:Other East Europe H:OPEC I:Asia J:Other LDCs K:Others





Question 163

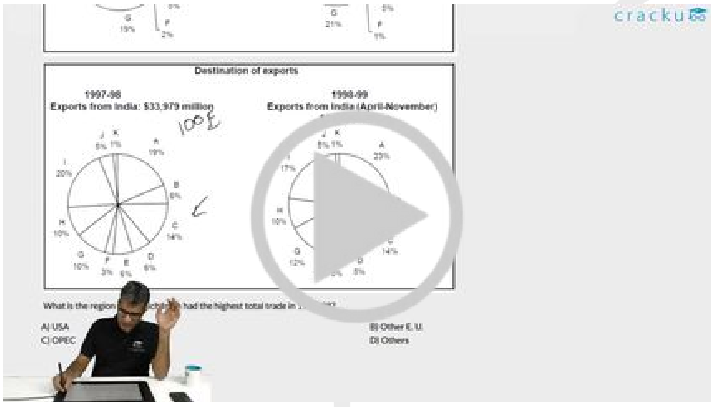
What is the region with the lowest trade deficit with India in 1997-98?

A USA

B Asia

C Others

D Other E. U.



VIDEO SOLUTION

Answers

1.**D**	2.**C**	3.**D**	4.**B**	5.**D**	6.**C**	7.**C**	8.**D**
9.**D**	10.**D**	11.**A**	12.**A**	13.**D**	14.**D**	15.**B**	16.**B**
17.**A**	18.**6**	19.**D**	20.**4**	21.**B**	22.**A**	23.**C**	24.**A**
25.**D**	26.**D**	27.**A**	28.**C**	29.**C**	30.**45**	31.**A**	32.**C**
33.**B**	34.**A**	35.**C**	36.**D**	37.**A**	38.**D**	39.**B**	40.**B**
41.**B**	42.**D**	43.**A**	44.**B**	45.**B**	46.**B**	47.**B**	48.**C**
49.**B**	50.**A**	51.**B**	52.**D**	53.**C**	54.**A**	55.**B**	56.**B**
57.**A**	58.**C**	59.**B**	60.**C**	61.**B**	62.**D**	63.**A**	64.**30**
65.**A**	66.**A**	67.**1**	68.**3**	69.**C**	70.**C**	71.**A**	72.**B**
73.**5**	74.**A**	75.**5**	76.**A**	77.**A**	78.**40**	79.**2**	80.**D**
81.**C**	82.**A**	83.**4**	84.**10**	85.**B**	86.**B**	87.**C**	88.**B**
89.**B**	90.**C**	91.**C**	92.**C**	93.**2**	94.**D**	95.**A**	96.**A**
97.**D**	98.**C**	99.**A**	100.**D**	101.**A**	102.**B**	103.**B**	104.**D**
105.**B**	106.**B**	107.**A**	108.**D**	109.**B**	110.**B**	111.**A**	112.**C**
113.**C**	114.**A**	115.**B**	116.**B**	117.**C**	118.**B**	119.**B**	120.**D**
121.**B**	122.**B**	123.**C**	124.**C**	125.**A**	126.**C**	127.**C**	128.**C**
129.**C**	130.**B**	131.**D**	132.**D**	133.**C**	134.**D**	135.**D**	136.**D**
137.**A**	138.**C**	139.**C**	140.**A**	141.**C**	142.**B**	143.**34**	144.**42**
145.**A**	146.**B**	147.**B**	148.**B**	149.**D**	150.**A**	151.**150**	152.**B**
153.**C**	154.**B**	155.**A**	156.**A**	157.**D**	158.**C**	159.**A**	160.**B**
161.**D**	162.**B**	163.**A**					

Explanations

1.**D**

Year	Plywood in Rs/Tonne	Plywood in Rs/meter cube	Saw Timber in Rs/Tonne	Saw Timber in Rs/meter cube	Difference
1989	4	2.8	12	9.6	6.8
1990	5	3.5	10	8	4.5
1991	4	2.8	13	10.4	7.6
1992	6	4.2	15	12	7.8

From the above diagram, we can say that difference is highest in 1992.

VIEW SOLUTION

2.**C**

In this case the average realization will be-

$$\frac{(4\times5.26\times1.05)+(3\times14.28\times1.01)+(3\times20\times1.1)}{4+3+3}\approx 13.15$$

VIEW SOLUTION

3.**D**

In the diagram, we can see the sudden increase in MAY, and there was no such increase again during the given period.

Hence, May is the answer.

VIEW SOLUTION

4. **B**
There were 30 naya mixture in 97' which will also work in 98' ,thus we have 50 new mixtures in 98' . Now in 99' 20% of 30 i.e. 6 are thrown in junk . So in 99' we have 24 + 50 = 74 naya mixtures of previous years. So naya mixture grinder purchased in 99' is 124-74 = 50. Hence option b.

VIEW SOLUTION

5. **D**
Since X axis gives expenditure we have to see if average is as right as possible. Average expenditure of dubey family is rightmost. Hence, option D.

VIEW SOLUTION

6. **C**
SInce the Y-axis gives the income, to find minimum average income see if any family has income in different months as below as possible in the graph.

From the graph we can see that coomar has lowest average income. Hence option C.

VIEW SOLUTION

7. **C**
Let the GDP of china and India be c and i respectively. According to given condition . We know that (Foreign inflows)china = 10*(Foreign inflows)India. And Foreign inflows = FEI * GDP . So we get , 4.8 * c = 10*0.72*i ; which is c=1.5*i . Hence , China's GDP in 1998 was 50% higher than that of India.

VIEW SOLUTION

8. **D**
The given values in the graph are ratio of FEI to GDP un %. So if entire value increase we can conculde that there can be increase in numerator and decrease in value of denominator and if entire value decrease we can conculde that there can be decrease in numerator and increase in value of denominator. hence only options 2 and 5 comply to these .

VIEW SOLUTION

9. **D**
Since we do not know the distribution of households in each region, we cannot determine the average income.

VIEW SOLUTION

10. **D**
In this question the total population does not change much in the whole span. So the per capita food grain production will be maximum when the total foodgrain production is maximum. It happens in 1995.

VIEW SOLUTION

11. **A**
The number of man-hours spent in coding = 425(offshore) + 100(onsite) = 525
The number of man-hours spent in offshore design and offshore coding together = 100+425 = 525
So, option a) is the correct answer

VIEW SOLUTION

12. **A**
The answer to this question is provided in the table.
The average income of middle income category is 30,000
The average income of upper-middle income category is 50,000 and
The average income of high income category is 75,000

Hence, the correct answer is 75,000

VIEW SOLUTION

13. **D**
Profitability in the year 1994-95 is 2.5/100 = 2.5%
Profitability in the year 1995-96 is 4.5/250 = 1.8%
Profitability in the year 1996-97 is 6/300 = 2%
Profitability in the year 1997-98 is 8.5/290 = 2.93%
Profitability in the year 1998-99 is 12/680 = 1.76%

Profitability increased/decreased in some years, hence the first three options are wrong.

So, the correct answer is option (d)

VIEW SOLUTION

14. **D**
Percentage growth in net profit in the year 1995-96 is 4.5/2.5 - 1 = 80%
Percentage growth in net profit in the year 1996-97 is 6/4.5 - 1 = 33.33%
Percentage growth in net profit in the year 1997-98 is 8.5/6 - 1 = 41.67%
Percentage growth in net profit in the year 1998-99 is 12/8.5 - 1 = 41.17%

Hence, the percentage growth in net profit is maximum in the year 1995-96

VIEW SOLUTION

15. **B**
We have been given details about the quarterly sales figures. Also, we have been given details about the sales figures every month. Some of the data are missing and some additional conditions have been given in the question. Let us try to complete the pie chart as much as possible with the data available to us.

It is known that the sales figures during the three months of the second quarter (April, May, June) of 2016 form an arithmetic progression.
We know that the sales in April is 40.
Let the sales in May be 40+x and the sales in June be 40+2x.
We know that the total sales in Q2 is 150.
=> 40 + 40 + x + 40 + 2x = 150
3x = 30
x = 10

Therefore, sales in May 2016 = 40 + 10 = 50
Sales in June 2016 = 40 + 20 = 60

Similarly, it has been given that the sales in October, November, and December 2016 form an arithmetic

progression.
Sales in October = 100
Sales in Q4 = 360

Let the sales in November be 100+y and the sales in December be 100+2y.
100 + 100 + y + 100 + 2y = 360
300 + 3y = 360
=> y = 20

Sales in November 2016 = 120 and Sales in December 2016 = 140
Sales in Q1 of 2016 = Sum of the sales in the months of January, February, and March 2016
= 80 + 60 + 100
= 240

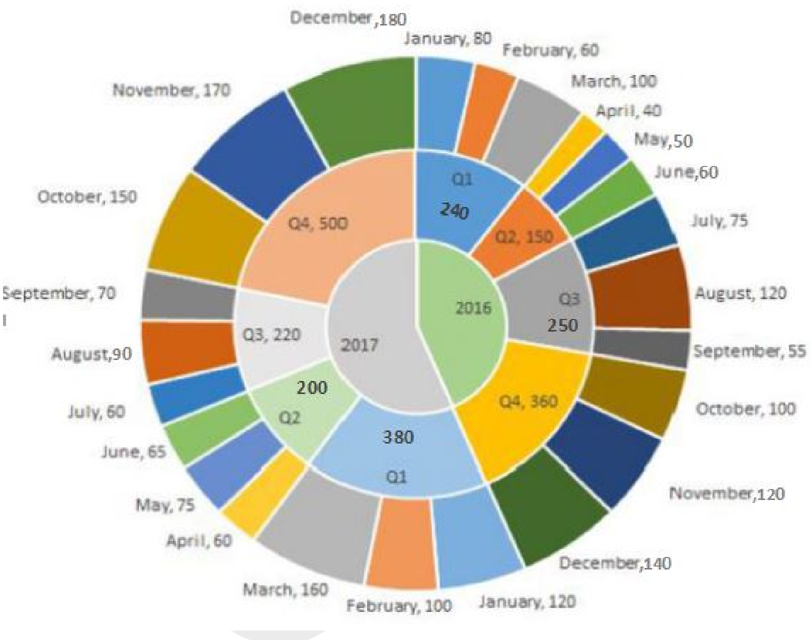
Sales in Q3 of 2016 = Sum of the sales in the months of July, August, and September 2016

= 75 + 120 + 55
= 250

Sales in Q1 of 2017 = 120 + 100 + 160 = 380
Sales in Q2 of 2017 = 65 + 75 + 60 = 200
We know that sales in Q3 of 2017 = 220

Let the sales in August of 2017 be ‘a’.
60 + 70 + a = 220
=> a = 90
Sales in August 2017 = 90
We know that sales in Q4 of 2017 = 500
Let the sales in December of 2017 be ‘d’.
150 + 170 + d = 500
=> d = 180

Sales in December 2017 = 180



March of 2017:
Sales in March of 2017 = 160
Sales in February of 2017 = 100
% increase = 60/100 = 60%

October of 2017:

Sales in October of 2017 = 150
Sales in September of 2017 = 70
As we can see, the sales has increased by more than 100%.

March of 2016:
Sales in March of 2016 = 100
Sales in February of 2016 = 60
% increase in sales is less than 100%.

October of 2016:
Sales in October of 2016 = 100
Sales in September of 2016 = 55
% increase is less than 100%

As we can see, the percentage increase in sale as compared to the previous month was highest in October of 2017 among the given options. Therefore, option B is the right answer.

VIDEO SOLUTION

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16. **B**
As it is given that profit = sales - cost

So profit for september = $\frac{3}{36} \times 100$ = 8% (Nearly)

profit for march = $\frac{3}{30} \times 100$ = 10% (nearly)

profit for july = $\frac{2}{35} \times 100$ < 10%

profit for may = $\frac{2}{36} \times 100$ <10%

So highest percentage profit is for march month.

VIEW SOLUTION

17. **A**
Simple calculations show difference in values of sales for karnataka over 2 successive pair of years - 1999-2000 and 2000-2001 - same.

The increase in 1999-2000 = 4839 - 4265 = 574

The increase in 2000-2001 = 5413 - 4839 = 574

VIEW SOLUTION

18. **6**
The frequency distribution is:

S: ~~3,3~~,3,4,4,4,5,5,6,7

F: ~~1,1~~,2,3,3,4,5,5,5,7

C: 1,2,2,2,~~3,3~~,3,3,4,6

or

S: 3,3,3,~~4,4~~,4,5,5,6,7

F: ~~1,1~~,2,3,3,4,5,5,5,7

C: 1,2,~~2,2~~,3,3,3,3,4,6

Zooma(Z) has a total score of 17 (comes under happy category), and other 2 countries, which are categorized as happy, have the highest score in exactly one parameter.

Suppose the other two countries are P and Q

Z have two possibilities for S, F, C : (6,7,4) & (6,5,6)

All the other cases are negated because "All the 3 countries, which are categorised as happy, have the highest score In exactly one parameter."

For Example : 7,7,3 is not possible because 7 being the highest score is there in two parameters.

So, it scored 6 in S in both the cases.

VIDEO SOLUTION

19. **D**

The data can be tabulated as follows(approximately):

States	IPC Crimes	SLL Crimes	Other Crimes	Total Crimes
Telangana	3-4	14-15	6-7	24-25
Puducherry	1	0	30	31
Kerala	7-8	15-16	10-11	33-34
Haryana	3-4	25-26	9-10	37-38
Maharashtra	15-16	35-36	5-6	55-56
Tamilnadu	2-3	25-26	35-36	62-64
Goa	25-26	35-36	18-19	80
Karnataka	15-16	48-49	25-26	91
Delhi	63-64	35-36	42-43	142-143
West Bengal	0	520	0	520

	IPC crimes	SLL crimes	OtherCrimes
Delhi	*	*	*
Goa	*	4	*
Haryana	8	6	*
Karnataka	3	2	*
Kerala	*	9	*
Maharashtra	3	4	8
Puducherry	13	29	*
Tamil Nadu	11	7	*
Telangana	6	9	8
WestBengal	17	*	16

The highest cases are registered in West Bengal and Delhi.

The total number of IPC crimes = 63-64

The total number of SLL crimes = 520+35-36 = 555-556

Hence the ratio = (63-64)/(555-556) = 0.11 (Approximately) = 1:9

VIDEO SOLUTION

20. **4**

Writing down the data with us in a table form.

	Asia	Europe	Rest of World	Total
Dheeraj	3	7	1	11
Samantha	0	9	4	13
Nitesh	1	6	12	19
Total	4	22	17	43

But this includes countries which have been visited by one person, two people and three people. Essentially, there is an overlap of countries.
It is given to us that 32 distinct countries were visited among the three people,
Using that we can write down the equations,

$I + II + III = 32$...(1)
$I + 2II + 3III = 43$...(2)
Where I, II, and III denote the number of countries visited exactly by 1 of them, 2 of them and all of them.
We are told that USA(ROW) is the only country which was visited by all three people, that means $III = 1$

Subtracting Equation 1 from 2 we get, $II + 2III = 11$
And since III=1 we get $II = 9$

So, we are looking for 9 countries visited by exactly 2 people, and 22 countries visited by exactly 1 person.
Using the information from the problem set, we can note down the following,

Combination	Asia	EU	ROW	Total
Dheeraj				
Samantha				
Nitesh				
Dheeraj and Nitesh	1(China)			
Dheeraj and Samantha		1(France)		
Samantha and Nitesh				
All Three	0	0	1(USA)	
Total				32

We can see that Dheeraj visited only 1 country in ROW so every other value in that column has to be zero for wherever Dinesh is present. And Samantha has not visited any country in Asia so every value with Samantha present in Asia should be zero as well.

Combination	Asia	EU	ROW	Total
Dheeraj			0	
Samantha	0			
Nitesh				
Dheeraj and Nitesh	1(China)	0	0	1
Dheeraj and Samantha	0	1(France)	0	1
Samantha and Nitesh	0			
All Three	0	0	1(USA)	
Total				32

We have the following distribution, we are looking for 9 countries visited by exactly 2 people. And we have identified 2 countries, that means, 7 were countries were visited by both Samantha and Nitesh only. And given that half the countries visited by both of them are in Europe, and total number of countries visited by both of them is 7(only two of them)+1(USA)=8, they visited 4 countries in Europe together and 3 countries in ROW together. And using that information, we can fill in the rest of the table as well. Since we know the number of countries visited by each of them in a particular continent.

Combination	Asia	EU	ROW	Total
Dheeraj	2	6	0	8
Samantha	0	4	0	4
Nitesh	0	2	8	10
Dheeraj and Nitesh	1(China)	0	0	1
Dheeraj and Samantha	0	1(France)	0	1
Samantha and Nitesh	0	4	3	7
All Three	0	0	1(USA)	1
Total	3	17	12	32

Number of countries in ROW visited by both Nitesh and Samantha is 3+1=4.

VIDEO SOLUTION

21. **B**

Total tonnes of transportation by both rail and road is about 31*12/100 = 3.72 million tonnes and total cost incured 18*30/100 = 5.4 .

Hence required value is 5.4/3.72 which is about 1.5.

Hence option B .

VIEW SOLUTION

22. **A**

Cheapest mode of transport will be the one which will have highest transport volume and comparatively lowest cost. We can figure out from the graph that Road have wide gap with very less costs.

For Road, Cost = 6/22

For Rail, Cost = 12/9

For Pipeline, = 65/49

For Ship, Cost = 10/9

Lowest cost is for road.

VIEW SOLUTION

23. **C**

Total increase in 4 years = 8640 - 5290 = 3350

% increase = $\frac{3350}{8640}$ = 63.3%

Average per-year growth = 63.3/4 = 15.9%

VIEW SOLUTION

24. **A**

To have highest amount of savings, income should be high and expenditure should be low. For ahuja family the gap is highest . Hence they have highest amount of savings

VIEW SOLUTION

25. **D**
The profit per employee will be hishest when the ratio of profit to the number of employees is the highest. It happens in march and the value is $\frac{33-30}{12} = \frac{1}{4}$

VIEW SOLUTION

26. **D**
From the graph we can clearly make out that for dubey family in both the months income is nearly equal to expenditure, thus hey have lowest savings.

VIEW SOLUTION

27. **A**
We know that average of the particular area would increase by 1 each year if there is no change in any member of the group.

So, now in marketing, ob and om, we can see increase in average by 1 in 3 successive years.

But in Finance we see that average decreases 2 times, once when a new teacher is recruited and other time when the 60 year old retires.

Hence option A.

VIEW SOLUTION

28. **C**
Percentage of students more in marketing jobs than in finance jobs are 14%, 31%, 20%, 18% and 0% in 1992, 1993, 1994, 1995 and 1996 respectively.

=> Number of people = 14*800/100 + 31*650/100 + 20*1100/100 + 18*1200/100 = 112 + 202 + 220 + 216 = 750

VIEW SOLUTION

29. **C**
The total number of male and female test cases in 1980 = 1000

Age group(In 1980)	Males alive in 1990	Males alive in 2000	Males alive in 2010	Males alive in 2020
10-20	230	180	150	140
20-30	235	205	165	125
30-40	210	160	90	50
40-50	190	100	25	5

Age group(In 1980)	Females alive in 1990	Females alive in 2000	Females alive in 2010	Females alive in 2020
10-20	240	210	160	100
20-30	225	175	145	105
30-40	190	150	100	60
40-50	220	120	30	18

The total number of males in the age group of 30-40(in 1980) alive till 2010 = 90

The total number of females in the age group of 30-40(in 1980) alive till 2010 = 100

Thus, the total number of people in the age group of 30-40(in 1980) alive till 2010 = 90 + 100 = 190

Thus, the correct option is C.

VIDEO SOLUTION

30. **45**

We can note down the data from the chart into the form of a table, that gives us,

Surfer/Blogger	A	B	C	D	Total
M	10	0			30
N	25	0			30
O	0	0			30
P	5	25			30
X	0	0			30
Y	5	20			30
Total	45	45	45	45	180

We are told that D receives more stars than C from Y. Considering Y has already given 25 stars, it will give 0 stars to C and 5 stars to D.
The only two surfers who have not given any stars to A or B is O and X, and these are the two surfers to give all of their stars to a single blogger.

We are also told that M gives different stars to the four bloggers,
Since he has already given 0 and 10, the remaining distinct stars should add up to 20. The only numbers that are remaining that add up to 20 are 5 and 15.
We know that X rewards one of C or D 30 stars and O rewards one of C or D 30 stars. Given that, M could not have rewarded D 15 stars, since Y rewarded D 5 stars, and D is also going to rewarded 30 stars by O or X, and since the total is same for all, which is 45. This is not possible.
This means that, M rewarded C 15 stars and D 5 stars.
This gives us two cases,
Case-1

Surfer/Blogger	A	B	C	D	Total
M	10	0	15	5	30
N	25	0	0	5	30
O	0	0	30	0	30
P	5	25	0	0	30
X	0	0	0	30	30
Y	5	20	0	5	30
Total	45	45	45	45	180

Case-2

Surfer/Blogger	A	B	C	D	Total
M	10	0	15	5	30
N	25	0	0	5	30
O	0	0	0	30	30
P	5	25	0	0	30
X	0	0	30	0	30
Y	5	20	0	5	30
Total	45	45	45	45	180

We can use these two cases to answer the questions,

D was rewarded 45 stars in total.

VIDEO SOLUTION

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31. **A**

Percentage increase from 2006 to 2007 is 120/3.8 = 31.5%. approximately

Now if the same % increase is considered over period 2007-2008, we get . 31.5 * 500 = x-500.

X is 657 approximately.

So ,the difference between the estimated subscription in Europe in 2008 and what it would have been if it were computed using the percentage growth rate of 2007 (over 2006), is closest to: 657-600 = 57. Hence option A.

VIEW SOLUTION

32. **C**

Growth in 2007 => $\frac{500-380}{380} * 100$ = 31.58

Growth in 2005 => $\frac{280-190}{190} * 100$ = 47.37%

Percentage change = $\frac{47.37-31.58}{47.37} * 100$ = 35% approximately.

=> 35% is the answer

VIEW SOLUTION

33. **B**

Since in 94 total production was 50% more than 89 , so total production = 150. Also we know that total production in given 4 sectors in 94 = 76.8. Hence other than the 4 sectors total production is 150 - 76.8 = 73.2 . Hence the percentage increase in production between 1989 and 1994 in sectors other than the four listed above is: (73.2-40) / 40 is approx 87.5% . Hence option B.

VIEW SOLUTION

34. **A**

Let first approximately calculate production in-the four sectors, manufacturing, mining & quarrying, electricity and chemicals in 1994 we get , 25.4 , 18 , 14.55 , 18.85 respectively . Hence overall growth = 76.8. Now The percentage increase of production in-the four sectors, taken together, in 1994, relative to 1989, is approximately = (76.8-60)/60 which is approx. 25 %. Hence option A.

VIEW SOLUTION

35. **C**
From the graph it is clearly visible that Electricity had the highest growth during the period 1989 and 1998. Hence option C.

VIEW SOLUTION

CAT Percentile Predictor

36. **D**
We cant find absolute value of FEI as GDP value is not given . Hence option D.

VIEW SOLUTION

37. **A**
From the given data we can clearly notice that FEI for India decreased (changed) by greatest percentage.

For India the value = (1.71-0.72)/1.71 = 57.894%

VIEW SOLUTION

38. **D**
The overall revenue has increased by a ratio of $\frac{49639}{29870}$ = 1.67.

So, the state whose revenue ratio increases by more than 1.67 is the answer.

Among the given options, only AP's sales tax revenue ratio has increased.

VIEW SOLUTION

39. **B**
We have to consider the ranking of that state that doesn't change throughout or changes only once.

The rankings of the states are as follows:

	1996 – 97	1997 – 98	1998 – 99	1999 – 00	2000 – 01
West Bengal	7	7	7	7	7
Uttar Pradesh	6	5	5	5	4
Tamil Nadu	2	2	2	2	2
Maharashtra	1	1	1	1	1
Karnataka	5	6	6	6	6
Gujarat	3	3	4	4	5
Andhra Pradesh	4	4	3	3	3

So, the states West Bengal, Tamil Nau, Maharashtra, Karnataka and Andhra Pradesh do not change their rankings more than once. There are 5 states in total under the criteria.

VIEW SOLUTION

40. **B**
Turkey has a value of 16% and a quantity of 15%.

The average price in Euro per kilogram for Turkey is (16*5.760)/(15*1.055) which is nearly equal to 5.6.

Hence option B.

VIEW SOLUTION

Important Verbal Ability Questions for CAT (Download PDF)

41. **B**
The country which has the highest average price would have low quality and high value when compared to other countries..

Switzerland fits the above description.

Hence, option B is the answer.

VIEW SOLUTION

42. **D**
Since 10 purana were disposed off in 1997, it means 50 new have been bought in 1995, which implies that 70 were already in use. The breakup of 70 purana cannot be determined. So the number of purana disposed in 2000 cannot be calculated.

VIEW SOLUTION

43. **A**
Now from B through F total work done is 81.7%. Dividing it in 5 we get 16.34. So difference in operation E is 28.6-16.34 = 12.3 %. Hence there is a reduction of 12.3 .

VIEW SOLUTION

44. **B**
There are 3 steel and 2 cement companies with a turnover more than 1000 and profit exceeding 10% of turnover. hence there are 5 choices for the investor.

VIEW SOLUTION

45. **B**
There are in total 7 companies - 3 steels , 2 cement , 2 textile for which the profit is more than 10% of the turnover.

VIEW SOLUTION

Data Interpretation for CAT Questions (download pdf)

46. **B**
Seeta's growth percentage = (57-50)/50 * 100% = 14%

Geeta's growth percentage = (61-49)/49 * 100% = 24.5%

Ram's growth percentage = (61-52)/52 * 100% = 17.30%

Shyam's growth percentage = (61 - 53.5)/53.5 * 100% = 14.01%

So, Seeta's growth percentage is the least.

VIEW SOLUTION

47. **B**

The rate of growth is the slope of the line. In the second and third months, there is an increase in the slope of the line. After the third month, there is a fall in the slope of the line i.e., the rate of growth is declining. Therefore, option b) is the correct answer.

VIEW SOLUTION

48. **C**
Max growth rate can be calculated by finding percentage growth over previous year.

We can see a significant increase in the year 1999-2000.

On calculating the ratio, we get $\frac{10284}{8067}$ = 1.275, which is more than any other year.

Hence, 1999-2000 is the answer.

VIEW SOLUTION

49. **B**
The raw materials used in 1991 was 55,000
The raw materials used in 1992 was 48,000 (decrease by 7,000)
The raw materials used in 1993 was 60,000 (increase by 12,000)
The raw materials used in 1994 was 65,000 (increase by 5,000)
The raw materials used in 1995 was 75,000 (increase by 10,000)

Hence, the increase in use of raw materials was maximum in 1993

VIEW SOLUTION

50. **A**
The population is increasing steadily, but the milk consumed decreased significantly in 1996.
Hence, the only two years we need to check for the least per capita milk consumption are the first year (1990) and 1996.

Milk consumed in 1990 is 5 million gallons.
Total population in 1990 is 34+35 = 69 million.
Hence, per capita consumption of milk in 1990 is 0.072 gallon per person.

Milk consumed in 1996 is 6 million gallons.
Total population in 1996 is 40+42 = 82 million.

Hence, per capita consumption of milk in 1996 is 0.073 gallon per person.

Hence, the year with the least per capita consumption of milk is 1990.

VIEW SOLUTION

51. **B**
Volatility for arhar is (2300-1500)/1900 = 42%
Volatility for pepper is (19500-17500)/18500 = 10.8%
Volatility for sugar is (1500-1410)/1450 = 6.2%
Volatility for gold is (4300-3800)/4150 = 12%

Hence, the highest volatility is around 40%

VIEW SOLUTION

52. **D**

The profit percentage equals 25%*(2105/1700-1) + 25%*(19250/18500-1) + 25%*(1430/1440-1) + 25%*(3850/4250-1)
This equals 25%*(23.82% + 4.05% -0.69% - 9.41%) = 25%*17.77 = 4.4%

The closest option to this is option (d)

VIEW SOLUTION

53. **C**
Volatility for arhar is (2300-1500)/1900 = 42%
Volatility for pepper is (19500-17500)/18500 = 10.8%
Volatility for sugar is (1500-1410)/1450 = 6.2%
Volatility for gold is (4300-3800)/4050 = 12%

Hence, the volatility is least for sugar.

VIEW SOLUTION

54. **A**
Price change of arhar = (2105-1700)/1700 = 23.82%
Price change of pepper = (19250-18500)/18500 = 4.05%
Price change of sugar = (1430 - 1440)/1430 = -0.69%
Price change of gold = (3850 - 4250)/3850 = -9.41%

Hence, the price change is maximum for arhar.

VIEW SOLUTION

55. **B**
In both imports and exports, India did the least amount of trade in 1997-98 with K (both equal to 1%).

Hence, India had the lowest total trade with K in the year 1997-98.

The amount of Indian exports equals 1%*33979 = 340 millions

VIDEO SOLUTION

56. **B**
Profitability in the year 1994-95 is 2.5/100 = 2.5%
Profitability in the year 1995-96 is 4.5/250 = 1.8%
Profitability in the year 1996-97 is 6/300 = 2%
Profitability in the year 1997-98 is 8.5/290 = 2.93%
Profitability in the year 1998-99 is 12/680 = 1.76%

Hence, the profitability is maximum in the year 1997-98

VIEW SOLUTION

57. **A**
Percentage growth in sales in the year 1995-96 is 250/100 - 1 = 150%
Percentage growth in sales in the year 1996-97 is 300/250 - 1 = 20%
Percentage growth in sales in the year 1997-98 is 290/300 - 1 = -3.33%
Percentage growth in sales in the year 1998-99 is 680/290 - 1 = 134%
Hence, the percentage growth in sales is maximum in the year 1995-96

VIEW SOLUTION

58. **C**
After an increase by 33%, the difference in the prices of one tonne of saw timber and one cubic metre of logs is given below according to the respective years.

The difference in 1987 is 15 - 10*133.33% = 1.67
The difference in 1988 is 16 - 10*133.33% = 2.67
The difference in 1989 is 17.5 - 12*133.33% = 1.5
The difference in 1990 is 15 - 10*133.33% = 1.67
The difference in 1991 is 17.5 - 12.5*133.33% = 0.833
The difference in 1992 is 18 - 15*133.33% = -2
The difference in 1993 is 20 - 20*133.33% = -6.67

Hence, the difference is least in the year 1991.

VIEW SOLUTION

59. **B**
The percentage increase in price of logs over the period of time equals 20/15-1 = 33.33%
The percentage increase in price of plywood over the period of time equals 7/3 - 1 = 133.33%
The percentage increase in price of sawtimber over the period of time equals 200/100 - 1 = 100%

Hence, the maximum percentage increase in price over the period of time happened with plywood.

VIEW SOLUTION

60. **C**
The maximum percentage increase in price of logs happened in the year 1991 (compared to 1990) and equaled 18/15 - 1 = 20%
The maximum percentage increase in price of plywood happened in the year 1992 (compared to 1991) and equaled 6/4 - 1 = 50%
The maximum percentage increase in price of sawtimber happened in the year 1993 (compared to 1992) and equaled 20/15 - 1 = 33.33%
Hence, the correct answer is 50%

VIEW SOLUTION

61. **B**
The answer is provided by the bar chart.
Look at the graph named North and the bar colored Orange.
This shows the percentage increase in the upper middle class households from the year 1987-88 to 1994-95 and it equals 200%

VIEW SOLUTION

62. **D**
We don't know the number of households in the upper-middle income class for Northern region and hence can't calculate the total household income for the same.

VIEW SOLUTION

63. **A**

For having the greatest percentage change, the slope of the line should be the highest. From the graph, we can see that slope is highest for A between Feb and March. So A is the correct answer.

VIEW SOLUTION

64. **30**

The total number of male and female test cases in 1980 = 1000

Age group(In 1980)	Males alive in 1990	Males alive in 2000	Males alive in 2010	Males alive in 2020
10-20	230	180	150	140
20-30	235	205	165	125
30-40	210	160	90	50
40-50	190	100	25	5

Age group(In 1980)	Females alive in 1990	Females alive in 2000	Females alive in 2010	Females alive in 2020
10-20	240	210	160	100
20-30	225	175	145	105
30-40	190	150	100	60
40-50	220	120	30	18

We are given that there are 250 females from age 20-30 in 1980 and in 2000 these females age are from 40-50 but only 175 are alive in 2000.

In 2000 there were 175 females from age 40-50. If we assume that out of these, 30 females were of age 48 years in 2000 and they died in 2005, then there are 30 females who died at the age of 53.

If we assume that out of the 175 females, 30 females were of age 42 years in 2000, and they died in 2005, then 30 females died at the age of 47. Now, if we assume that there are 15 females of age 42 and 15 females of age 48 in the year 2000, and they all died in 2005, then we have 15 females who died at the age of 47 and 15 females who died at the age of 53.

So we can see that there are many cases possible. We are given that there were 250 females aged 20-30 in 1980, and in 2010, these females ages are from 50-60, but only 145 are alive in 2010.

In 2010 there were 145 females from age 50-60. If we assume that out of these, 40 females were of age 58 years in 2010 and they died in 2015, then there are 40 females who died at the age of 63.

If we assume that out of the 145 females, 40 females are of age 52 years age in 2010, and they died in 2015, then 40 females died at the age of 57. Now, if we assume that there are 15 females of age 52 and 25 females of age 58 in the year 2010, and they all died in 2015, then we have 15 females who died at the age of 57 and 25 females who died at the age of 63.
So we can see that again, there are many cases possible. In the first case, the range of values possible is from 0 to 30. In the
second case, the range of values possible is from 0 to 40. So in total, we get a range of possible values from 0 to 70.

Thus, only one possible value of this question is not possible.

VIDEO SOLUTION

65. **A**

In order to find the trading partner with the highest trade deficit, we have to look for countries from whom India imported a lot but exported little.

The two countries which are important here are G and H. However, the exports to both are the same. But the imports from H are more than the imports from G.

So, the trading partner with whom India has the highest trade deficit is H.

The amount of trade deficit equals 23%*40779 - 10%*33979 = 5981 mio ~ 6 bn

VIDEO SOLUTION

66. **A**

We have been given details about the quarterly sales figures. Also, we have been given details about the sales figures every month. Some of the data are missing and some additional conditions have been given in the question. Let us try to complete the pie chart as much as possible with the data available to us.

It is known that the sales figures during the three months of the second quarter (April, May, June) of 2016 form an arithmetic progression.
We know that the sales in April is 40.
Let the sales in May be 40+x and the sales in June be 40+2x.
We know that the total sales in Q2 is 150.
=> 40 + 40 + x + 40 + 2x = 150
3x = 30
x = 10

Therefore, sales in May 2016 = 40 + 10 = 50
Sales in June 2016 = 40 + 20 = 60

Similarly, it has been given that the sales in October, November, and December 2016 form an arithmetic progression.
Sales in October = 100
Sales in Q4 = 360

Let the sales in November be 100+y and the sales in December be 100+2y.
100 + 100 + y + 100 + 2y = 360
300 + 3y = 360
=> y = 20

Sales in November 2016 = 120 and Sales in December 2016 = 140
Sales in Q1 of 2016 = Sum of the sales in the months of January, February, and March 2016
= 80 + 60 + 100
= 240

Sales in Q3 of 2016 = Sum of the sales in the months of July, August, and September 2016

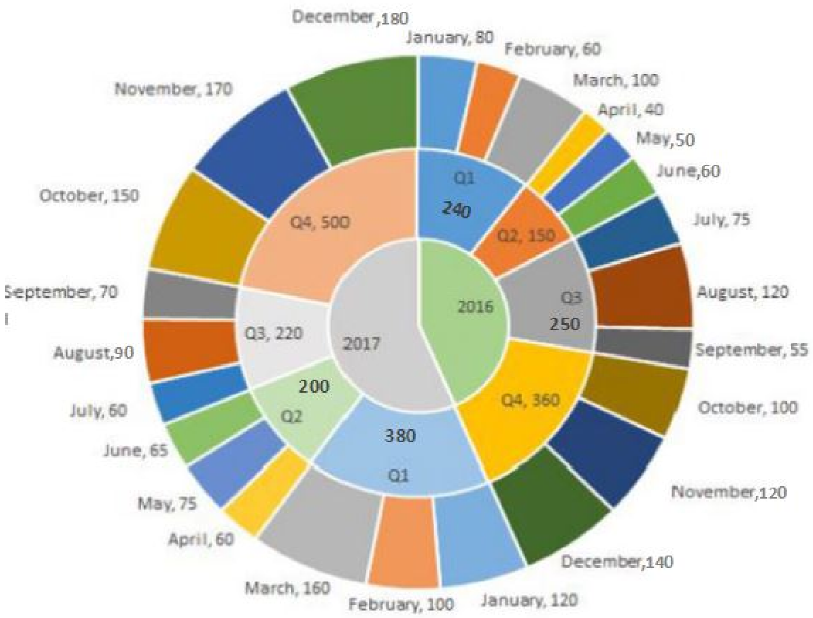
= 75 + 120 + 55
= 250

Sales in Q1 of 2017 = 120 + 100 + 160 = 380
Sales in Q2 of 2017 = 65 + 75 + 60 = 200
We know that sales in Q3 of 2017 = 220

Let the sales in August of 2017 be 'a'.
60 + 70 + a = 220
=> a = 90
Sales in August 2017 = 90
We know that sales in Q4 of 2017 = 500

Let the sales in December of 2017 be 'd'.
150 + 170 + d = 500
=> d = 180

Sales in December 2017 = 180



Q2 of 2017:

Sales in Q2 of 2017 = 200
Sales in Q1 of 2017 = 380
% decrease = 180/380

Q4 of 2017:
We can eliminate this option since the sales has increased in Q4 of 2017 as compared to the previous quarter.

Q2 of 2016:
Sales in Q2 of 2016 = 150
Sales in Q1 of 2016 = 240
% decrease = 90/240

Q1 of 2017:
Sales in Q1 of 2017 has increased as compared to sales in the previous quarter. We can eliminate this option as well.

180/380 is very close to 50%. 90/240 is closer to 33.33%. Therefore, option A is the right answer.

VIDEO SOLUTION

67. **1**
The frequency distribution is:

S: ~~3,3~~,3,4,4,4,5,5,6,7

F: ~~1,1~~,2,3,3,4,5,5,5,7

C: 1,2,2,2,~~3,3~~,3,3,4,6

or

S: 3,3,3,~~4,4~~,4,5,5,6,7

F: ~~1,1~~,2,3,3,4,5,5,5,7

C: 1,2,~~2,2,~~3,3,3,3,4,6

Given Amda and Cadella score is 7 each with identical in all parameters. So it can score either 3,1,3 in S,F,C respectively or 4,1,2 in S,F,C respectively. In both the cases, its score in F is 1.

VIDEO SOLUTION

68. **3**

Writing down the data with us in a table form.

	Asia	Europe	Rest of World	Total
Dheeraj	3	7	1	11
Samantha	0	9	4	13
Nitesh	1	6	12	19
Total	4	22	17	43

But this includes countries which have been visited by one person, two people and three people. Essentially, there is an overlap of countries.

It is given to us that 32 distinct countries were visited among the three people,

Using that we can write down the equations,

$I + II + III = 32$...(1)

$I + 2II + 3III = 43$...(2)

Where I, II, and III denote the number of countries visited exactly by 1 of them, 2 of them and all of them.

We are told that USA(ROW) is the only country which was visited by all three people, that means $III = 1$

Subtracting Equation 1 from 2 we get, $II + 2III = 11$

And since III=1 we get $II = 9$

So, we are looking for 9 countries visited by exactly 2 people, and 22 countries visited by exactly 1 person.

Using the information from the problem set, we can note down the following,

Combination	**Asia**	**EU**	**ROW**	**Total**
Dheeraj				
Samantha				
Nitesh				
Dheeraj and Nitesh	1(China)			
Dheeraj and Samantha		1(France)		
Samantha and Nitesh				
All Three	0	0	1(USA)	
Total				32

We can see that Dheeraj visited only 1 country in ROW so every other value in that column has to be zero for wherever Dinesh is present. And Samantha has not visited any country in Asia so every value with Samantha present in Asia should be zero as well.

Combination	Asia	EU	ROW	Total
Dheeraj			0	
Samantha	0			
Nitesh				
Dheeraj and Nitesh	1(China)	0	0	1
Dheeraj and Samantha	0	1(France)	0	1
Samantha and Nitesh	0			
All Three	0	0	1(USA)	
Total				32

We have the following distribution, we are looking for 9 countries visited by exactly 2 people. And we have identified 2 countries, that means, 7 were countries were visited by both Samantha and Nitesh only. And given that half the countries visited by both of them are in Europe, and total number of countries visited by both of them is 7(only two of them)+1(USA)=8, they visited 4 countries in Europe together and 3 countries in ROW together. And using that information, we can fill in the rest of the table as well. Since we know the number of countries visited by each of them in a particular continent.

Combination	Asia	EU	ROW	Total
Dheeraj	2	6	0	8
Samantha	0	4	0	4
Nitesh	0	2	8	10
Dheeraj and Nitesh	1(China)	0	0	1
Dheeraj and Samantha	0	1(France)	0	1
Samantha and Nitesh	0	4	3	7
All Three	0	0	1(USA)	1
Total	3	17	12	32

How many countries in Asia were visited by at least one of Dheeraj, Samantha and Nitesh is 3.

VIDEO SOLUTION

69. **C**

As the average age for OM area decreases in 2001, we can infer that the new faculty member joined OM on 1st April 2001. As his age was 25 on that date, his age on 1st April 2003 would be 25+2 = 27 years. Thus, the new faculty member would be 27 years old. Hence option C.

VIEW SOLUTION

70. **C**

The revenue of 50 units is 1000. So, the revenue per unit is 20.

The variable cost of 50 units is 700. So, the variable cost per unit is 14.

As fixed cost went up by 40 units, the total fixed cost (when number of units is less than 34) is 50+40 = 90

Let the minimum number of units that are needed to be produced to ensure that there is no loss be X.

So, $20X \leq 14X + 90$

Or $X \leq 15$

So, the least possible number of units to be produced to ensure no loss is 15.

VIDEO SOLUTION

71. **A**
The rate of growth is the slope of the line. During the third month, Geeta has a negative slope whereas all the others have a positive slope. So, the rate of growth of Geeta is the least. Option a) is the correct answer.

VIEW SOLUTION

72. **B**
The set's starting point is the conclusion that the temperature rises only when the AC is turned off.
When the AC is turned off, the temperature rises linearly, reaching the temperature at the time it was turned off in one hour.

The temperature rises in two instances,
From 0:00 to 0:30, it rises from 26 to 31, so by 1:00, it would have reached 36 degrees, meaning that the temperature at 0:00 would have been 36 degrees Celsius.

From 1:00 to 1:30, it rises from 26 to 30, so by 2:00, it would have reached 34 degrees, meaning the temperature at 1:00 would have been 34 degrees Celcius.

Calculating either of these would have been enough as we have the starting temperature at 23:00 at 38 degree celsius. Since the temperature decreases linearly, we can find the temperature at every instance using another data point.

Drawing out the temperature outside, we get:

Time	Temperature outside	Temperature inside	Mode of AC used
23:00	38	38	
23:30	37	32	
0:00	36	26	-
0:30	35	31	
1:00	34	26	-
1:30	33	30	
2:00	32	28	-

From this alone, we can answer the first question:
The AC was turned off for only two instances between 11:01 pm and 1:59 am:
Once at 0:00 and once at 1:00

Therefore, Option B is the correct answer.

VIDEO SOLUTION

73. **5**
The data can be tabulated as follows(approximately):

States	IPC Crimes	SLL Crimes	Other Crimes	Total Crimes
Telangana	3-4	14-15	6-7	24-25
Puducherry	1	0	30	31
Kerala	7-8	15-16	10-11	33-34
Haryana	3-4	25-26	9-10	37-38
Maharashtra	15-16	35-36	5-6	55-56
Tamilnadu	2-3	25-26	35-36	62-64
Goa	25-26	35-36	18-19	80
Karnataka	15-16	48-49	25-26	91
Delhi	63-64	35-36	42-43	142-143
West Bengal	0	520	0	520

	IPC crimes	SLL crimes	OtherCrimes
Delhi	*	*	*
Goa	*	4	*
Haryana	8	6	*
Karnataka	3	2	*
Kerala	*	9	*
Maharashtra	3	4	8
Puducherry	13	29	*
Tamil Nadu	11	7	*
Telangana	6	9	8
WestBengal	17	*	16

Rank of Delhi in IPC crimes category = 1, The rank of Karnataka and Maharashtra is 3(from table), then the rank of Goa can only be 2.

The rank of Telangana is 6 which has less |IPC crimes than Kerala, which means the rank of Kerala can be less than or equal to 5.

Now, there are two states with 3 ranks, so there will be no rank 4, there can only be rank 5 which is Kerala.

VIDEO SOLUTION

74. **A**

The data can be tabulated as follows(approximately):

States	IPC Crimes	SLL Crimes	Other Crimes	Total Crimes
Telangana	3-4	14-15	6-7	24-25
Puducherry	1	0	30	31
Kerala	7-8	15-16	10-11	33-34
Haryana	3-4	25-26	9-10	37-38
Maharashtra	15-16	35-36	5-6	55-56
Tamilnadu	2-3	25-26	35-36	62-64
Goa	25-26	35-36	18-19	80
Karnataka	15-16	48-49	25-26	91
Delhi	63-64	35-36	42-43	142-143
West Bengal	0	520	0	520

	IPC crimes	SLL crimes	OtherCrimes
Delhi	*	*	*
Goa	*	4	*
Haryana	8	6	*
Karnataka	3	2	*
Kerala	*	9	*
Maharashtra	3	4	8
Puducherry	13	29	*
Tamil Nadu	11	7	*
Telangana	6	9	8
WestBengal	17	*	16

From the table, the rank of Tamilnadu in other crimes is 2. The states which are not in the table will have crimes less than Telangana(i.e 24-25)

From the table the rank of Pudducherry in other crimes is 3.

VIDEO SOLUTION

75. **5**

The data can be tabulated as follows(approximately):

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States	IPC Crimes	SLL Crimes	Other Crimes	Total Crimes
Telangana	3-4	14-15	6-7	24-25
Puducherry	1	0	30	31
Kerala	7-8	15-16	10-11	33-34
Haryana	3-4	25-26	9-10	37-38
Maharashtra	15-16	35-36	5-6	55-56
Tamilnadu	2-3	25-26	35-36	62-64
Goa	25-26	35-36	18-19	80
Karnataka	15-16	48-49	25-26	91
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West Bengal	0	520	0	520

	IPC crimes	SLL crimes	OtherCrimes
Delhi	*	*	*
Goa	*	4	*
Haryana	8	6	*
Karnataka	3	2	*
Kerala	*	9	*
Maharashtra	3	4	8
Puducherry	13	29	*
Tamil Nadu	11	7	*
Telangana	6	9	8
WestBengal	17	*	16

The rank of Delhi in IPC crimes should be 1 because the states which are not in table cannot crime more than that of Telangana which is 24-25.

Similarly Delhi Rank in Other crimes will be 1.

Now in SLL crimes clearly West Bengal has rank 1. It is given that Karnataka has rank 2. The rank 3 can go to either Goa, Delhi and Maharashtra but Goa and Maharashtra already have rank 4. So Delhi will have rank 3. Also no state outside of the table can be ranked 3 in SLL crimes as maximum number of crime should be less than that of Telangana(24-25). Here the number of SLL crimes is 35-36.

Hence the sum of the ranks = 1+3+1=5

VIDEO SOLUTION

Cracku CAT Success Stories

76. **A**
After 1991 each year there is a positive growth rate change for manufacturing sector. Hence highest level of production would be in the year 1998.

VIEW SOLUTION

77. **A**
The rate of growth is the slope of the line.

From the graphs, we can see that the slope of the line is maximum for Geeta (red line) in the first two months. So, option a) is the correct answer.

VIEW SOLUTION

78. **40**
The total number of male and female test cases in 1980 = 1000

Age group(In 1980)	Males alive in 1990	Males alive in 2000	Males alive in 2010	Males alive in 2020
10-20	230	180	150	140
20-30	235	205	165	125
30-40	210	160	90	50
40-50	190	100	25	5

Age group(In 1980)	Females alive in 1990	Females alive in 2000	Females alive in 2010	Females alive in 2020
10-20	240	210	160	100
20-30	225	175	145	105
30-40	190	150	100	60
40-50	220	120	30	18

The total number of males between 20 and 30 years of age in 1980 who died in 2000 = 205

The total number of males between 20 and 30 years of age in 1980 who died in 2010 = 165

Thus, the total number of males between 20 and 30 years of age in 1980 who died in the period 2000 to 2010 = 205 - 165 = 40

VIDEO SOLUTION

79. **2**

Writing down the data with us in a table form.

	Asia	Europe	Rest of World	Total
Dheeraj	3	7	1	11
Samantha	0	9	4	13
Nitesh	1	6	12	19
Total	4	22	17	43

But this includes countries which have been visited by one person, two people and three people. Essentially, there is an overlap of countries.
It is given to us that 32 distinct countries were visited among the three people,
Using that we can write down the equations,
$I + II + III = 32$...(1)
$I + 2II + 3III = 43$...(2)
Where I, II, and III denote the number of countries visited exactly by 1 of them, 2 of them and all of them.
We are told that USA(ROW) is the only country which was visited by all three people, that means $III = 1$

Subtracting Equation 1 from 2 we get, $II + 2III = 11$
And since III=1 we get $II = 9$

So, we are looking for 9 countries visited by exactly 2 people, and 22 countries visited by exactly 1 person.
Using the information from the problem set, we can note down the following,

Combination	Asia	EU	ROW	Total
Dheeraj				
Samantha				
Nitesh				
Dheeraj and Nitesh	1(China)			
Dheeraj and Samantha		1(France)		
Samantha and Nitesh				
All Three	0	0	1(USA)	
Total				32

We can see that Dheeraj visited only 1 country in ROW so every other value in that column has to be zero for wherever Dinesh is present. And Samantha has not visited any country in Asia so every value with Samantha present in Asia should be zero as well.

Combination	Asia	EU	ROW	Total
Dheeraj			0	
Samantha	0			
Nitesh				
Dheeraj and Nitesh	1(China)	0	0	1
Dheeraj and Samantha	0	1(France)	0	1
Samantha and Nitesh	0			
All Three	0	0	1(USA)	
Total				32

We have the following distribution, we are looking for 9 countries visited by exactly 2 people. And we have identified 2 countries, that means, 7 were countries were visited by both Samantha and Nitesh only. And given that half the countries visited by both of them are in Europe, and total number of countries visited by both of them is 7(only two of them)+1(USA)=8, they visited 4 countries in Europe together and 3 countries in ROW together. And using that information, we can fill in the rest of the table as well. Since we know the number of countries visited by each of them in a particular continent.

Combination	Asia	EU	ROW	Total
Dheeraj	2	6	0	8
Samantha	0	4	0	4
Nitesh	0	2	8	10
Dheeraj and Nitesh	1(China)	0	0	1
Dheeraj and Samantha	0	1(France)	0	1
Samantha and Nitesh	0	4	3	7
All Three	0	0	1(USA)	1
Total	3	17	12	32

Number of countries visited by only Nitesh is 2.

VIDEO SOLUTION

80. **D**

Since one was born in 1947 and the other one in 1950 and have birthday on 20th November, on 1st April 2000 they will have age 49 and 52 respectively.

Average age of all three on 1st April 2000 = 49.33 => Total age = 49.33*3 =148

Hence, age of the third member on 1st April 2000 = 148 - 49 - 52 = 47

Age of the third member on 1st April 2005 = 47+5 = 52.

Hence option D.

VIEW SOLUTION

81. **C**

The difference is maximum when one quantity is maximum and other quantity is minimum. The percentage increase in production of food grains is maximum in 1995 and the percentage production increase in production of milk is negative for 1995.

VIEW SOLUTION

82. **A**

Purana mixer-grinders were purchased in 1999 = 236 - 222 +Disposed in 1999

= 236-222 + 20%(Purana mixer-grinders were purchased in 1997)

= 236 - 222 + 20%(182 - 162 + 10) = 236- 222 + 6 = 20.

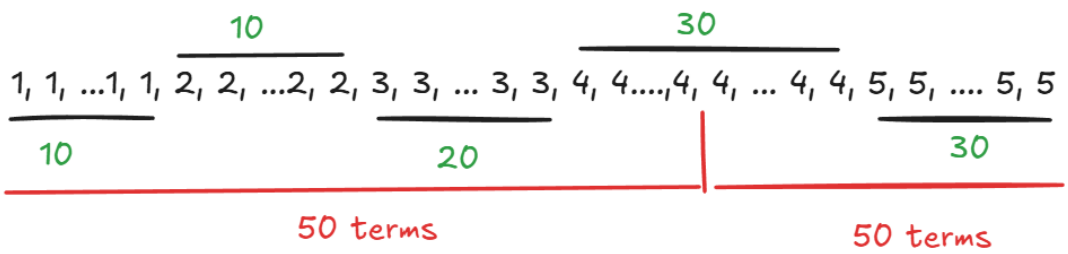
Hence option D.

VIEW SOLUTION

83. **4**

As deduced in the previous question, there were 10 1 ratings, 10 2 ratings, 20 3 ratings, 30 4 ratings, and 30 5 ratings.

The median would be the 50th and 51st ratings' average when arranged in ascending/descending order. Both of which would be 4



Therefore, 4 is the correct answer

VIDEO SOLUTION

84. **10**

The set's starting point is the conclusion that the temperature rises only when the AC is turned off.
When the AC is turned off, the temperature rises linearly, reaching the temperature at the time it was turned off in one hour.

The temperature rises in two instances,
From 0:00 to 0:30, it rises from 26 to 31, so by 1:00, it would have reached 36 degrees, meaning that the temperature at 0:00 would have been 36 degrees Celsius.

From 1:00 to 1:30, it rises from 26 to 30, so by 2:00, it would have reached 34 degrees, meaning the temperature at 1:00 would have been 34 degrees Celcius.

Calculating either of these would have been enough as we have the starting temperature at 23:00 at 38 degree celsius. Since the temperature decreases linearly, we can find the temperature at every instance using another data point.

Drawing out the temperature outside, we get:

Time	Temperature outside	Temperature inside	Mode of AC used
23:00	38	38	
23:30	37	32	
0:00	36	26	-
0:30	35	31	
1:00	34	26	-
1:30	33	30	
2:00	32	28	-

(Side note: We cannot consider the temperatures outside and inside at 23:00 and 2:00 since the time limit does not include those.)

The maximum difference between the elements of the outside and inside is maximum at 0:00 when the outside temperature is 36, and the inside temperature is 26.

Giving the maximum difference as 10

Therefore,10 is the correct answer.

VIDEO SOLUTION

85. **B**
Since the increase in imports and exports is at average, the exports and imports in the next 4months will be half of their respective first eight months.

Exports in first 8 months of 1998-1999 is $21436 so, in next 4 months it will be 21436/2 = 10718

Total exports= 21346+10718=32154.

Similarly, total imports= 28126+ 28126/2 = 42189

Trade deficit = 42189-32064= 10035

Trade deficit in 1997-98= 40779-33979=6800.

Increase = (10035-6800)/6800 = 47.5%

VIDEO SOLUTION

86. **B**

Till 34 units, the fixed cost is constant and the revenue increasing per every unit made.
So, to calculate the number of units that need to be produced to maximize the profit, we have to check only at two points: 34 units and 40 units.

The revenue of 50 units is 1000. So, the revenue per unit is 20.
The variable cost of 50 units is 700. So, the variable cost per unit is 14.

If number of units produced is 34,
Fixed cost is 50
Variable cost is 14*34 = 476
Revenue is 20*34 = 680.
So, profit is 680-476-50 = 154

If number of units produced is 40,
Fixed cost is 100
Variable cost is 14*40 = 560
Revenue is 20*40 = 800.
So, profit is 800-560-100 = 140

So, the profit is maximized when the number of units produced is 34

VIDEO SOLUTION

87. **C**
If the costs per tonne of transport by ship, air and road are represented by P, Q and R respectively.

P = 10/9

Q = 7/11

R = 6/22

We can see that P>Q and Q>R and P>R. Hence option C.

VIEW SOLUTION

88. **B**
If suppose effort allocation is inter-changed between operations B and C, then C and D, and then D and E then operation E will have value of B . So arranging effort in E in ascending order we have company 4 ,company 5 , company 3, ... Hence company 3 is ranked third. Hence option B.

VIEW SOLUTION

89. **B**
Total number of hours needed for testing = 300+150 = 450
New number of hours spend offshore for testing = 300/2 = 150
So, required percentage = 150/450 * 100% = 33.33%
From the options, the closest answer is option b)

VIEW SOLUTION

90. **C**
Total number of man-hours spent onsite = 75+100+150 = 325
The sum of the estimated and actual effort for offshore design = 200 (approx)
The estimated man-hours of offshore coding = 425
The actual man-hours of offshore testing = 300 (approx)
Half of the man-hours of estimated offshore coding = 210(approx)
So, option c) is the closest

VIEW SOLUTION

91. **C**
Total number of man-hours needed for the entire work to be done = 100+75+425+100+300+150 = 1150
Man-hours spent onsite = 75+100+150 = 325
So, required percentage = 325/1150 * 100% = 13/46 * 100% = 28.26% = 30% approx

VIEW SOLUTION

92. **C**
There are 2 steel companies with turnover of more than 2000 and having profit less than 300.

VIEW SOLUTION

93. **2**
We can note down the data from the chart into the form of a table, that gives us,

Surfer/Blogger	A	B	C	D	Total
M	10	0			30
N	25	0			30
O	0	0			30
P	5	25			30
X	0	0			30
Y	5	20			30
Total	45	45	45	45	180

We are told that D receives more stars than C from Y. Considering Y has already given 25 stars, it will give 0 stars to C and 5 stars to D.
The only two surfers who have not given any stars to A or B is O and X, and these are the two surfers to give all of their stars to a single blogger.

We are also told that M gives different stars to the four bloggers,
Since he has already given 0 and 10, the remaining distinct stars should add up to 20. The only numbers that are remaining that add up to 20 are 5 and 15.
We know that X rewards one of C or D 30 stars and O rewards one of C or D 30 stars. Given that, M could not have rewarded D 15 stars, since Y rewarded D 5 stars, and D is also going to rewarded 30 stars by O or X, and since the total is same for all, which is 45. This is not possible.
This means that, M rewarded C 15 stars and D 5 stars.
This gives us two cases,
Case-1

Surfer/Blogger	A	B	C	D	Total
M	10	0	15	5	30
N	25	0	0	5	30
O	0	0	30	0	30
P	5	25	0	0	30
X	0	0	0	30	30
Y	5	20	0	5	30
Total	45	45	45	45	180

Case-2

Surfer/Blogger	A	B	C	D	Total
M	10	0	15	5	30
N	25	0	0	5	30
O	0	0	0	30	30
P	5	25	0	0	30
X	0	0	30	0	30
Y	5	20	0	5	30
Total	45	45	45	45	180

We can use these two cases to answer the questions,

M distributed among 3 bloggers, N among 2 bloggers, O among 1, P among 2, X among 1, Y among 3

Hence 2 surfers distribute their stars among 2 bloggers.

VIDEO SOLUTION

94. **D**

In 1993, price of logs = Rs. 20 per cubic metre.

volume sales of plywood, saw timber and logs were 40%, 30% and 30%

In 1993, price of plywood = Rs. 7 per tonne

In 1993, price of plywood = Rs. 20 per tonne

Weight of one cubic metre of saw dust and plywood both = 800 kg

1 tonne=1000kg

Price of 1000kg of plywood= Rs 7

Price of 800kg(1 m^3)= 7 x 0.8 = Rs 5.6 per cubic metre.

Price of saw timber = 20 x 0.8 = Rs 16 per cubic metre.

We know that the sales volume of plywood, saw timber and logs are in the ratio 4 : 3 : 3.
Thus, the average realization must be the weighted avg of these three-

$\frac{(4\times5.6)+(3\times16)+(3\times20)}{4+3+3}$

=13.1

= Rs. 13 (Approximately)

VIEW SOLUTION

95. **A**
To answer this question, let us calculate the highest percentage increase in market value of each share.

For share A, the highest percentage increase in market value is from Feb to March and equals 110/90 -1 = 22.22%
For share B, the highest percentage increase in market value is from Jan to Feb and equals 75/70 -1 = 7.14%
For share C, the highest percentage increase in market value is from Feb to March and equals 60/55 -1 = 9.09%
For share D, the highest percentage increase in market value is from Jan to Feb and equals 50/40 -1 = 25%

Hence, the required answer is February and the required share is share D

VIEW SOLUTION

96. **A**
Suppose there are 100 subscribers from europe in the year 2003.

Out of which 60 are men and 40 are women.

According to given condition in the year 2010 no. of male subscribers would be 60*1.05^7 = 84 and no. of female subscribers would be 40*1.1^7 = 78.

So total equal to 162.

Hence % increase is 100*(162-100)/100 = 62.

Hence option A.

VIEW SOLUTION

97. **D**
Assume total base to be 100 (index = 100).
Contribution of manufacturing sector = 20% = 20 (in 1989).
In 1990, it has grown by 9%.
=> Contribution in 1990 = 1.09*20 = 20 + 1.8 = 21.8.

Hence, the growth of manufacturing sector in 1990 is 21.8 . Similarly for other sectors the growth rates fpr mining 15.6 , electricity 10.9 and chemicals is 16.2 . So total overall groth is 64.5 %. </br>
Growth of manufacturing sector in 199 is 21.8 - 0.01*21.8 = 21.6 . Similarly for other sectors the growth rates fpr mining 15.7 , electricity 11.88 and chemicals is 16.36 . So total overall groth is 65.54 %. Hence the growth rate change in 1991 is (65.54-64.5) / 64.5 = 1.5%.
Hence option D.

VIEW SOLUTION

98. **C**
We know that the Excise duty is levied on the total value of liquor produced by the 5 distilleries. Since the excise duty remains constant, the excise duty will be in the same order as the order of the amount of the liquor produced by them.
Thus, the correct order is DCEBA.

VIEW SOLUTION

99. **A**

Loss from the transaction from Ram's standpoint is Sum of share prices of C and D - Share price of A.

So, loss from the transaction at the end of January is (40+60) - 100 - = 0
So, loss from the transaction at the end of February is (50+55) - 90= 15
So, loss from the transaction at the end of March is (60+50)-110 = 0
So, loss from the transaction at the end of April is (65+40) - 105 = 0
So, loss from the transaction at the end of May is (60+45) -100 = 5
So, loss from the transaction at the end of June is (55+45) - 110 = 0

So, the correct answer is 15

VIEW SOLUTION

100. **D**
Loss from the transaction from Ram's standpoint is Sum of share prices of C and D - Share price of A.

So, loss from the transaction at the end of January is (40+60) - 100 - = 0
So, loss from the transaction at the end of February is (50+55) - 90= 15
So, loss from the transaction at the end of March is (60+50)-110 = 0
So, loss from the transaction at the end of April is (65+40) - 105 = 0
So, loss from the transaction at the end of May is (60+45) -100 = 5
So, loss from the transaction at the end of June is (55+45) - 110 = 0

So, the correct answer is February.

VIEW SOLUTION

101. **A**
To answer this question, let us calculate the highest increase in market value of each share.

For share A, the highest increase in market value is from Feb to March and equals 110 - 90 = 20
For share B, the market value increases constantly over all the months and is equal to 75-70 = 5
For share C, the highest increase in market value is from Feb to March and equals 60 - 55 = 5
For share D, the highest increase in market value is from Jan to Feb and equals 50 - 40 = 10

Hence, the correct answer is March and the required share is share A

VIEW SOLUTION

102. **B**
S: ~~3~~,~~3~~,3,4,4,4,5,5,~~6~~,7

F: ~~1~~,~~1~~,2,3,3,4,5,5,5,~~7~~

C: 1,2,2,2,~~3~~,~~3~~,3,3,~~4~~,6

Benga and Delma, two countries categorized as happy, are tied with the same total score.

The best numbers remaining are 7,5,6 which adds upto 18, If Benga scores 18, then Delma can't score 18.

Similarly both can't score 17 and 16. Both can score 15 and their distribution will be:

Benga: 7,5,3

Delma: 4,5,6 or 5,4,6

VIDEO SOLUTION

103. **B**
S: ~~3~~,~~3~~,3,4,4,4,5,5,~~6~~,7

F: ~~1~~,~~1~~,2,3,3,4,5,5,5,~~7~~

C: 1,2,2,2,~~3,3~~,3,3,~~4~~,6

Given that Benga scores 16, and Delma scores 15.

The possibility is Benga: 5,5,6 and Delma: 7,5,3

If Benga's distribution is 7,3,6 then Delma can't score 15.

Strike off those numbers.

S: ~~3,3~~,3,4,4,4,5,~~5,6,7~~

F: ~~1,1~~,2,3,3,4,5,~~5,5,7~~

C: 1,2,2,2,~~3,3~~,3,~~3,4,6~~

We have to maximum number of countries with score 13. This score does not comes under the category of happy. So to score 13, the distribution can be 5,5,3. Hence, maximum 1 country is possible.

VIDEO SOLUTION

104. **D**
The set's starting point is the conclusion that the temperature rises only when the AC is turned off.
When the AC is turned off, the temperature rises linearly, reaching the temperature at the time it was turned off in one hour.

The temperature rises in two instances,
From 0:00 to 0:30, it rises from 26 to 31, so by 1:00, it would have reached 36 degrees, meaning that the temperature at 0:00 would have been 36 degrees Celsius.

From 1:00 to 1:30, it rises from 26 to 30, so by 2:00, it would have reached 34 degrees, meaning the temperature at 1:00 would have been 34 degrees Celcius.

Calculating either of these would have been enough as we have the starting temperature at 23:00 at 38 degree celsius. Since the temperature decreases linearly, we can find the temperature at every instance using another data point.

Drawing out the temperature outside, we get:

Time	Temperature outside	Temperature inside	Mode of AC used
23:00	38	38	
23:30	37	32	
0:00	36	26	-
0:30	35	31	
1:00	34	26	-
1:30	33	30	
2:00	32	28	-

We can see that the AC was turned on at 0:30, 1:30 and 23:00.
We do not know about the status at 23:30

We are to count the times after 11:01 pm; hence, turning on the AC at 23:00 does not count.

Hence, the answer should be 2 or 3.

But in the final clue, we are given that POWER mode was used longer than REGULAR mode.
Since there are 4 periods of 30 minutes each, the combinations possible are POWER-3, REGULAR-1, and POWER-4, REGULAR-0

If the AC were on REGULAR mode at 23:00, it would have functioned for 1 hour, and the power mode would have also functioned for 1 hour (total), which would go against the final clue.

The following would be the cases.

	Case-1	Case - 2	Case - 3	Case - 4	Case - 5
23:00	Regular	Power	Power	Power	Power
23:30	Power	Regular	Power	Power	Power
0:30	Power	Power	Regular	Power	Power
1:30	Power	Power	Power	Regular	Power

But there were three instances of activation between 11:01 and 1:59

Hence, Option D would be the correct answer.

VIDEO SOLUTION

105. **B**
%increase in profit in year 1995 = 16.66%
With this increment, profit in year 1996 = 70(1+$(16.66 \div 100)$) = 82 (nearly)

VIEW SOLUTION

106. **B**
Number of Naya mixer-grinders were disposed off by the end of 1999 are 6 and at the end of 2000 are 10. So in total 16 naya mixer disposed off.

VIEW SOLUTION

107. **A**
We are given that on day 3, a total of 100 ratings came in

The modes were 4 and 5, meaning that an equal number of 4 and 5 ratings came in; let it be 10b (since we are given that all ratings were non-zero multiples of 10)

Let's take the number of 1 and 2 ratings as 10a each, giving the number of 3 ratings as 20a

Adding all of these up: 10a+10a+20a+10b+10b = 100
40a+20b = 100
2a+b = 5

The only integer combination for a and b, without them being zero, is a being 1 and b being 3 or a=2 and b=1

But taking b as 1 would make 3 as the mode rating, so we can not consider the latter case.

We are giving 10 - 1 ratings, 10 - 2 ratings, 20 - 3 ratings, 30 - 4 ratings, and 30 - 5 ratings.

The average would be $\frac{(10\times 1)+(10\times 2)+(20\times 3)+(30\times 4)+(30\times 5)}{100} = \frac{360}{100} = 3.6$

Therefore, Option A is the correct answer.

VIDEO SOLUTION

108. **D**
Every year, there was a comparative increase in both US and Europe subscriptions.

But in 2009, the increase in US subscriptions is very marginal and in Europe subscriptions is similar to that of any other year.

Hence, the percentage change in 2009 over 2008 would be more than any other year.

VIEW SOLUTION

109. **B**
If she drives at 40 kmph, the fuel consumed = 200/40 * 2.5 = 12.5 liters
If she drives at 60 kmph, the fuel consumed = 13.33 liters
If she drives at 80 kmph, the fuel consumed = 200/80 * 7.9 = 19.75 liters
To minimize fuel consumption, Manasa has to decrease the speed.

VIEW SOLUTION

110. **B**
Till 34 units, the fixed cost is constant and the revenue increasing per every unit made.

So, to calculate the number of units that need to be produced to maximize the profit, we have to check only at two points: 34 units and 45 units.

The revenue of 50 units is 1000. So, the revenue per unit is 20.
The variable cost of 50 units is 700. So, the variable cost per unit is 14.

If number of units produced is 34,
Fixed cost is 50
Variable cost is 14*34 = 476
Revenue is 20*34 = 680.
So, profit is 680-476-50 = 154

If number of units produced is 45,
Fixed cost is 100
Variable cost is 14*45 = 630
Revenue is 20*45 = 900.
So, profit is 900-630-100 = 170

So, the profit per unit is maximized when the number of units produced is 34

VIDEO SOLUTION

111. **A**
The total number of male and female test cases in 1980 = 1000

Age group(In 1980)	Males alive in 1990	Males alive in 2000	Males alive in 2010	Males alive in 2020
10-20	230	180	150	140
20-30	235	205	165	125
30-40	210	160	90	50
40-50	190	100	25	5

Age group(In 1980)	Females alive in 1990	Females alive in 2000	Females alive in 2010	Females alive in 2020
10-20	240	210	160	100
20-30	225	175	145	105
30-40	190	150	100	60
40-50	220	120	30	18

The total number of males alive in 2000 = 180 + 205 + 160 + 100 = 645

Thus, the number of dead males in 2000 = 1000 - 645 = 355

Similarly, the total number of dead females in 2000 = 1000 - (210 + 175 + 150 + 120) = 1000 - 655 = 345

Thus, the required ratio = 355 : 345 = 71 : 69.

Thus, the correct option is A.

VIDEO SOLUTION

112. **C**

The total number of male and female test cases in 1980 = 1000

Age group(In 1980)	Males alive in 1990	Males alive in 2000	Males alive in 2010	Males alive in 2020
10-20	230	180	150	140
20-30	235	205	165	125
30-40	210	160	90	50
40-50	190	100	25	5

Age group(In 1980)	Females alive in 1990	Females alive in 2000	Females alive in 2010	Females alive in 2020
10-20	240	210	160	100
20-30	225	175	145	105
30-40	190	150	100	60
40-50	220	120	30	18

The total number of males less than 30 years (in 1980) survived until 2020 = 140 + 125 = 265

The total number of females less than 30 years (in 1980) survived until 2020 = 100 + 105 = 205

Thus, The total number of people less than 30 years (in 1980) survived until 2020 = 205 + 265 = 470

Thus, the correct option is C.

VIDEO SOLUTION

113. **C**

The average age in the Finance area dropped twice - once due to the retirement of a member and once due to the joining of a new member. We see that the average age of the faculty is around 45-50. This number is closer to the retirement age of 60 than to the joining age of 25. Hence, the new joining will produce a greater drop in average age than the retirement. Hence, the new faculty member must have joined in 2002 and the old faculty member must have retired in 2001. Hence option C.

VIEW SOLUTION

114. **A**

Average of numbers 100.5 , 67, 141, 143.9 and 65 will be

$$100 + \frac{41+0.5+43.9-33-35}{5} = 103.48$$

VIEW SOLUTION

115. **B**

Production rate in 1990 = 15.6 ; Production rate in 1991 = 15.7 ; Production rate in 1992 = 15.85 ; Production rate in 1993 = 15.46. So clearly we can see that production rate would be lowest in the year 1993. Hence option B.

VIEW SOLUTION

116. **B**

Ranking of UP changed 2 times which is highest among other states.

VIEW SOLUTION

117. **C**

The four major trading partners with India in 1997-98 are A, G, H and I.
From the numbers, we can ignore the others for this questions.

Total trade with A in the year 1997-98 is 9%*40779 + 19%*33979 = 10,126
Total trade with G in the year 1997-98 is 19%*40779 + 10%*33979 = 11,146
Total trade with H in the year 1997-98 is 23%*40779 + 10%*33979 = 12,777
Total trade with I in the year 1997-98 is 14%*40779 + 20%*33979 = 12,505
Hence, the correct option is block H which is OPEC

VIDEO SOLUTION

118. **B**

Decline occur in years 1985 and 1988 respectively.
In year 1985, decline was nearly 33%
Whether in year 1988 decline was nearly 54%
Hence answer will be B.

VIEW SOLUTION

119. **B**

Average simple annual growth of the distilleries are-

A-> $\frac{12.89-6.41}{2\times6.41} \times 100 = 50.54$%
B-> $\frac{12.07-3.15}{2\times3.15} \times 100 = 141.58$%
C-> $\frac{11.92-1.64}{2\times1.64} \times 100 = 313.41$%
D-> $\frac{5.79-1.05}{2\times1.05} \times 100 = 225.71$%

E-> $\frac{4.21-2.45}{2\times2.45} \times 100 = 35.91$%

We can see that the highest growth rate was in C and lowest in E.
So, if E would have grown by 313.41% in 2 years, its supply would have been-

$\frac{2.45\times313.41\times2}{100} = 15.11$ lts

VIEW SOLUTION

120. **D**

In the question we are only provided with the data of five states. We don't know the excise duty rates for other states. Thus, the answer cannot be determined.

VIEW SOLUTION

121. **B**
The difference in the excise duty as a percentage is least amongst AP and Maharashtra and it equals (60-52)/52 * 100
This equals 8/52*100 = 200/13 = 15.38% which is approximately equal to 15%

VIEW SOLUTION

122. **B**
Ratio in 1991 = $\frac{10}{120}$ = 0.833

Ratio in 1992 = $\frac{25}{125}$ = 0.2

Ratio in 1993 = $\frac{20}{145}$ = 0.138

Ratio in 1995 = $\frac{20}{160}$ = 0.125

1992 has maximum profit per unit cost.

VIEW SOLUTION

123. **C**
Total Profit = 10 + 25 + 20 + 0 + 20 = 75

Total Cost = 170 + 175 + 195 + 200 + 210 = 950

% of Profit = $\frac{75}{950}$ = 8%

VIEW SOLUTION

124. **C**
Ratio in 1992 = $\frac{25}{50}$ = 0.5

Ratio in 1993 = $\frac{25}{60}$ = 0.4167

Ratio in 1994 = $\frac{30}{70}$ = 0.428

Ratio in 1995 = $\frac{30}{80}$ = 0.375

Hence, overheads, as a percentage of raw material, is maximum in 1992.

VIEW SOLUTION

125. **A**
Interest has remained constant over the given years attaining the value of 50-60 in all the year. Only in the year 1994, the value of interest is little less.

VIEW SOLUTION

126. **C**

The profits of the company became zero only in one year (1994)
Hence, the change in profit is maximum in the period 1993-94 and equals 100%.
Note that there was a decrease in profits in other years also, but the maximum possible change is 100% and it was occurred in 1993-94

VIEW SOLUTION

127. **C**
Ratio of nutrient in milk(1 gallon) and foodgrains(1 tonne) is 3:2

There is a huge spike in the consumption of food grains in 1995.

Total nutrients consumed in 1995 = 30*80 + 7*120 = 3240 million gms.

This is more than any other year.

=> 1995 is the answer

VIEW SOLUTION

128. **C**
The maximum availability of the nutrient will be either in 1994 or 1995. because the cumulative production of milk and food grains is maximum in these years.

Availability in 1994 = (120*8) + (25*80) = 2960

Availability in 1995 = (6*120) + (31*80) = 3200

Hence the availability is maximum in 1995.

VIEW SOLUTION

129. **C**
The consumption of milk is comparatively lesser than that of food grains and also the consumption of milk does not change by much during the given period.

Hence, the change in food grains is the key to estimate the calorie count.

We can see the spike in consumption of food grains in 1995. Hence, percapita consumption is highest in 1995.

VIEW SOLUTION

130. **B**
Number of students in 1994 passed out = 1100.

Number of students in 1994 from finance = 23% of 1100 = 23*11

Number of students in 1994 from software = 21% of 1100 = 21*11

Difference in salary = 12(23*11*7550-21*11*7050) = 3379200 = 33.8 lakhs

VIEW SOLUTION

131. **D**
As we do not know the average salary for students who did not pursue Finance, Marketing or Software jobs, we cannot find the average salary of all the students in 1993.

Hence, a unique value cannot be determined.

VIEW SOLUTION

132. **D**

Finance salary increase = 9810 - 5450 = 4360

% increase = $\frac{4360}{5450} * 100$ = 80%

VIEW SOLUTION

133. **C**

We can see that the end value is a little more than 15000 but not more than 16000.

So the increase in number of employees is between 5000 to 6000.

According to the options, it is 5800.

VIEW SOLUTION

134. **D**

This percentage is maximum in May which is equal to $\frac{36-32}{32} = 1/8/$

VIEW SOLUTION

135. **D**

%increase in expenditure in year 1990 = above 7 but lower than 8
in year 1991 = near to 4
in year 1992 = above 8
in year 1993 = 8%
in year 1994 = lower than 4
in year 1995 = above 7 but lower than 8
Hence highest percentage increase is in year 1992

VIEW SOLUTION

136. **D**

percentage increase in profit in year 1990 = 25%
in 1991 = 20%
in 1992 = 33.33%
in 1993 = 25%
in 1994 = 20%
in 1995 = 16.66%

So highest %increase is in year 1992.

VIEW SOLUTION

137. **A**

Total expenditure = 102 + 110 + 115 + 125 + 135 + 140 + 150 = 877
Total revenue = 120 + 130 + 145 + 165 + 185 + 200 + 220 = 1165
%expenditure to revenue = $\frac{877}{1165} \times 100$ = 75 (nearly)

VIEW SOLUTION

138. **C**

The cumulative average rating at the end of day 3 would be $\frac{(3.1\times 200)+(3.6\times 100)}{200+100}$

[the cumulative average rating on day 2= 3.1
Number of ratings received by day 2= 200]

$\frac{360+620}{300} = \frac{980}{300} = 3.266$

The increase in the cumulative average from day 2 to day 3 can be calculated as $\frac{3.266-3.1}{3.1} \times 100 \approx 5.34$

Which aligns with the statement given in option C.

Therefore, Option C is the correct answer.

VIDEO SOLUTION

139. **C**

Reading the bar graphs, we can find the distribution of the subscribers as:

	Kids	Elders	Others
2023	15%	20%	65%
2024	20%	30%	50%

Taking clue one into consideration, let's take the total number of subscribers in 2023 as 100x and in 2204 as 110x
This would give the split of subscribers as follows:

	Kids	Elders	Others	
2023	15x	20x	65x	100x
2024	22x	33x	55x	110x

Clue 2 says that half of 22x, which is 11x and 2/3 of 33x, which is 22x, used one app, given that in 2024, out of 55x, 33x used one app.

Clues 3 and 4 are to be used with each other.
Clue 4 says that out of 15x kids, 10,000 used one app, and out of 20x Elders, 15,000 used multiple apps.
Clue 3 says that the kids who used multiple apps (15x - 10000) were the same as Elders who used one app (20x - 15000)

Equating these two we get:
15x- 10000 = 20x - 15000
5x = 5000
x = 1000

The first question asks about the number of others in 2024, which is 55x, or simply 55,000

Therefore, Option C is the correct answer.

VIDEO SOLUTION

140. **A**

Reading the bar graphs, we can find the distribution of the subscribers as:

	Kids	Elders	Others
2023	15%	20%	65%
2024	20%	30%	50%

Taking clue one into consideration, let's take the total number of subscribers in 2023 as 100x and in 2204 as 110x
This would give the split of subscribers as follows:

	Kids	Elders	Others	
2023	15x	20x	65x	100x
2024	22x	33x	55x	110x

Clue 2 says that half of 22x, which is 11x and 2/3 of 33x, which is 22x, used one app, given that in 2024, out of 55x, 33x used one app.

Clues 3 and 4 are to be used with each other.
Clue 4 says that out of 15x kids, 10,000 used one app, and out of 20x Elders, 15,000 used multiple apps.
Clue 3 says that the kids who used multiple apps (15x - 10000) were the same as Elders who used one app (20x - 15000)

Equating these two, we get:
15x- 10000 = 20x - 15000
5x = 5000
x = 1000

There are 15x, that is, 15,000 kids in 2023, of which we are given that 10,000 use one app.
So, 5,000 use multiple apps.
The percentage of kids using multiple apps in 2023 would hence be $\frac{5000}{15000} \times 100 = \frac{100}{3} = 33.33\%$

Therefore, Option A is the correct answer.

VIDEO SOLUTION

141. **C**
Reading the bar graphs, we can find the distribution of the subscribers as:

	Kids	Elders	Others
2023	15%	20%	65%
2024	20%	30%	50%

Taking clue one into consideration, let's take the total number of subscribers in 2023 as 100x and in 2204 as 110x
This would give the split of subscribers as follows:

	Kids	Elders	Others	
2023	15x	20x	65x	100x
2024	22x	33x	55x	110x

Clue 2 says that half of 22x, which is 11x and 2/3 of 33x, which is 22x, used one app, given that in 2024, out of 55x, 33x used one app.

Clues 3 and 4 are to be used with each other.
Clue 4 says that out of 15x kids, 10,000 used one app, and out of 20x Elders, 15,000 used multiple apps.
Clue 3 says that the kids who used multiple apps (15x - 10000) were the same as Elders who used one app (20x - 15000)

Equating these two, we get:
15x- 10000 = 20x - 15000
5x = 5000
x = 1000

In 2023, there were 20,000 elders; in 2024, there were 33,000 elders.
The increase in percentage would be $\frac{33000-20000}{20000} \times 100 = \frac{13}{20} \times 100 = 65\%$

Therefore, Option C is the correct answer.

VIDEO SOLUTION

142. **B**
Reading the bar graphs, we can find the distribution of the subscribers as:

	Kids	Elders	Others
2023	15%	20%	65%
2024	20%	30%	50%

Taking clue one into consideration, let's take the total number of subscribers in 2023 as 100x and in 2204 as 110x
This would give the split of subscribers as follows:

	Kids	Elders	Others	
2023	15x	20x	65x	100x
2024	22x	33x	55x	110x

Clue 2 says that half of 22x, which is 11x and 2/3 of 33x, which is 22x, used one app, given that in 2024, out of 55x, 33x used one app.

Clues 3 and 4 are to be used with each other.
Clue 4 says that out of 15x kids, 10,000 used one app, and out of 20x Elders, 15,000 used multiple apps.
Clue 3 says that the kids who used multiple apps (15x - 10000) were the same as Elders who used one app (20x - 15000)

Equating these two, we get:
15x- 10000 = 20x - 15000
5x = 5000
x = 1000

In clue 2, we were given that 33000 kids and elders out of 55000 kids and elders use one app, showing that 22000 use multiple apps. To minimize the total usage of multiple apps, we can assume that all 55000 other subscribers use one app.

Giving the percentage of multiple app users as: $\frac{22,000}{110,000} \times 100 = \frac{100}{5} = 20\%$

Therefore, Option B is the correct answer.

VIDEO SOLUTION

143. **34**
The set's starting point is the conclusion that the temperature rises only when the AC is turned off.
When the AC is turned off, the temperature rises linearly, reaching the temperature at the time it was turned off in one hour.

The temperature rises in two instances,
From 0:00 to 0:30, it rises from 26 to 31, so by 1:00, it would have reached 36 degrees, meaning that the temperature at 0:00 would have been 36 degrees Celsius.

From 1:00 to 1:30, it rises from 26 to 30, so by 2:00, it would have reached 34 degrees, meaning the temperature at 1:00 would have been 34 degrees Celcius.

Calculating either of these would have been enough as we have the starting temperature at 23:00 at 38 degrees Celsius. Since the temperature decreases linearly, we can find the temperature at every instance using another data point.

Drawing out the temperature outside, we get:

Time	Temperature outside	Temperature inside	Mode of AC used
23:00	38	38	
23:30	37	32	
0:00	36	26	-
0:30	35	31	
1:00	34	26	-
1:30	33	30	
2:00	32	28	-

From the table, we can determine that the temperature outside at 1 am was 34 degrees Celsius.
Therefore, 34 is the correct answer.

VIDEO SOLUTION

144. **42**
The set's starting point is the conclusion that the temperature rises only when the AC is turned off.
When the AC is turned off, the temperature rises linearly, reaching the temperature at the time it was turned off in one hour.

The temperature rises in two instances,
From 0:00 to 0:30, it rises from 26 to 31, so by 1:00, it would have reached 36 degrees, meaning that the temperature at 0:00 would have been 36 degrees Celsius.

From 1:00 to 1:30, it rises from 26 to 30, so by 2:00, it would have reached 34 degrees, meaning the temperature at 1:00 would have been 34 degrees Celcius.

Calculating either of these would have been enough as we have the starting temperature at 23:00 at 38 degree celsius. Since the temperature decreases linearly, we can find the temperature at every instance using another data point.

Drawing out the temperature outside, we get:

Time	Temperature outside	Temperature inside	Mode of AC used
23:00	38	38	
23:30	37	32	
0:00	36	26	-
0:30	35	31	
1:00	34	26	-
1:30	33	30	
2:00	32	28	-

Extrapolating this linear chain of temperature drop (2 degrees drop every hour), we can see that the temperature at 22:00 would be 40 degrees Celsius, and the temperature at 21:00 or 9 pm would be 42 degrees Celsius.

Therefore, 42 is the correct answer.

VIDEO SOLUTION

145. **A**
We can note down the data from the chart into the form of a table, that gives us,

Surfer/Blogger	A	B	C	D	Total
M	10	0			30
N	25	0			30
O	0	0			30
P	5	25			30
X	0	0			30
Y	5	20			30
Total	45	45	45	45	180

We are told that D receives more stars than C from Y. Considering Y has already given 25 stars, it will give 0 stars to C and 5 stars to D.
The only two surfers who have not given any stars to A or B is O and X, and these are the two surfers to give all of their stars to a single blogger.

We are also told that M gives different stars to the four bloggers,
Since he has already given 0 and 10, the remaining distinct stars should add up to 20. The only numbers that are remaining that add up to 20 are 5 and 15.
We know that X rewards one of C or D 30 stars and O rewards one of C or D 30 stars. Given that, M could not have rewarded D 15 stars, since Y rewarded D 5 stars, and D is also going to rewarded 30 stars by O or X, and since the total is same for all, which is 45. This is not possible.
This means that, M rewarded C 15 stars and D 5 stars.
This gives us two cases,
Case-1

Surfer/Blogger	A	B	C	D	Total
M	10	0	15	5	30
N	25	0	0	5	30
O	0	0	30	0	30
P	5	25	0	0	30
X	0	0	0	30	30
Y	5	20	0	5	30
Total	45	45	45	45	180

Case-2

Surfer/Blogger	A	B	C	D	Total
M	10	0	15	5	30
N	25	0	0	5	30
O	0	0	0	30	30
P	5	25	0	0	30
X	0	0	30	0	30
Y	5	20	0	5	30
Total	45	45	45	45	180

We can use these two cases to answer the questions,

D received 5 stars from Y.

VIDEO SOLUTION

146. **B**

We can note down the data from the chart into the form of a table, that gives us,

Surfer/Blogger	A	B	C	D	Total
M	10	0			30
N	25	0			30
O	0	0			30
P	5	25			30
X	0	0			30
Y	5	20			30
Total	45	45	45	45	180

We are told that D receives more stars than C from Y. Considering Y has already given 25 stars, it will give 0 stars to C and 5 stars to D.

The only two surfers who have not given any stars to A or B is O and X, and these are the two surfers to give all of their stars to a single blogger.

We are also told that M gives different stars to the four bloggers,

Since he has already given 0 and 10, the remaining distinct stars should add up to 20. The only numbers that are remaining that add up to 20 are 5 and 15.

We know that X rewards one of C or D 30 stars and O rewards one of C or D 30 stars. Given that, M could not have rewarded D 15 stars, since Y rewarded D 5 stars, and D is also going to rewarded 30 stars by O or X, and since the total is same for all, which is 45. This is not possible.

This means that, M rewarded C 15 stars and D 5 stars.

This gives us two cases,

Case-1

Surfer/Blogger	A	B	C	D	Total
M	10	0	15	5	30
N	25	0	0	5	30
O	0	0	30	0	30
P	5	25	0	0	30
X	0	0	0	30	30
Y	5	20	0	5	30
Total	45	45	45	45	180

Case-2

Surfer/Blogger	A	B	C	D	Total
M	10	0	15	5	30
N	25	0	0	5	30
O	0	0	0	30	30
P	5	25	0	0	30
X	0	0	30	0	30
Y	5	20	0	5	30
Total	45	45	45	45	180

We can use these two cases to answer the questions,

There are two cases formed since we cannot identify for certain two whom did O reward the stars to,
We can identify the number of stars rewarded by M to C.

Hence only statement 1 can be identified uniquely.

VIDEO SOLUTION

147. **B**
Since the increase in imports and exports is at average, the exports and imports in the next 4months will be half of their respective first eight months.

Exports in the first 8 months of 1998-1999 is $21436 so, in the next 4 months it will be 21436/2 = 10718

Total exports= 21346+10718=32154.

Although total imports have been increased the percentage share for all is decreased expect A and G. This means surely one of A and G has the highest increase.

Percentage growth in exports of country A is (23%*32154 - 19%*33979)/19%*33979 = 14.5%
Percentage growth in exports of country G is (12%*32154 - 10%*33979)/10%*33979 = 13.5 %
Hence, the region to which Indian exports witnessed the highest percentage growth is region A i.e USA

VIDEO SOLUTION

148. **B**
We have been given details about the quarterly sales figures. Also, we have been given details about the sales figures every month. Some of the data are missing and some additional conditions have been given in the question. Let us try to complete the pie chart as much as possible with the data available to us.

It is known that the sales figures during the three months of the second quarter (April, May, June) of 2016 form an arithmetic progression.
We know that the sales in April is 40.
Let the sales in May be 40+x and the sales in June be 40+2x.
We know that the total sales in Q2 is 150.
=> 40 + 40 + x + 40 + 2x = 150
3x = 30
x = 10

Therefore, sales in May 2016 = 40 + 10 = 50
Sales in June 2016 = 40 + 20 = 60

Similarly, it has been given that the sales in October, November, and December 2016 form an arithmetic progression.
Sales in October = 100
Sales in Q4 = 360

Let the sales in November be 100+y and the sales in December be 100+2y.
100 + 100 + y + 100 + 2y = 360
300 + 3y = 360
=> y = 20

Sales in November 2016 = 120 and Sales in December 2016 = 140
Sales in Q1 of 2016 = Sum of the sales in the months of January, February, and March 2016
= 80 + 60 + 100
= 240

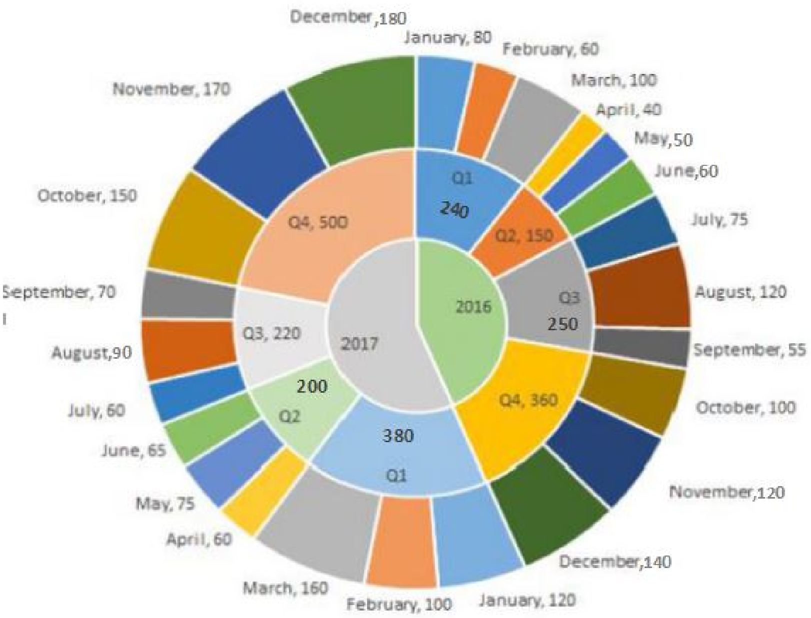
Sales in Q3 of 2016 = Sum of the sales in the months of July, August, and September 2016

= 75 + 120 + 55
= 250

Sales in Q1 of 2017 = 120 + 100 + 160 = 380
Sales in Q2 of 2017 = 65 + 75 + 60 = 200
We know that sales in Q3 of 2017 = 220

Let the sales in August of 2017 be ‘a’.
60 + 70 + a = 220
=> a = 90
Sales in August 2017 = 90
We know that sales in Q4 of 2017 = 500
Let the sales in December of 2017 be ‘d’.
150 + 170 + d = 500
=> d = 180

Sales in December 2017 = 180



Among the given 4 options, we have to find the quarter in which the increase in sale from the previous quarter was the highest.

Q2:
Sales in 2017 = 200
Sales in 2016 = 150

Q1:
Sales in 2017 = 380
Sales in 2016 = 240

Q3:
Sales in 2017 = 220
Sales in 2016 = 250

Q4:
Sales in 2017 = 500
Sales in 2016 = 360

We can eliminate Q3 since the sales has decreased.
Growth in Q2 sales = 50/150 = 1/3 = 33.33%

Growth in Q1 sales = (380-240)/240 = 140/240 = 58.33%

Growth in Q4 sales = (500-360)/360 = 140/360

140/240 > 140/360

Therefore, Q1 has recorded the highest growth in sales and hence, option B is the right answer.

VIDEO SOLUTION

149. **D**
Writing down the data with us in a table form.

	Asia	Europe	Rest of World	Total
Dheeraj	3	7	1	11
Samantha	0	9	4	13
Nitesh	1	6	12	19
Total	4	22	17	43

But this includes countries which have been visited by one person, two people and three people. Essentially, there is an overlap of countries.
It is given to us that 32 distinct countries were visited among the three people,
Using that we can write down the equations,
$I + II + III = 32$...(1)
$I + 2II + 3III = 43$...(2)
Where I, II, and III denote the number of countries visited exactly by 1 of them, 2 of them and all of them.
We are told that USA(ROW) is the only country which was visited by all three people, that means $III = 1$

Subtracting Equation 1 from 2 we get, $II + 2III = 11$
And since III=1 we get $II = 9$

So, we are looking for 9 countries visited by exactly 2 people, and 22 countries visited by exactly 1 person.
Using the information from the problem set, we can note down the following,

Combination	Asia	EU	ROW	Total
Dheeraj				
Samantha				
Nitesh				
Dheeraj and Nitesh	1(China)			
Dheeraj and Samantha		1(France)		
Samantha and Nitesh				
All Three	0	0	1(USA)	
Total				32

We can see that Dheeraj visited only 1 country in ROW so every other value in that column has to be zero for wherever Dinesh is present. And Samantha has not visited any country in Asia so every value with Samantha present in Asia should be zero as well.

Combination	Asia	EU	ROW	Total
Dheeraj			0	
Samantha	0			
Nitesh				
Dheeraj and Nitesh	1(China)	0	0	1
Dheeraj and Samantha	0	1(France)	0	1
Samantha and Nitesh	0			
All Three	0	0	1(USA)	
Total				32

We have the following distribution, we are looking for 9 countries visited by exactly 2 people. And we have identified 2 countries, that means, 7 were countries were visited by both Samantha and Nitesh only. And given that half the countries visited by both of them are in Europe, and total number of countries visited by both of them is 7(only two of them)+1(USA)=8, they visited 4 countries in Europe together and 3 countries in ROW together. And using that information, we can fill in the rest of the table as well. Since we know the number of countries visited by each of them in a particular continent.

Combination	Asia	EU	ROW	Total
Dheeraj	2	6	0	8
Samantha	0	4	0	4
Nitesh	0	2	8	10
Dheeraj and Nitesh	1(China)	0	0	1
Dheeraj and Samantha	0	1(France)	0	1
Samantha and Nitesh	0	4	3	7
All Three	0	0	1(USA)	1
Total	3	17	12	32

Number of countries in EU visited by exactly 1 person is 12.

VIDEO SOLUTION

150. **A**

The revenue of 50 units is 1000. So, the revenue per unit is 20.
The variable cost of 50 units is 700. So, the variable cost per unit is 14.

Let the minimum number of units that are needed to be produced to ensure that there is a profit of at least 50.
So, total revenue is 20X and total cost is 14X+50.
So, Profit is 20X - (14X+50) = 6X-50 > 50
Hence, 6X > 100
Or X > 16
So, the least possible number of units to be produced is 17.

VIDEO SOLUTION

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151. **150**

From the given chart, we can find the average rating on day 2:

$\frac{(5\times 1)+(10\times 2)+(5\times 3)+(20\times 4)+(10\times 5)}{5+10+5+20+10} = \frac{170}{50} = 3.4$

We are given the cumulative average of day 1 and day 2 as 3.1, and the average at the end of day 1 is 3.

Let's take the number of ratings received on day 1 as x; using this overall average and the average on day 2, we get the equation:

$\frac{3x+(50\times 3.4)}{x+50} = 3.1$
$3x + 170 = 3.1x + 155$
$15 \ = \ 0.1x$
$x = 150$

Therefore, the number of ratings received on day 1 is 150.

VIDEO SOLUTION

152. **B**

Average will be = $(120 + 130 + 145 + 165 + 185 + 200 + 220) \div 7 = 166$ (nearly)

VIEW SOLUTION

153. **C**

We have been given details about the quarterly sales figures. Also, we have been given details about the sales figures every month. Some of the data are missing and some additional conditions have been given in the question. Let us try to complete the pie chart as much as possible with the data available to us.

It is known that the sales figures during the three months of the second quarter (April, May, June) of 2016 form an arithmetic progression.
We know that the sales in April is 40.
Let the sales in May be 40+x and the sales in June be 40+2x.
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Let the sales in November be 100+y and the sales in December be 100+2y.
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Sales in November 2016 = 120 and Sales in December 2016 = 140
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Sales in Q3 of 2016 = Sum of the sales in the months of July, August, and September 2016

= 75 + 120 + 55
= 250

Sales in Q1 of 2017 = 120 + 100 + 160 = 380
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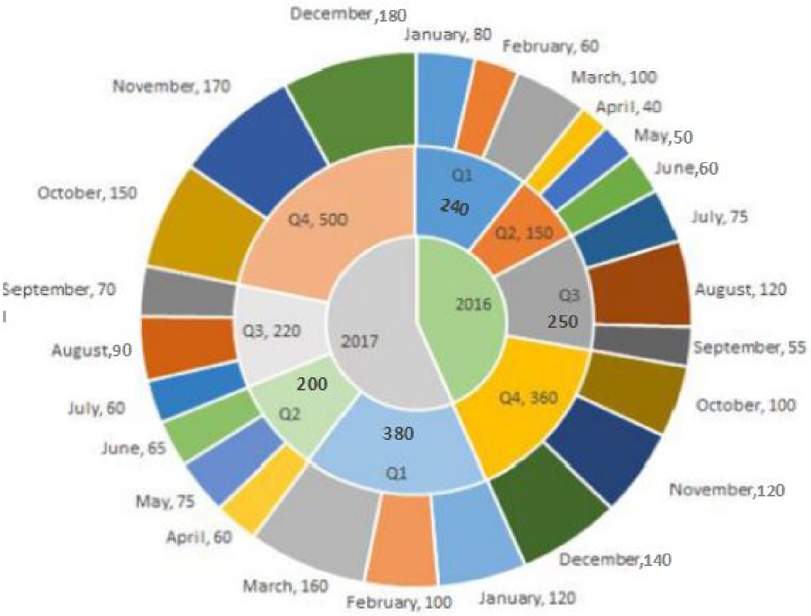
Let the sales in August of 2017 be ‘a’.
60 + 70 + a = 220
=> a = 90
Sales in August 2017 = 90
We know that sales in Q4 of 2017 = 500
Let the sales in December of 2017 be ‘d’.
150 + 170 + d = 500
=> d = 180

Sales in December 2017 = 180



Sales in December 2016 = 140
Sales in December 2017 = 180

Percentage change = (180-140)/140 = 40/140 = 28.57%

Therefore, option C is the right answer.

VIDEO SOLUTION

154. **B**
At 60 kmph, time taken to reach the destination = 200/60 = 3.33 hours.
Fuel consumed per hour at 60 kmph = 4 liters.
Therefore, total fuel consumed = 3.33*4 = 13.33 liters.

VIEW SOLUTION

155. **A**
Total working hours = 1150
So, we have to find the task where the number of working hours is approximately 575
Coding = 425 + 100 = 525
Design = 100 + 75 = 175
Offshore testing = 300
Offshore testing + design = 300 + 100 = 400
So, option a) is the correct answer

VIEW SOLUTION

156. **A**
Total offshore work = 100+420+300 = 820. If 50% of the offshore work were to be carried out onsite i.e 410 carried out onsite.

So new work carried out onsite is in design is [80 + (100*410)/(825)] which is around 130. Also new work carried out onsite is in coding is [100 + (425*410)(825)] which is around 315.

And new work carried out onsite is in testing is [150+(300*410)(825)] which is around 300. Hence the amount of coding done is greater than that of testing.

VIEW SOLUTION

157. **D**
Operations B, C and D have more effort weightage in companies 1,3,5,6.

Also we notice that among them operation E has highest already, and for E company 5 has already the highest percentage. So redistributing the share of E would increase further more. Hence option D.

VIEW SOLUTION

158. **C**
Tamilnadu has maintaines the ranking of second highest tax revenue throughout the five years.

Hence, option C is the answer.

VIEW SOLUTION

159. **A**
Base value must be taken as 100 => 1985 is 100

1984 => $\frac{100.5}{67} * 100$ = 150

1986 => $\frac{141}{67} * 100$ = 211

Only option A has the same values => Option A is the answer

VIEW SOLUTION

160. **B**
Percentage growth of all households South region showed the maximum increase in all households in the given period. (100% + 340%+ 425% = 865%)

VIEW SOLUTION

161. **D**
Since we do not know the region-wise population of 1987-1988, we cannot find the increase in the number of households.
Hence, we cannot find the total income for the year 1995-96.
So the ratio cannot be determined.

VIEW SOLUTION

162. **B**
The revenue of 50 units is 1000. So, the revenue per unit is 20.
The variable cost of 50 units is 700. So, the variable cost per unit is 14.
Let the minimum number of units that are needed to be produced to ensure that there is no loss be X.

So, 20X > 14X + 50
Or X > 8
So, the least possible number of units to be produced is 9.

VIDEO SOLUTION

163. **A**

In order to find the country which has the least trade deficit with India, we have to look for countries whom India exported a lot and imported less.

The only two countries which fit the bill are countries A and I.

Trade deficit of A with India in 1997-98 is 9%*40779 - 19%*33979   = -2.78 billion
Trade deficit of I with India in 1997-98 is 14%*40779 - 20%*33979   = -1.08 billion

Hence, the country with the least trade deficit is country A or USA

VIDEO SOLUTION